Purging Monetary Policy Surprises: Necessary but Not Sufficient[1]

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Abstract

High-frequency monetary policy surprises are predictable from information available before the announcement, and orthogonalising them on pre-announcement controls has become the standard remedy. We ask what that purge achieves. It is *necessary*: the raw surprise implies that output *rises* after a tightening, and every defensible control set restores the contraction. It is *not sufficient*, in a stronger sense than the language of set identification conveys. The estimated response varies substantially across equally defensible control sets, but by less than the sampling uncertainty around any one of them, so the range of published answers restates imprecision rather than recording a disagreement about the world. Two features of the design explain this. The purged instrument barely moves the policy rate once aggregated to the frequency of the VAR, and the shocks generating that range do not describe a common experiment: their implied policy paths diverge sharply and most reverse sign within a year. We propose a cheap, scale-invariant screen on that path. On mechanism, the restriction that the gap between the true and the perceived policy rule is the same in every state is rejected once the state is measured by financial conditions dated before the announcement. This is the footprint of the Fed put, the tendency of the Federal Reserve to ease when equity prices fall without tightening symmetrically when they rise. Its cost is visible in the first stage, where one common slope strips most of the instrument's relevance from tightening surprises while leaving easing surprises intact. It changes which estimator is appropriate.

Keywords: monetary policy shocks; high-frequency identification; orthogonalisation; Fed put; information effect; weak instruments; proxy VAR.

JEL classification codes: E43, E52, E58, C22, C36

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1 Introduction

Over the past two decades, high-frequency identification has become the workhorse for measuring the macroeconomic effects of monetary policy. The idea, from Kuttner (2001) and Gürkaynak et al. (2005), is that the change in interest-rate futures in a tight window around an FOMC announcement isolates the surprise component of policy, which can then instrument a VAR (Gertler and Karadi, 2015; Stock and Watson, 2018) or a local projection (Nakamura and Steinsson, 2018). Identification rests on a single assumption: the surprise is orthogonal to everything known just before the announcement.

Bauer and Swanson (2023a, hereafter BS) show that this assumption fails. Surprises are predictable from pre-announcement macro and financial data (a "Fed response to news"), so they are contaminated by a systematic, non-policy component. The same mechanism, they argue, rather than a genuine information effect, explains the rate–stock co-movements that Nakamura and Steinsson (2018) read as evidence that the Fed reveals private information. BS propose to remove the contamination by projecting the surprise on six pre-FOMC predictors and keeping the residual; Bauer and Swanson (2023b) carry that correction through to VARs and local projections, and the orthogonalisation is now standard practice. Yet it raises questions it does not settle. How much of the identified shock survives the purge: its sign, its magnitude? Does the choice of controls matter, and by how much? And is a single symmetric projection the right correction for the contamination that is actually there?

Let s_τ be the high-frequency surprise around announcement τ. If s_τ were a pure policy shock it would be orthogonal to any variable in the information set $\mathcal{I}_{\tau^-}$ dated strictly before the announcement. It is not: s_τ is partly ex-post forecastable, and that forecastable component reflects the market reaction to the Fed responding to non-policy developments. Predictability of s_τ with respect to $\mathcal{I}_{\tau^-}$ is therefore the empirical footprint of contamination. The BS remedy projects the surprise on pre-announcement controls $X_{\tau^-} \in \mathcal{I}_{\tau^-}$ and keeps the residual,

$$s_\tau \;=\; X'_{\tau^-}\beta + s_\tau^\perp, \qquad \widehat{s}_\tau^{\,\perp} \;=\; s_\tau - X'_{\tau^-}\widehat{\beta}. \tag{1}$$

We assemble an extended panel of strictly pre-announcement controls (macro-announcement surprises, a broad Bloomberg high-frequency financial block, public FRED series, and real-time vintages), identify the shock in a monthly *internal-instrument* VAR, in which the surprise is ordered first and the impulse responses are read off the recursively-identified system (Plagborg-Møller and Wolf, 2021), and then vary the orthogonalisation along every dimension that matters: the control set, the estimator, the frequency, the components of the surprise, the state, and the ordering of purging and aggregation.

The first and organising result is that the purge is *necessary but not sufficient*. It is necessary: the raw surprise yields a wrong-signed, information-contaminated response (output *rises* after a tightening), and orthogonalising on any reasonable control set restores the textbook contraction. But it is not sufficient, and in a stronger sense than the usual set-identification language conveys. Across control sets equally predictive of the surprise the peak output response varies by a factor of two, but a wild bootstrap shows that spread is smaller than the sampling error around any one of them. The variance of the point estimates across control sets is 0.13; the mean bootstrap variance *within* a control set is 0.90. In the units of the responses, the estimates run from -1.0% to -2.3%, a spread of 1.3 points, against a standard error of about 0.9 on each. The design does not deliver an informative magnitude, and the leading source of uncertainty is not which defensible purge one adopts but the estimate's own imprecision.

This is easy to read as the opposite of what it says, so we state it carefully. The estimate certainly *does* depend on the control set: change the six controls and the peak output response moves by a factor of two. What the comparison establishes is that nothing can be *learned* from that dependence, because resampling the same data moves the estimate by more. A researcher reporting -1.0% and one reporting -2.3% have not discovered that the magnitude depends on the controls; they have discovered that neither of them has an informative estimate of it. That is the sense in which the purge is *not sufficient*, and it is stronger than set identification, which would concede that every point in the band is a defensible answer. Here no point in the band is a defensible answer about magnitude, whichever controls produced it.

Where the control set demonstrably *does* matter is elsewhere, and Section 4.3 is about precisely that. Different control sets leave instruments of very different strength and imply policy paths that differ in kind rather than in degree, and those differences are far too large to be sampling noise. The practical lesson is therefore not that one should choose controls carefully because the magnitude depends on them, but that one should choose them so that what has been identified is a monetary experiment at all, and should stop reading the magnitude as though the choice of controls were the binding uncertainty.

Two features of the design explain that, and both are reportable in two lines by anyone using this machinery. The purged monthly instrument moves the two-year rate by one to three basis points on impact: an instrument that weak cannot produce a precisely estimated output elasticity, however carefully it is purged. And the shocks whose responses form the band do not describe a common policy experiment: scaled to a common impact, their implied two-year-rate paths at six months range over a factor of five hundred, and six of ten have the policy rate wrong-signed at twelve months. The purpose of these two facts is to distinguish between the two reasons a band of estimates can be wide. A band is wide either because the underlying experiment is well measured but genuinely ambiguous, or because it is badly measured. Both apply here: the instrument is weak, which caps the precision of any single estimate, and the objects being compared are not the same experiment, which makes the width of the band uninterpretable even in principle. We therefore propose a screen on the implied path, asking of a candidate shock only that the policy rate rise on impact, not explode, and still be on the same side of zero after a year. It is invariant to every normalisation choice in this literature and uses two numbers any VAR already produces. Applied to the ten shocks it rejects eight, the benchmark among them.

BS give the contamination a microstructure. Writing a_τ for the Fed's true responsiveness to intermeeting news x_τ and $\bar{a}_\tau$ for the responsiveness markets price, the surprise is $MPS_\tau = \phi_\tau x_\tau + \varepsilon_\tau$ with $\phi_\tau = a_\tau - \bar{a}_\tau$ the *reaction gap*. Their reading is that agents *persistently underestimate* the rule: $\phi_\tau = \delta > 0$ and roughly constant, which biases the estimated output response *upward* and is corrected by a single symmetric projection. That restriction is testable, and we test it on an *ex-ante* state, the pre-announcement financial state, rather than on the realised sign of the surprise. Interacting the control vector with that state rejects a common slope ($W = 17.15$ on six restrictions, wild-bootstrap $p = 0.030$), and a penalised state-dependent projection raises the cross-validated R^2 of the surprise from zero to 0.07 where the pooled projection has no out-of-sample power at all. The reaction gap is state-dependent.

Our reading of *which* state is the Cieslak and Vissing-Jorgensen (2021) *Fed put*: the systematic component the purge removes is a one-sided easing response to financial deterioration (when markets fall the Fed is expected to cut, and does), so it loads almost entirely on the easing surprises, while tightenings, which typically follow ordinary good-news data, carry little contamination. It is worth being explicit about what an "easing surprise" is here. It is a negative value of MPS_τ, which arises whenever policy turns out easier than priced. That includes a cut larger than expected and a cut where none was expected, but also a *tightening* smaller than the one the market had priced. What defines the easing state is the sign of the surprise relative to expectations, not the direction of the policy move itself. Section 2 shows that a Fed put in the *rule* translates into a one-sided gap in the *surprise* even when agents are fully rational, provided the instrument through which they price the rule is linear, and derives the three implications on which the two hypotheses part company. Section 5 tests them. The sharpest is the direction of the raw bias: a state-invariant $\delta > 0$ predicts an upward-biased output response on *both* halves of the sample. It is present on only one. Identified from easing surprises alone, the *raw* instrument produces the entire output puzzle, so there is an upward bias for the purge to remove. Identified from tightening surprises alone it already produces a textbook contraction, so on that half there is no bias to correct, and a symmetric purge that corrects it anyway is removing something that was not there. Identified from tightening surprises alone the *raw* instrument already gives a textbook contraction (industrial production to about -1.6%); identified from easing surprises alone it gives the entire puzzle (output *rises*, to about $+0.8\%$ at a year), which orthogonalisation then removes.

This has a consequence the standard purge overlooks. A single symmetric projection, estimated mostly off the high-variance easing and crisis observations, is applied to *every* surprise: it under-cleans the easings and over-cleans the near-uncontaminated tightenings. The clearest evidence is in the first stage rather than in any impulse response, and is invariant to how the responses are scaled: purging removes 29% of the instrument's impact on the two-year rate on the easing half and 77% on the tightening half. One pooled slope strips three-quarters of the relevance from the half of the sample that carried little contamination to begin with.

What does *not* follow is a different magnitude, and the reason is instructive. A projection conditioned on the *realised sign* of the surprise appears to moderate the peak response substantially; the same projection conditioned on the *ex-ante* state does not, giving -2.6% against a pooled -2.3%. Same estimator, same sample: the apparent moderation is a property of conditioning on the dependent variable, which truncates the shock within each half and attenuates the instrument built from it. What survives is the specification result (one slope is the wrong object), and, by Section 4.2, the reason correcting it does not visibly move the answer: the sampling error is larger than the differences at stake.

Three further findings round out the argument. First, the purge is *complementary* to the information-effect corrections rather than a substitute: the Jarociński and Karadi (2020) stock-price sign restriction fixes what the purge cannot (a third of orthogonalised surprises still lie in the information-effect quadrant), while the purge fixes the output sign the sign restriction alone does not, and an informationally-robust Miranda-Agrippino and Ricco (2021) instrument remains significantly predictable by the BS controls.

Second, the contamination is uneven along the *other* dimensions too: it lives disproportionately in the forward-guidance (path) component of the surprise, which is twice as predictable as the target, and peaks in the zero-lower-bound regime; and its coefficient is outlier-driven, so a robust projection roughly halves the spread across purges. A two-state Markov-switching model that lets the *data* date the regimes confirms that the high-predictability state is persistent and endogenously timed, and that the path component sits in it three-quarters of the sample against roughly a third for the target. How one *summarises* the pre-announcement panel (a minimal test-based projection, or its principal components) matters more than which six controls one picks.

Third, the purge has a mechanical cost at the monthly frequency. Because the BS controls are asset prices measured over multi-week windows, in a month with more than one announcement the window used to purge a later surprise *contains* the earlier one, so purging can drain genuine monetary variation: a *leakage* that is the mirror image of the aggregation bias in Amodeo (2026). The two results concern different objects. Amodeo (2026) studies the *order* of orthogonalisation and aggregation and shows that summing before purging is biased; we hold his recommended order fixed and study the *timing of the control window*, showing that it leaks genuine monetary variation and weakens the first stage. A control window starting the day after the previous announcement should resolve the trade-off, though confirming that requires intraday control data we have not yet assembled.

Our contribution is a test of the mechanism the standard purge assumes, and the estimator that follows from rejecting it. Relative to BS we take their reaction-gap decomposition as given, show that their constant-gap restriction is the nested null, reject it, and quantify what a symmetric projection costs. Relative to Jarociński and Karadi (2020) and Miranda-Agrippino and Ricco (2021) we show their corrections are complementary to, not substitutes for, the purge. Relative to Amodeo (2026), who shows that summing surprises before purging is biased so each surprise should be purged first, we show that even under his recommended ordering the *timing of the control window* still leaks the previous shock. Closest in framework is Bolhuis et al. (2024), who adopt the same rule/information decomposition to build a cross-country shock dataset and read a near-zero orthogonalisation R^2 as evidence that little genuine variation is removed; we take that decomposition as given but read a small R^2 the other way. A near-zero orthogonalisation R^2 does not establish that there was little to remove. It establishes that the residual instrument is weak, and a weak instrument cannot pin down a magnitude however clean it is. Contamination is a property of the population projection and biases the impulse response whether or not the sample R^2 is large.

In short: the purge fixes the sign but does not deliver an informative magnitude, and the spread across defensible purges is smaller than the sampling error around any of them; a small orthogonalisation R^2 signals weak identification rather than a clean shock; the purge cleans only the rule channel, so a sign restriction is still needed; the reaction gap is state-dependent, which changes the right estimator without changing the answer; and purging before aggregating leaks genuine variation.

Related Literature

The paper builds on four literatures. The first is *high-frequency identification* of monetary policy. Kuttner (2001) measures policy surprises from fed-funds futures; Gürkaynak et al. (2005) extend this to the whole expected path and distinguish a target from a path factor; Gertler and Karadi (2015) and Stock and Watson (2018) use such surprises as external instruments in proxy VARs and Nakamura and Steinsson (2018) in local projections; Ramey (2016) surveys the approach. We adopt this machinery (identifying the shock via the Plagborg-Møller and Wolf (2021) internal-instrument equivalence, which is stable where the external-instrument first stage is weak), and study what happens to the identified responses as the surprise is cleaned.

The second is the *predictability* of the surprises and the "response to news." Bauer and Swanson (2023a) document the predictability and propose the purge this paper anatomises, though the predictability of rate expectations over the business cycle is older (Cieslak, 2018). The reaction-gap reading requires a measure of what agents actually believe the rule to be: Bauer et al. (2024) construct exactly that from surveys, and Schmeling et al. (2022) document the survey underprediction of rate changes that BS invoke.

The reaction-gap reading also has a directly measurable ingredient, and one paper measures it. Bauer et al. (2024) elicit from surveys what forecasters believe the Fed's response coefficients to be, the object we write as $\bar{a}_\tau$, and document that those perceptions move over time and depart from estimated rule coefficients. That is the empirical counterpart of $\phi_\tau = a_\tau - \bar{a}_\tau$, and it matters for the argument here in two ways. It makes the reaction gap an observable rather than a residual, so the constant-gap null of (5) is in principle testable on survey data as well as on surprises; and, because their perceptions vary with the state of the economy, it supplies independent evidence for the state dependence that Section 5 infers from the surprises alone. A joint test (does the survey-measured gap widen in the states where the surprise becomes predictable?) is beyond this paper's data but is the natural next step, and would discriminate between our reading and Bauer and Swanson's (2023a) more sharply than anything available from high-frequency data by itself. Chen (2025) finds that financial conditions alone account for most of the predictability and that the standard predictors operate through market pricing rather than the Fed's private information (evidence, like ours, for a response-to-news reading), and Cotton (2025) documents the same predictability across countries. More broadly, Brennan et al. (2024) show that the choice of construction *method* can move the resulting shock series as much as the choice of data, and Bu et al. (2021) pursue an alternative remedy through identification by heteroskedasticity. We show that two conditioning sets, financial-and-macro controls versus Greenbook forecasts, capture complementary contamination, and that the predictability is state-dependent and outlier-driven.

The third is the *information effect*. Nakamura and Steinsson (2018) and Jarociński and Karadi (2020) argue that policy announcements also reveal the central bank's assessment of the economy; Jarociński and Karadi (2020) separate the two by the sign of the simultaneous stock-price move. We show this correction is orthogonal to the predictability purge, and that the two are needed together. Hoesch et al. (2023) show the information channel is itself time-varying, a variation that parallels the state-dependence we find in the *predictability*.

The fourth is the *structure of the systematic component and of weak identification*. Cieslak and Vissing-Jorgensen (2021) document the Fed put (a one-sided easing response to equity declines, operating through downgrades to the growth outlook), which is the economic content of our asymmetry. Hack et al. (2024) show that a constant-coefficient projection on expected fundamentals cannot deliver a clean shock when the systematic response varies; our state-dependent decomposition is the correctable special case of their critique. Staiger and Stock (1997), Stock and Yogo (2005) and Montiel Olea and Pflueger (2013) formalise weak-instrument inference, which is exactly the regime the purge induces at meeting frequency, and Montiel Olea et al. (2021) and Jentsch and Lunsford (2022) the corresponding inference in proxy VARs; the target/path (Gürkaynak et al., 2005; Swanson, 2021) and multi-shock (Jarociński, 2024; Lewis, 2021) decompositions locate where the predictable variation resides. Closest to our leakage result is Amodeo (2026); methodologically closest is Bolhuis et al. (2024).

The remainder of the paper proceeds as follows. Section 2 sets out the decomposition of the surprise, derives the Fed put as a state-dependent reaction gap, and states the nested hypothesis that organises the empirics. Section 3 describes the control panel and the VAR. Section 4 establishes the necessary-but-not-sufficient result and shows that the magnitude is not informatively estimated. Section 5 is the heart of the paper: it tests the constant-gap null and states what follows from rejecting it. Section 6 locates the contamination along the remaining dimensions (leverage, target versus path, and regime), and Section 7 asks how the pre-announcement panel should be summarised. Section 8 adds the information-effect dimension and Section 9 the purge-then-aggregate leakage. Section 10 concludes.

2 A state-dependent decomposition of the surprise

2.1 Two channels and one coefficient

Let the central bank set the expected path of policy as a systematic function of the information available at the announcement plus an exogenous policy shock. The high-frequency surprise s_τ, the revision to the expected path in a tight window around announcement τ, then admits a three-way decomposition,

$$s_\tau = \underbrace{\varepsilon_\tau}_{\text{policy shock}} + \underbrace{r_\tau}_{\text{rule channel}} + \underbrace{\iota_\tau}_{\text{information channel}}, \tag{2}$$

with three channels that we will refer to by number for the rest of the paper. *(1)* ε_τ, the structural monetary shock: the committee did something the rule did not call for. *(2)* r_τ, the *rule* channel: the committee did what its rule called for, but the market's estimate of that rule was wrong, so the correct response arrives as a surprise: this is the "response to news" that Bauer and Swanson identify. *(3)* ι_τ, the *information* channel: the announcement reveals the central bank's own assessment of the economy, and the market revises its outlook rather than its view of the rule. A concrete case separates (1) from (2). In June 2003 the Fed cut rates by less than markets had priced (Bauer and Swanson, 2023a), which registers as a *tightening* surprise; the committee was not tightening, and nothing about the economy was revealed that markets did not know: the surprise was a revision to how aggressively the Fed was thought to respond. That is channel (2) with $\varepsilon_\tau \approx 0$. Only ε_τ is a policy shock; (2) and (3) are the two contaminations, and they are removed by different instruments.

Doing so *sequentially* is valid under a *separability* assumption. By sequential we mean the following two steps. First orthogonalise the surprise on pre-announcement controls, which removes r_τ. Then apply the Jarociński and Karadi (2020) poor-man sign restriction to the residual, discarding or re-signing the announcements at which the purged surprise and the statement-window equity return move together, which removes what is left of ι_τ. Section 8 runs both steps and compares the resulting responses with the raw surprise and with the orthogonalised-only surprise (Figure 11). Separability is what licenses doing them in sequence rather than jointly. The rule channel is spanned by pre-announcement observables, $r_\tau = \beta' X_\tau$ with X_τ dated strictly before the announcement; the information channel is orthogonal to X_τ and identified instead by its *contemporaneous* sign, ι_τ co-moving positively with the statement-window equity return while ε_τ co-moves negatively. Under separability the population projection of s_τ on X_τ recovers the rule channel without touching the information channel,

$$s_\tau = \beta' X_\tau + s_\tau^\perp, \qquad \beta \neq 0 \text{ iff the rule channel is present}, \qquad \widehat{s}_\tau^\perp = s_\tau - \widehat{\beta}' X_\tau, \tag{3}$$

so the orthogonalised surprise is by construction uncorrelated with X_τ yet still carries ι_τ. Separability is exactly what makes the two steps compose: each acts on a distinct subspace, so the order is immaterial and neither undoes the other. Section 8 verifies that the two corrections are in fact complementary; the rest of this section concerns the rule channel alone.

Bauer and Swanson (2023a) give the rule channel its microstructure. Write the Fed's reaction to the intermeeting news x_τ as $a_\tau x_\tau$ and the reaction priced by markets as $\bar{a}_\tau x_\tau$. Because x_τ is realised inside the pre-announcement window while a_τ is not observed, the surprise is

$$MPS_\tau = \underbrace{(a_\tau - \bar{a}_\tau)}_{\phi_\tau} x_\tau + \varepsilon_\tau, \qquad \mathbb{E}[\varepsilon_\tau \mid x_\tau, \phi_\tau] = 0. \tag{4}$$

The single object ϕ_τ, the *reaction gap* between the Fed's true responsiveness and the one agents price, carries the whole rule-channel contamination. It is also the object on which this paper and Bauer and Swanson (2023a) differ.

2.2 The Bauer–Swanson conjecture and its nested alternative

Bauer and Swanson (2023a) conjecture that agents *persistently underestimate* the rule: $\bar{a}_\tau < a_\tau$ on average, so $\phi_\tau = \delta > 0$ and roughly constant. Their supporting evidence is of a piece with that reading: recursive estimates of the policy rule trending up since the 1980s, Greenspan's 2001 remark that the Fed had come to respond "more aggressively than had been our wont in earlier decades," and survey forecasts that underpredict funds-rate changes (Cieslak, 2018; Schmeling et al., 2022). Under $\phi_\tau \equiv \delta$ a single symmetric projection is the right correction and the purge is complete.

We propose the alternative that the reaction gap is *state-dependent*, and in a specific direction. Cieslak and Vissing-Jorgensen (2021) document a *Fed put*: policy accommodation follows poor equity returns, and the response is strongly one-sided: the loadings of the funds-rate change on *negative* intermeeting excess returns sum to 5.35 against −1.80 on positive returns, the latter individually insignificant. Their preferred mechanism is not market support but the growth outlook: equity declines transmit to consumption and investment, the staff forecast is downgraded, and the Fed eases. That is a *rule* channel in the sense of (2), not an information channel: the Fed is responding to news the market can also see.[2]

Let the state be the sign of the intermeeting excess equity return, $s_\tau = \mathbf{1}\{rx_\tau < 0\}$ (an *ex-ante observable*, which will matter), and write the asymmetric rule $a(s) = a_+ + (a_- - a_+)\mathbf{1}\{s = -\}$ with $a_- > a_+ \geq 0$. If markets priced the kink correctly, $\bar{a}(s) = a(s)$ and there would be no predictability in either state. Two departures generate one:

(i) **Symmetric pricing of an asymmetric rule.** Agents price a single slope $\bar{a}$, the average responsiveness, because the instruments through which they express it, the futures curve and survey point forecasts, are linear in observables. Then $\phi_- = a_- - \bar{a} > 0 > a_+ - \bar{a} = \phi_+$: agents appear to underestimate the easing response and to *over*estimate the tightening response. Appendix D shows this requires no irrationality, and that $|\phi_+| \ll \phi_-$ because the stress state carries most of the news variance.

(ii) **Uniform underpricing.** Agents price $\bar{a}(s) = a(s) - \delta$ for a constant $\delta > 0$. Then $\phi_- = \phi_+ = \delta$: this is Bauer and Swanson's (2023a) benchmark.

The two are nested in ϕ, which makes them testable against each other:

$$H_{\text{BS}}: \ \phi_- = \phi_+ = \delta > 0 \qquad \text{versus} \qquad H_{\text{FP}}: \ \phi_- > \phi_+, \ \ \phi_+ \approx 0. \tag{5}$$

[2] This also supplies a consistency check we return to in Section 8. Cieslak and Vissing-Jorgensen (2021) find that Greenbook growth revisions absorb most of the equity return's predictive power for the funds rate. Accordingly, the predictability we find *beyond* the Greenbook should not be carried by the equity control, and it is not; it is carried by trailing payroll growth.

2.3 Three implications that separate them

The first implication, (T1), is about the symmetry of the projection *slope* rather than of its fit. Under H_{BS} the state-conditional slopes are equal; under H_{FP} they are not. The natural test interacts the controls with the state and tests the interaction block. It matters that the test be on the *slope*, because the state-conditional fit is uninformative: from (4),

$$R^2(s) \;=\; \frac{\phi_s^2\ \mathrm{Var}(x \mid s)}{\phi_s^2\ \mathrm{Var}(x \mid s) + \sigma_\varepsilon^2(s)}, \tag{6}$$

so an R^2 gap arises either from $\phi_- > \phi_+$ or, at a common ϕ, from $\mathrm{Var}(x \mid \mathrm{stress}) > \mathrm{Var}(x \mid \mathrm{normal})$. Stress states have the larger news variance, so a raw R^2 comparison is biased *towards* H_{FP} even when H_{BS} is true. We therefore report state-conditional R^2s as description and the interaction test as evidence.

The second implication, (T2), concerns the direction of the raw bias, and it is the sharpest cut. Embedding (4) in an outcome equation $y_{t+h} = \theta_h \varepsilon_t + \psi_h(s_t) x_t + u_{t+h}$ and taking the proxy-IV estimand, H_{BS} (in which $\phi \equiv \delta$ and, with no state dependence in the instrument, ψ_h enters only through its mean) gives

$$b_h^{\mathrm{raw}} - \theta_h = \frac{\delta\,(\psi_h - \delta\,\theta_h)\,\sigma_x^2}{\sigma_\varepsilon^2 + \delta^2 \sigma_x^2}, \tag{7}$$

The left-hand side is the probability limit of the proxy-IV estimator at horizon h minus the structural response it is meant to recover, so (7) is an asymptotic bias rather than a finite-sample one. That it depends on θ_h is a feature and not a defect. The contaminating term ϕx_t enters the outcome equation twice, directly through ψ_h and indirectly through its correlation with the policy shock, and the second channel is proportional to the object being estimated. The bias therefore vanishes exactly when $\psi_h = \delta\theta_h$, which is the knife-edge case in which non-policy news happens to move output in the same proportion the policy shock does. Appendix D derives (7) and its state-dependent counterpart (32) from the same primitives. which for small δ has the sign of $\psi_h > 0$: persistent underestimation biases the estimated output response *upward*, and (because δ does not depend on the state) it does so *by the same amount on both halves of the sample*. That is a prediction about where the output puzzle should be found, and it is directly checkable: identify the shock from tightening surprises alone and from easing surprises alone and ask whether both are biased upward. H_{FP} predicts instead that the upward bias is confined to the easing half, since $\phi_+ \approx 0$ leaves the tightenings uncontaminated.

The third implication, (T3), is real-time detectability. A persistent gap $\delta > 0$ is a standing forecast error: agents would have to be wrong in the same direction, meeting after meeting, in a way an econometrician with their information set could have exploited in real time. An expanding-window, real-time-vintage projection of the surprise on the controls is the test. We record at the outset that this implication is the weakest of the three: with δ small and σ_ε^2 large, $\delta > 0$ is compatible with negligible out-of-sample R^2, so a failure to forecast is suggestive rather than decisive. (T1) and (T2) carry the weight.

2.4 What the two hypotheses imply for the purge

The reason (5) is worth testing is that the two hypotheses call for different estimators. Pooled least squares returns one slope,

$$b = \frac{\mathbb{E}[\phi(s)x^2]}{\mathbb{E}[x^2]} = \sum_s \phi_s w_s, \qquad w_s = \frac{\mathbb{E}[x^2 \mid s]\,\Pr(s)}{\mathbb{E}[x^2]}, \tag{8}$$

a variance-weighted average with $\phi_- > b > \phi_+$, and leaves the residual $\varepsilon_t + [\phi(s_t) - b]x_t$.

Proposition 1 (A state-dependent projection is the right purge)**.** *The pooled purge subtracts the single slope b of (8) and so leaves $\varepsilon_t + [\phi(s_t) - b]x_t$: it under-cleans the high-ϕ state and over-cleans the low-ϕ state. A projection whose slope is conditioned on the state, $b(s_t) = \phi(s_t)$, removes the systematic term exactly and returns ε_t. The pooled purge is the correct estimator if and only if H_{BS} holds, $\phi_- = \phi_+$.*

Proposition 2 (The impulse-response bias of a symmetric purge)**.** *Let the outcome respond to the shock and, because x_t is genuine news, to the state as well, $y_{t+h} = \theta_h \varepsilon_t + \psi_h(s_t)\, x_t + u_{t+h}$. The state-specific purge identifies θ_h exactly, whereas the pooled purge carries a bias*

$$b_h^{\text{pool}} - \theta_h = \frac{(\psi_h^- - \psi_h^+)\, W - \theta_h V_\phi}{\sigma_\varepsilon^2 + V_\phi},$$

Both expressions are probability limits, and both carry a term in θ_h for the reason given after (7). where $W > 0$ is the state-conditional covariance the pooled purge leaves uncleaned and $V_\phi \geq 0$ the x^2-weighted variance of ϕ across states. The bias vanishes if and only if the reaction gap is state-invariant ($\phi_- = \phi_+$, so $W = V_\phi = 0$) or *the news transmits to output symmetrically ($\psi_h^- = \psi_h^+$).*

The derivation of the asymmetric rule from a kinked outlook response, and the extension to a continuously time-varying ϕ_t are in Appendix D. Three readings are worth stating. First, Proposition 2 makes Bauer and Swanson (2023a) a corner rather than a rival: under H_{BS} their purge is exactly right, and the question is empirical, not conceptual. Second, both conditions must fail for the symmetric purge to be biased: the reaction gap must be state-dependent (an instrument-side fact) *and* the news must transmit asymmetrically (an outcome-side fact, since financial deterioration need not move output like strong activity); Section 5 shows both fail. Third, the state-dependence in (5) is a disciplined special case of the time-varying $\tilde{\phi}_t$ of Hack et al. (2024). Their point is that a constant-coefficient projection on expected fundamentals cannot deliver a clean shock when the systematic response varies; ours is that when the variation is indexed by an *observable* state it can be corrected by conditioning, which is why we define the state on the intermeeting equity return rather than on the realised surprise.

Two further facts, developed below, qualify what the purge can deliver even when it is correctly specified. A small orthogonalisation R^2 is not reassurance but precisely the *weak-instrument* regime: contamination is a population fact ($\beta \neq 0$), and even a small population β can sign-flip the impulse responses, so "little predictability" and "little contamination" are different statements (Section 7). And β is not unique: many equally-predictive choices of X_τ span nearly the same predictable direction yet imply peak magnitudes that differ by a factor of two, so the purge is *set-identifying* rather than point-identifying (Section 4).

3 Data and design

Every control uses only information dated strictly before the announcement. For a meeting on date τ the window ends on the last weekday before τ and begins a fixed number of trading days earlier; the Bauer–Swanson benchmark uses a sixty-six-trading-day window, about thirteen weeks, and the panel also carries the same controls measured over one-, two-, three-, six- and twelve-month windows so that the window length can itself be varied. The first block is sixteen macro-announcement surprises, each entering both raw and standardised by the pre-2019 standard deviation of that release's surprises; the second is fifty-three Bloomberg high-frequency financial changes, with prices, indices, exchange rates and volatility measures entering as log-changes and yields, spreads, policy rates and financial-conditions indices as level changes; the third is public FRED series entered as window changes and levels; and the fourth duplicates the revisable series with a real-time, as-of-the-meeting ALFRED vintage, so that any predictability cannot be a mechanical look-ahead artefact. Coverage is ragged (market prices reach back to 1988, macro surprises begin in August 1997, and several financial series start only in the 2000s or 2010s), so every test uses a complete-case filter or a matched sample start, and because the window grid is nested by construction it is used with subset search or LASSO rather than a single joint F-test. Table 13 in Appendix A lists every series with its source, transform and coverage.

We embed each candidate shock in a monthly, five-variable vector autoregression whose variables are, in order, the shock itself, one hundred times the log of industrial production, one hundred times the log of the consumer price index, the Gilchrist–Zakrajšek excess bond premium (Gilchrist and Zakrajšek, 2012), and the two-year Treasury rate. The system is estimated with twelve lags over 1990m1–2023m12, with exogenous dummies for 2020m3–m8 to neutralise the pandemic outliers, and the shock is the surprise orthogonalised on a control set as in (1) and summed to the calendar month. The shock is ordered first and the system is identified recursively by a Cholesky factorisation: this is the Plagborg-Møller and Wolf (2021) internal-instrument counterpart of a proxy (external-instrument) SVAR. Because the purge removes exactly the systematic component of the surprise that co-moves with realised rates, the orthogonalised surprise is a weak monthly instrument; a standard external-instrument VAR would then divide by a near-zero first stage and produce explosive responses, whereas the internal-instrument recursion targets the same object while remaining numerically stable. The identified policy-rate response is normalised to a 25 basis-point increase.[3]

Two facts about this design should be stated plainly at the outset, because they condition everything that follows. First, the purged monthly surprise is a *weak* proxy: its robust first-stage F against the two-year-rate innovation is 0.54, against 5.8 for the raw surprise. Table 14 in Appendix B reports, for every shock used below, the monthly and meeting-frequency first stages and the Montiel Olea and Pflueger (2013) effective F.

The natural objection is that this measures the wrong thing: a surprise built from fed-funds and Eurodollar futures need not load most strongly on a *two-year* yield, so weakness against GS2 might be an artefact of the policy variable rather than a property of the instrument. It is not. Re-estimating the same system with the effective funds rate, the three-month bill, and the one-, two-, five- and

[3] *Reporting conventions.* Unless stated otherwise, every impulse response is reported as the *peak* and the *twelve-month* value, in that order, with the estimation sample named; magnitudes are not comparable across samples, and several of the differences reported below between the full-sample and post-1998 windows are sample effects rather than purge effects. The two-year normalisation slope is estimated on the same events, so the *ranking* of magnitudes across purges is more trustworthy than their absolute levels.

ten-year Treasury yields in the policy slot, and with the **?** shadow rate to allow for the lower bound, all on a common 1990–2023 sample, the purged instrument's first stage never rises above $F = 1.05$. The best of the seven is the one-year rate, at $\iota = 1.09$ bp and $F = 1.05$. Every one of them is an order of magnitude below any conventional threshold, so the weakness belongs to the purged instrument and not to the choice of policy variable (Table 17).

The shape of that comparison is a check on the design rather than a caveat, and it is worth reporting for its own sake. A monetary surprise constructed from fed-funds and Eurodollar futures should move the short end of the curve hardest and lose its grip as maturity rises. It does. The first stage is hump-shaped in maturity for both the raw and the purged surprise, peaking at the one-year rate, and then declining monotonically: for the raw surprise ι falls from 3.09 bp at one year to 2.42 at five and 1.23 at ten, and for the purged surprise from 1.09 bp to 0.51 and finally to -0.08 bp, an impact statistically indistinguishable from zero ($F = 0.00$). The ten-year yield is the natural placebo here, and the instrument fails to move it. Had it done so we would be looking at a term-premium or risk factor rather than a policy shock, and the whole design would be in question; that it does not is evidence the surprise is picking up what it is supposed to.

One asymmetry between the raw and the purged surprise deserves recording. On the *raw* surprise the choice of policy variable does matter: the one-year rate gives $F = 10.2$ against 5.8 for the two-year, and it passes the path screen of Section 4.3. On the purged surprise no choice helps, because none of the seven clears $F = 1.1$. We keep the two-year rate as the baseline for comparability with Bauer and Swanson (2023b) and report the one-year results alongside. With a first stage this weak, identification comes substantially from the recursive timing assumption rather than from the instrument, and the sensitivity of the magnitude to the purge, documented in Section 4, is the signature of that weakness rather than evidence against it. Second, and consequently, inference calls for weak-instrument-robust confidence sets whose coverage is constant in instrument strength, rather than a moving-block bootstrap, which is valid only under strong-proxy asymptotics (Jentsch and Lunsford, 2022) and would under-cover in this regime. The natural tool is an Anderson–Rubin-type interval that is uniformly valid even when the proxy is weak (Montiel Olea et al., 2021). We overlay such a band on the identified set of Section 4; elsewhere responses are point estimates and should be read as such. The decomposition is reported in Table 1 and the band in Figure 3.

Because the claim that the purge is *necessary* rests on the raw surprise being wrong-signed, Table 16 in Appendix B repeats the raw and `BS6` responses across lag length (6 and 12), sample (1990–2007, 1990–2019, full), policy indicator (one- and two-year rate) and identification (internal versus external instrument).

3.1 Specification and stability diagnostics

This design is informative about the *sign* and the *shape* of an impulse response and much less informative about its *level*. That is a measured property of the estimated system rather than a concession made after the fact, and we establish it before any impulse response rather than in a robustness appendix, because two of the three diagnostics below bound what the responses can afterwards be asked to mean.

The purged monthly surprise is a *weak* proxy. The quantity that matters is the impact response of the policy variable to the ordered-first shock,

$$\iota \;\equiv\; \left[\mathrm{chol}(\Sigma)\right]_{\mathrm{GS2},\,1} \;=\; \frac{\mathrm{Cov}\left(u_t^{\mathrm{GS2}},\, u_t^{z}\right)}{\sigma_{u^z}}, \tag{9}$$

where $\mathrm{chol}(\Sigma)$ is the lower-triangular Cholesky factor of the reduced-form covariance Σ with the instrument ordered first, and u^z is the instrument's own reduced-form innovation. In words, ι is the covariance between the policy variable's innovation and the instrument's, per standard deviation of the latter. Equation (9) is the monthly first stage of the instrument for the two-year rate, expressed per standard deviation of the instrument. It answers a concrete question: in a month whose purged surprise is one standard deviation larger than usual, how much further does the two-year rate move? For the instrument to speak directly about a conventional 25 bp policy move, the answer would have to be of that order. Across the ten candidate shocks it ranges from 0.9 to 3.0 basis points.

The consequence for normalisation is easy to miss, and it is arithmetic before it is econometrics. Scaling a response so that the policy rate moves a fixed 25 bp on impact means multiplying the entire response by $25/\iota$. For the strongest of the ten shocks that multiplier is 8.2; for the weakest it is 29.2, so two shocks the convention is meant to place on a common footing are magnified by factors differing three and a half times over, and the one magnified most is magnified *because* it was weakest. The reason is that ι is not a unit chosen by the econometrician but an estimate of how much of the policy rate the instrument actually moves. Dividing by it awards the largest scale factor to the shock about which the data are least informative, with a sampling error that enters squared, and orders output responses by first-stage weakness rather than by transmission, which is what Section 4.3 finds, correlating the peak output response with the reciprocal of ι at $+0.79$.

We therefore normalise each shock by a *common* announcement-window slope, so that all shocks are the same size in surprise units, and report ι alongside as the first-stage diagnostic it is (Table 14). Responses are then comparable across control sets without the scale of any one of them being set by its own weak first stage, and Section 4.2 shows that the choice also determines which inference is available.

Two further properties of the estimated system bear on how far the responses can be read, and are reported in Appendix B.1: the dynamics sit at a unit root, which is why we read shapes rather than levels beyond two years, and a non-trivial share of the monthly instrument is predictable from the system's own lags, which is why a purge validated at meeting frequency is not automatically a purge at the frequency of the VAR.

Sections 4 and 4.3 are organised around the division of labour these diagnostics imply, the second of them showing why the shape of the policy path, not its level, is the object that survives. Table 15 in Appendix B reports ι, ς and the largest companion root for every shock, and Table 16 re-estimates the system with $\Delta \log \mathrm{IP}$ and $\Delta \log \mathrm{CPI}$ and with six lags, so that the reader can see which conclusions survive a stationary specification.

4 Necessary but not sufficient: the identified set

Against the common design of Section 3 we compare the BS shock (`BS6`) and the un-purged surprise (`RAW`) with two groups of alternative six-control sets (Table 14 of Appendix B). Both hold the *amount* of predictability at the `BS6` level (in-sample $R^2 \approx 0.11$–0.15), so each removes a comparable slice of the surprise and leaves an instrument of comparable strength; they differ in *what* they remove.

Group A's five sets span the *same* predictable factor as BS from different variables, canonical correlation 0.94–0.99 with the six BS controls, while Group B's three sets are deliberately BS-*distinct*, at canonical correlation 0.64–0.71.[4]

Bauer and Swanson (2023a) condition on six pre-announcement controls (the nonfarm-payrolls surprise, trailing payroll growth, the three-month S&P 500 return, the change in the Treasury yield-curve slope, a commodity-price change, and Treasury-yield skewness), so the predictability they remove lives in a financial-market block alongside a labour-market surprise. Bauer and Swanson (2023b) develop the same diagnostic across a wider set of surprise measures and samples and reach the same conclusion about the need for the purge; the control set below is theirs. The Group A sets reach that predictability from different variables but cannot help re-spanning the same block: every one contains a broad commodity index and a Treasury-rate or yield-curve variable, and all but one adds an equity index and a credit-spread or rate-volatility measure. That common financial-market core is why their canonical correlation with BS stays near one, and, as Section 7 shows, why a financial-market control is the one variable a defensible purge should always include. This is not an artefact of the five tabulated sets: across the 150 equally-predictive draws that build the identified set below, *every one* contains at least one financial-market control, and even when the twenty macro-announcement surprises are added to the draw pool all 150 still include one against only 85% that include a macro control. What varies across the sets is only *which* instrument stands in for each block; the predictable factor itself is held fixed.

The purge is first *necessary*. The un-purged surprise (`RAW`, grey dashed in Figure 1) gives a contaminated, wrong-signed response: industrial production is shallow and non-monotone, bouncing to $+0.14$ a month after the shock, the excess bond premium barely moves, and there is no durable contraction, the signature of a surprise still loaded with non-policy shocks that offset the policy effect. Purging on *any* set, in either group, restores the textbook pattern: a persistent output decline, a sustained rise in the excess bond premium, falling prices, and a two-year rate that rises on impact and eases as the economy weakens. This is robust to the design choices that might be thought to drive it (Table 16). One cannot get the signs right without the purge.

It is not, however, *sufficient*. Purging leaves a weak instrument, and weak instruments are exactly the regime in which point estimates are sensitive to specification. Merely relabelling the predictable factor then moves the magnitude: across Group A (Figure 1), sets that are interchangeable by the predictability criterion and 0.94–0.99 correlated with BS still imply a peak industrial-production response from about -1.0% (`A2`) to -2.0% (`BS6`), and a CPI response from -1.0% to -1.6%: a factor-of-two spread in both the real and the price effect for the same 25 bp shock, even though every set restores the same qualitative contraction.

Group B changes not the labelling but the *target*. Built from macro-announcement surprises rather than from financial-market variables, it removes the Fed's systematic response to *macro news* rather than to financial conditions: a genuinely different, lower-overlap slice of the non-policy contamination. Holding its in-sample R^2 and instrument strength at the BS level while pushing the canonical correlation down to 0.64–0.71 is what makes the comparison clean: any difference in the responses can be assigned to *which* component is removed, not to how much. Its responses (Figure 15 in Appendix C) are shallower and far less persistent than the BS shock's: industrial production recovers to roughly zero by two years instead of remaining near -2%. That

[4] Canonical correlation is the largest attainable correlation between a linear combination of one control set and a linear combination of the other: near one the two span the same predictive direction and differ only in labelling; lower, they load on genuinely different predictors.

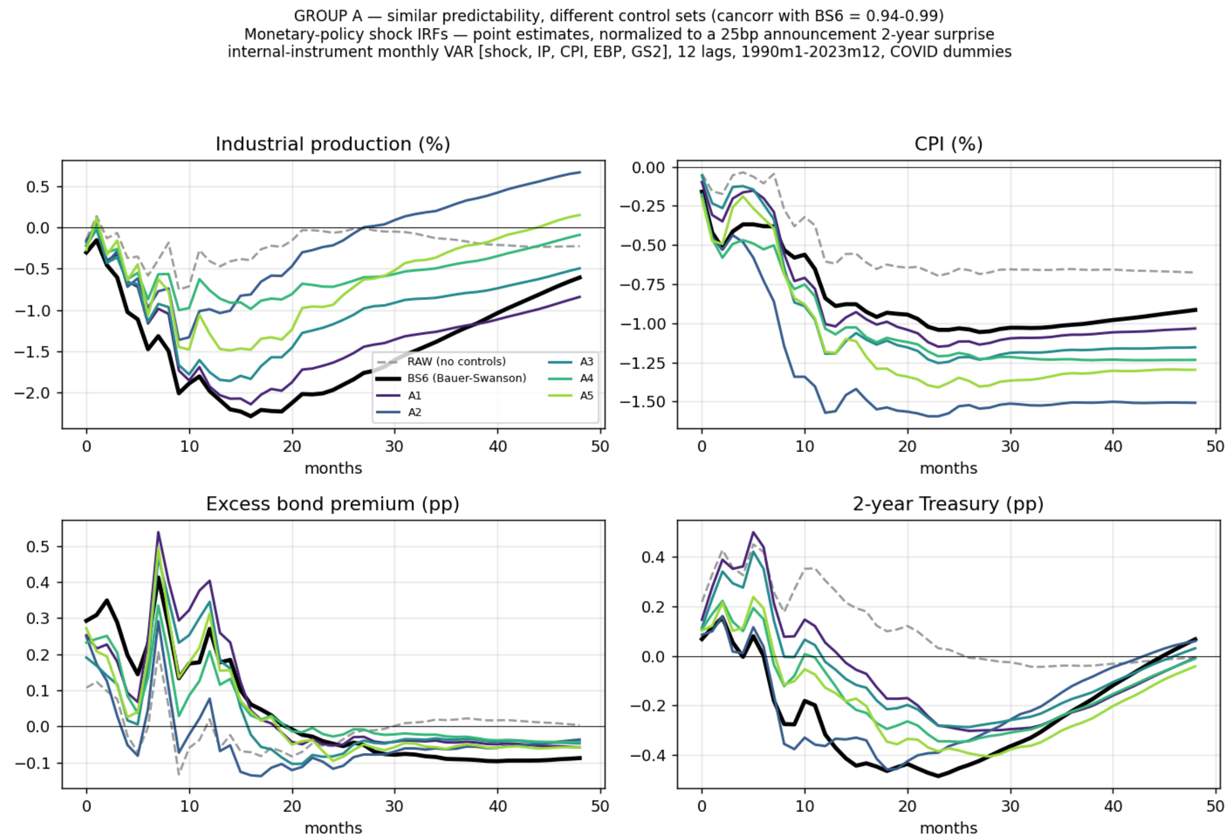

Figure 1: **Group A: similar predictability, different control sets** (canonical correlation with BS 0.94–0.99). Point-estimate IRFs to a 25 bp announcement two-year surprise, full sample. Alternative bases for the same predictable factor give the same qualitative response but magnitudes that differ by up to a factor of two.

an equally-predictive *financial* purge and an equally-predictive *macro* purge give materially different magnitudes both widens the set and suggests the surprise carries distinct systematic responses to different non-policy shocks, only one of which any single six-control set removes.

4.1 The magnitude is not informatively estimated

Two control sets equally defensible against the predictability diagnostic imply different answers to "how much does output fall after a 25 bp tightening?" The question this section settles is what that disagreement is evidence *of*. Drawing 150 six-control purges at least as predictive of the surprise as `BS6` and running each through the same VAR (construction in Appendix C), the peak industrial-production response is confined to a band (Figure 2, Table 3). `BS6` sits inside it; `RAW` lies outside it for output, prices and credit, so purging moves identification into a qualitatively different region.

It is tempting to read the width of that band as an identified set: the range of magnitudes the predictability diagnostic alone cannot narrow. Section 4.2 shows that reading is not available here, and Section 4.3 shows that even if it were, the objects inside the band are not comparable

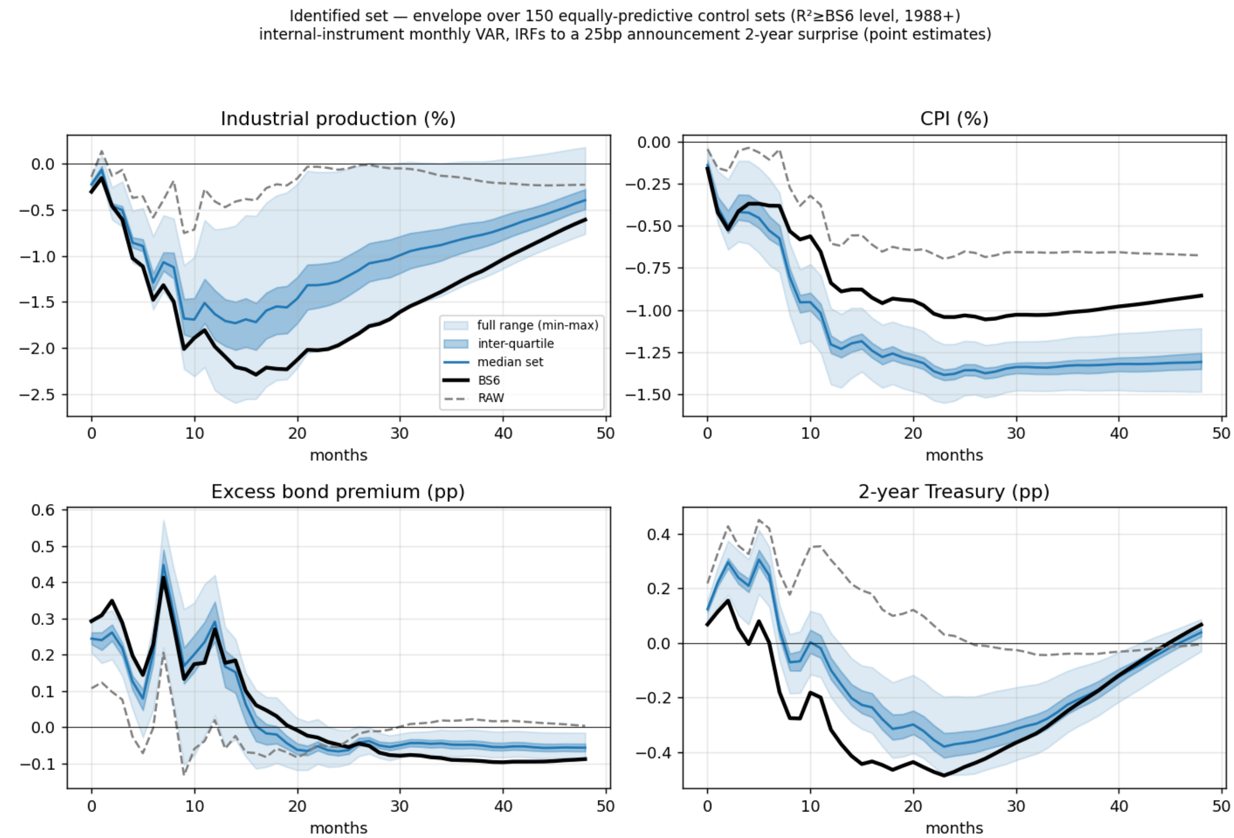

Figure 2: **Responses across equally-predictive purges** (150 six-control sets, $R^2 \geq$ `BS6` level, 1988+). Shaded: pointwise min–max (light) and inter-quartile (dark) envelope; solid blue the median. `BS6` sits inside the collection, `RAW` outside it. Section 4.2 shows the width of this envelope is smaller than the sampling error around any one line inside it, which is why we do not describe it as an identified set.

experiments. We therefore state the conclusion of this section in the weaker and more defensible form: *the purge recovers the sign of the output response and does not deliver an informative estimate of its magnitude.* The two reasons are logically independent and both bind.

Before turning to them, two candidate explanations for the width can be set aside, because ruling them out is what leaves sampling error and incomparability as the operative ones. The first is overfitting in the selection criterion. Section 7.1 shows that choosing subsets on *in-sample* R^2 selects the least stable projections, so the band might be populated by draws whose coefficients would not survive resampling. Rebuilding the set on a cross-validated criterion instead (Table 18) leaves the band and its interquartile range essentially unchanged, and varying the number of controls per draw from three to ten moves the median by less than the width of a single bootstrap standard error. The width is not an artefact of the selection rule.

The second is the normalisation. Under the common-slope convention of Section 3.1 every shock is the same size in surprise units by construction, so the spread cannot reflect shocks of different sizes; the residual variation in ι across sets is a first-stage *fact* to be reported (Table 15), not a scale to be divided out. Section 5.5 gives the sharpest illustration of what happens when one does.

The exercise also reads Chen (2025)'s Section 7 through the same lens. Chen purges the Nakamura and Steinsson (2018) surprise on financial conditions in a proxy VAR and finds conventional, puzzle-free responses: better, he argues, than purging on the BS predictors. Running the same idea through our machinery (Figure 18 in Appendix C), every financial purge flips the output sign the raw surprise gets wrong and all land inside the band, with depth monotone in how much of the surprise the proxy removes ($R^2 = 0.05$, 0.10 and 0.19 giving twelve-month responses of -0.4%, -0.6% and -1.6%). Chen's "`FCorth` is better" is therefore not a sharper identification but a lightly-cleaned draw from the same collection, and, on the evidence of the next two sections, the differences among such draws are not statistically resolvable.

4.2 The spread across purges is smaller than the sampling error

A band of estimates across specifications is informative only if the movement it records is large relative to the sampling uncertainty around any one of them. If the confidence band around `BS6` alone already spans the interval, the band is a restatement of imprecision and nothing in its width is attributable to the choice of controls. This section shows that is the case here, which is why Section 4.1 states its conclusion as a negative rather than as a set.

Write the observed dispersion of the peak response across control sets as the sum of a specification and a sampling component,

$$\underbrace{\mathrm{Var}_j\left(\hat{\theta}_j^{\mathrm{peak}}\right)}_{\text{observed across draws}} = \underbrace{\mathrm{Var}_j\left(\theta_j^{\mathrm{peak}}\right)}_{\text{specification: what the control set changes}} + \underbrace{\mathbb{E}_j\left[\mathrm{Var}\left(\hat{\theta}_j^{\mathrm{peak}} \mid j\right)\right]}_{\text{sampling: what one more draw of the data would change}}, \tag{10}$$

where j indexes control sets, and define $\lambda \equiv \mathrm{Var}_j(\theta_j^{\mathrm{peak}}) / \mathrm{Var}_j(\hat{\theta}_j^{\mathrm{peak}})$ as the share of the observed spread that is genuinely about which controls one picks.

This is where the normalisation choice of Section 3.1 pays a second dividend. Under the common slope the impulse response is a smooth function of the reduced-form parameters (B, Σ) (identification runs through the recursive ordering, not through a first-stage division), so a recursive-design wild bootstrap delivers valid pointwise bands. Divide by ι instead and the estimand becomes a ratio with a near-zero denominator, percentile bands understate the uncertainty, and the decomposition below could not be computed at all.

The sampling component dominates, and not marginally. Table 1 reports a variance of the point estimates across control sets of 0.133 against a mean bootstrap variance *within* sets of 0.910, a ratio of about one to seven, so λ is zero to three decimal places. Put in the units of the responses themselves: the peak estimates run from -0.99 to -2.27, a spread of 1.28 percentage points, while the bootstrap standard error of any single peak is about 0.95. *The entire collection of "equally defensible" answers spans about one and a third standard errors of one of them.*

It is worth asking what would have to change for that verdict to reverse, because the answer is not "a better purge". Standard errors fall with the square root of the sample, so to make the observed 1.28-point spread two standard errors wide, the point at which two control sets could be told apart at conventional levels, the standard error would have to fall to 0.64, which takes about 2.2 times as many observations. On monthly data spanning thirty-five years, that is another forty-odd years of FOMC meetings, at an unchanged monetary regime. The identified set is not narrow enough to be informative and cannot be made so by any amount of care in choosing controls; the binding constraint is the number of monetary announcements that have happened.

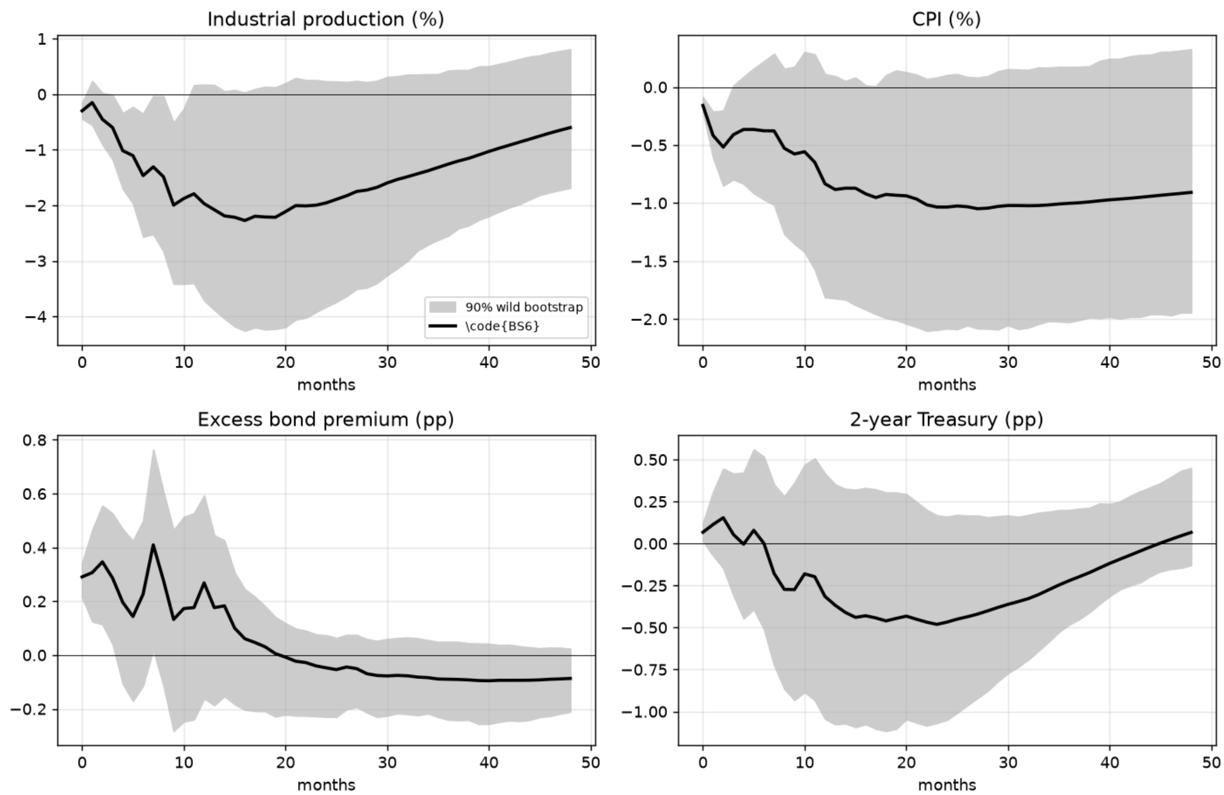

Figure 3: **Sampling uncertainty around a single purge.** Impulse responses to the BS6 shock under the common-slope normalisation, with a 90% recursive-design wild-bootstrap band (500 replications, Rademacher weights). The width of this band is the benchmark against which the collection of responses in Figure 2 must be read: the band measures what one more draw of the data would change, the collection what a different defensible control set would change. The first is the larger by a factor of seven in variance.

Table 1: Decomposing the observed spread in the peak output response, equation (10). The within-set component is the mean, across control sets, of the bootstrap variance of the peak; the between-set component is the variance of the point estimates across sets. λ is the specification share.

Component	Value
Across-control-set variance of the peak IP response	0.1329
Mean within-set sampling variance (wild bootstrap)	0.9103
λ, specification share of the observed spread	0.000

Three readings were possible before the computation and we set them out in advance: a large λ would have made the band a specification result; a small λ would mean the magnitude is not informatively estimated; unbounded individual bands would confine the paper to signs and shapes. The second obtained. Three consequences follow.

First, the paper's magnitude claims have to be stated as what they are. Reporting a peak response of -2.3% for `BS6` and -1.0% for `A4` as though the gap were a finding about the controls is not supportable; the two are within one standard error of each other. This applies with equal force to comparisons the paper makes elsewhere, and Section 5.5 flags where.

Second, and this is the constructive part, the conclusion is *stronger* than set-identification, not weaker. A set-identified magnitude would say the data pin the answer to a range. What the data say instead is that this design does not pin the magnitude at all: the leading source of uncertainty is not which defensible purge one adopts but the estimate's own sampling error, which the weak first stage of Section 3.1 makes large. A reader who wants a number from a high-frequency-identified monetary VAR of this form should be told that the design does not support one, and told why: an instrument that moves the two-year rate by one to three basis points on impact cannot deliver a precisely estimated output elasticity, however carefully it is purged.

Third, the sign result is untouched. Sampling error large enough to swamp a 1.3-point spread in peak magnitudes is not large enough to make the `RAW` response (shallow, non-monotone, and outside the purged collection for output, prices and credit) indistinguishable from the purged ones. Necessity survives; sufficiency fails in a stronger sense than the phrase "set-identified" conveys.

4.3 Do these purges describe the same policy experiment?

A band of impulse responses is only a set of competing answers to one question if the objects inside it are the same object. Section 4.2 showed that the spread across purges is smaller than the sampling error around any one of them. This section shows something logically independent and, for practice, just as damaging: even setting precision aside, the shocks that generate the band do not all deliver anything recognisable as a 25 bp tightening.

The natural check is the *shape* of the implied policy path, because shape is invariant to every normalisation choice discussed above. Define

$$\rho_h \equiv \frac{\mathrm{IRF}_h(\mathrm{GS2})}{\mathrm{IRF}_0(\mathrm{GS2})}, \tag{11}$$

the response of the policy variable at horizon h relative to its own impact response. Any common rescaling of a shock cancels from (11), so ρ_h is a property of the estimated system and of the identifying vector, not of the units in which the answer is reported. For a monetary tightening one expects $\rho_h > 0$ over the first year, rates rise and then revert, with a hump that need not be monotone but should not reverse sign while the output response is still building.

Table 2 reports the implied path in basis points, scaled to a common 25 bp impact. The dispersion is not a matter of degree.

Three readings of Table 2 matter. First, ρ_6 spans 0.01 to 6.04. The same nominal 25 bp shock leaves the two-year rate essentially unchanged after six months under one control set and 151 bp higher under another. Second, six of the ten have the policy rate *wrong-signed* at twelve months; the benchmark `BS6` moves from +25 bp on impact to −117 bp a year later while industrial production sits at −7.3% and stays there. A rate path that reverses by more than four times its impact is not a tightening that the economy is responding to; it is a system whose identified vector points somewhere other than monetary policy. It is worth saying out loud what such a series asserts. Read as a monetary experiment, `BS6` claims that the Federal Reserve raised the two-year rate by a quarter of a point and then, within twelve months, lowered it by more than a full percentage

Table 2: Implied policy path by control set, in basis points, scaled to a common 25 bp impact. ρ_h is defined in (11) and is invariant to the normalisation convention. The final column applies the screening criterion (12). Full sample, per-shock coverage.

Shock	h=0	h=6	h=12	ρ_6	ρ_{12}	Coherent?
`RAW`	25	48	35	1.92	1.39	yes
`BS6`	25	0	-117	0.01	-4.67	**no** — sign reversal
`A1`	25	75	9	3.01	0.37	yes
`A2`	25	11	-106	0.43	-4.25	**no** — sign reversal
`A3`	25	77	-5	3.09	-0.21	**no** — sign reversal
`A4`	25	34	-15	1.38	-0.61	**no** — sign reversal
`A5`	25	47	-31	1.87	-1.25	**no** — sign reversal
`B1`	25	127	-1	5.07	-0.03	**no** — explosive
`B2`	25	151	44	6.04	1.76	**no** — explosive
`B3`	25	101	33	4.03	1.32	**no** — explosive

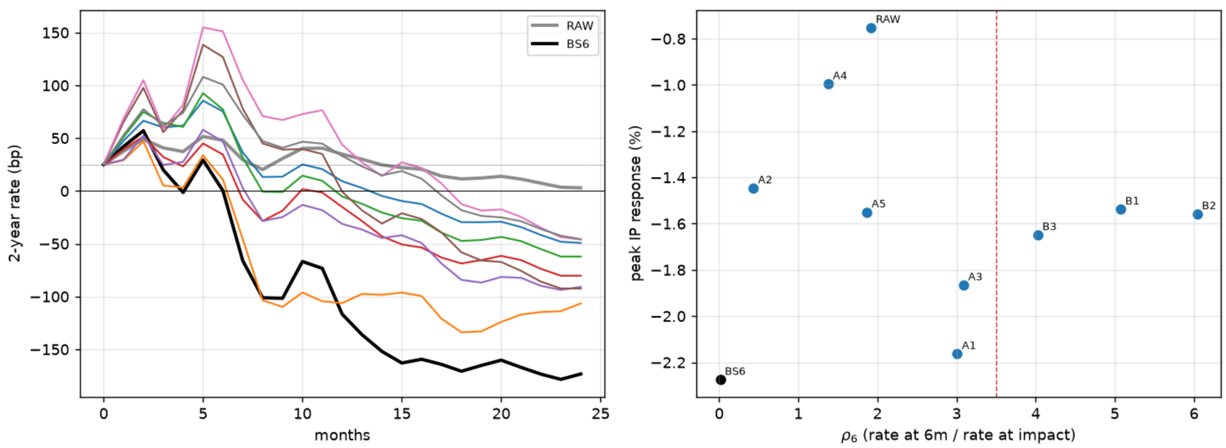

Figure 4: **The implied policy paths are not the same experiment.** *Left:* the two-year rate response of each candidate shock over two years, scaled to a common 25 bp impact; the ratio plotted is ρ_h of (11) and is invariant to the normalisation convention. `BS6` is in black and `RAW` in grey. *Right:* the peak output response against ρ_6, with the screening threshold of (12) marked. If these shocks were competing estimates of one magnitude the scatter would slope downward, a more persistent tightening depressing output by more, and it does not.

point (an easing of 2008 proportions set off by a 25 bp tightening), while output fell and stayed down. No committee has ever behaved that way, and no reader shown that path on its own would call it monetary policy. What gets reported from this shock is its peak output response; the path that produced it is not reported at all, which is why the incoherence has gone unremarked. Third, and this is the diagnostic that ties the section to the rest of the paper, the estimated output response is essentially *uncorrelated* with the policy path it is supposedly a response to. Across the ten shocks the correlation between the peak output response and ρ_6 is -0.01, where a coherent set of monetary experiments would deliver a large negative value: a tightening that persists longer should depress output by more. What the output response does track is the reciprocal of the impact ι, at a correlation of $+0.79$. The ordering of magnitudes in the identified set is the ordering of first-stage weakness, not of transmission.

The remedy is to require of a candidate shock what one would require of any monetary experiment before comparing its output response with another's:

$$\iota > 0, \qquad 0 < \rho_6 \leq 3.5, \qquad \rho_{12} > 0. \tag{12}$$

The bounds are deliberately loose: (12) admits paths that decay almost to zero by six months and paths that triple, and asks only that the rate be on the same side of zero as its impact after a year. On the full sample it admits `RAW` and `A1` and rejects the other eight, `BS6` included. Applying it to the draws that build the identified set of Section 4.1 is the natural way to report it. Table 3 reports the band before and after the screen, together with the number of draws rejected. The comparison is the operative one for Section 4.1: the unscreened band is the object the paper has been calling the identified set, and the screened band is the range of answers among shocks that at least describe a tightening.

The outcome is instructive and not the one we expected. Of 150 random six-control draws that meet the predictiveness threshold, *all* 150 pass (12): the band is $[-2.14, -0.68]$ with median -1.50 before the screen and identically after it. The screen therefore rejects eight of the ten named control sets and none of the random draws. The two populations differ in a way that explains this. The draws are all estimated on the single complete-case sample on which the extended panel is available, and are all required to reach $R^2 \geq 0.12$; the named sets are each estimated on their own complete-case sample and include sets whose first stage is weak. What fails (12), in other words, is not a particular choice of controls but the combination of a weak first stage with a sample that differs from its neighbours', which is the same diagnosis Sections 4.2 and 3.1 reach by other routes.

Table 3: The identified set before and after the coherence screen (12). Draws are six-control subsets at least as predictive of the surprise as `BS6`; the screen requires a positive impact on the policy rate, a six-month path within 3.5 times impact, and a twelve-month path of the same sign.

Set	N	peak IP band	median
All equally-predictive draws	150	$[-2.14, -0.68]$	-1.50
Passing the coherence screen (12)	150	$[-2.14, -0.68]$	-1.50

Section 4.1 read the width of the band as the ambiguity the predictability diagnostic leaves behind. Table 2 says the band is not only wide but heterogeneous in kind: it mixes shocks that describe a persistent tightening, shocks whose rates revert within two quarters, and shocks whose rates reverse. Reporting a single interval over that mixture overstates how much of it is a magnitude disagreement about one experiment. The honest object is the band computed over the screened subset, with the count of rejected draws reported, and, as Section 3.1 makes clear, with the caveat that the screen is applied to a system estimated at a unit root, so it disciplines shape rather than certifying levels.

We are careful about what the screen therefore delivers. On the draws it does not narrow the band at all, so it is not a device for shrinking the identified set, and we do not claim it as one. Its use is diagnostic and it is at the level of the individual published shock: given a candidate series, (12) says in two numbers whether the object being compared with other estimates describes a monetary tightening. That eight of the ten sets in Table 2 fail it, while the underlying design passes it whenever the sample and the first stage are held fixed, is the point. The heterogeneity documented in this section is not intrinsic to orthogonalisation; it is what happens when a literature compares series estimated on different samples with first stages of very different strength, and reports the resulting spread as a range of estimates of one quantity.

Two features of this argument are worth emphasising because they are unusual in this literature. It is *scale-invariant*, so it is untouched by the normalisation debate that occupies much of the applied discussion of high-frequency instruments; and it is *cheap*, requiring only two numbers already produced by any proxy or internal-instrument VAR. We know of no paper in the high-frequency identification literature that reports the implied policy path of its candidate instrument, and Table 2 suggests the omission is not innocuous.

It is worth restating what Sections 4.2 and 4.3 establish together, because each is incomplete without the other. Section 4.1 showed that control sets equally predictive of the surprise deliver different impulse responses. The natural conclusion is that the choice of controls matters for the magnitude and should be made carefully. Section 4.2 shows that this conclusion is not available: the differences are smaller than the sampling error, so the spread cannot be read as evidence about magnitudes at all. Without it, a reader would over-interpret the band; with it alone, a reader would conclude that the choice of controls is harmless. Section 4.3 supplies the missing half. The choice does matter, but along dimensions the sampling error cannot explain: the strength of the resulting instrument and the coherence of the policy path it implies, on which the ten candidates differ by orders of magnitude and eight of ten fail a minimal requirement. The advice that follows is therefore not to pick controls carefully in the hope of a better magnitude, which no choice delivers, but to summarise the pre-announcement information set in a way that does not leave the choice open at all, and to screen whatever comes out. Section 7.2 makes the case for principal components of the full panel on exactly this ground: they remove the control-subset degree of freedom rather than exercising it.

5 The Fed put: testing the constant-underestimation null

Section 2 reduced the difference between a complete and an incomplete purge to a single restriction, $\phi_- = \phi_+$. This section tests it on the three margins of (5) and finds it rejected on the two that carry weight.

The economics is worth stating before the tests, because it fixes the sign of everything that follows. Suppose the committee responds to a falling equity market more forcefully than to a rising one: it eases into financial deterioration, and does not symmetrically lean against a rally. A market that prices one state-invariant slope is then under-prepared in exactly one state: the one in which equities have already fallen. In that state the announced policy is easier than priced, so the surprise is systematically negative and systematically related to the pre-meeting equity return; in the other state the priced and the actual response roughly agree, and little systematic is left in the surprise to find. Each of the three implications of Section 2.3 is that chain read at a different point. The projection slope on pre-meeting equity returns should be steeper in the easing state (T1). A single pooled slope must therefore fit one state too aggressively and the other too timidly, which shows up not as a bias in any reported coefficient but as unequal destruction of the instrument's first stage across the two states (T2). And because the asymmetry is a property of the rule rather than a standing forecast error, it should be invisible in real time (T3).

5.1 The reaction gap is state-dependent (T1)

Implication (T1) is a statement about a projection *slope*, and it has to be tested as one. We proceed in three steps: the specification that matches Cieslak and Vissing-Jorgensen's (2021) own, the interaction test that nests the two hypotheses, and the by-sign description that motivated the exercise but cannot serve as its evidence. The two tests give different answers, and the difference is informative rather than awkward.

Cieslak and Vissing-Jorgensen (2021) identify the Fed put from a piecewise-linear regression of the realised funds-rate change on the intermeeting excess equity return, split at zero. Their asymmetry is stark: the loadings on negative returns sum to 5.35 against -1.80 on positive returns, with the positive-return loadings individually insignificant. If the reaction gap of Section 2.2 inherits that kink, the same piecewise structure should appear in the high-frequency *surprise*, because the part of the easing response that a linearly-priced rule fails to anticipate is exactly what the surprise measures. We estimate

$$MPS_\tau = \alpha + \gamma_- \, rx_\tau^- + \gamma_+ \, rx_\tau^+ + \eta' Z_\tau + e_\tau, \qquad rx_\tau^- = \min(rx_\tau, 0),\;\; rx_\tau^+ = \max(rx_\tau, 0), \tag{13}$$

where rx_τ is the pre-meeting excess return on the S&P 500 and Z_τ collects the remaining BS controls. H_{FP} predicts $\gamma_- \gg \gamma_+ \approx 0$; H_{BS} predicts $\gamma_- = \gamma_+$. Because rx_τ is dated strictly before the announcement, (13) conditions on an *ex-ante observable* state, which matters for two reasons that Section 5.5 will show are not academic: it avoids conditioning on the sign of the dependent variable, and, by Proposition 1, it is what makes the resulting correction implementable in real time rather than only ex post.

The general form interacts the full control vector with the state indicator $s_\tau = \mathbf{1}\{rx_\tau < 0\}$,

$$MPS_\tau = \alpha + \delta\, s_\tau + \beta' X_\tau + \left(\beta^{\text{int}}\right)' (s_\tau \cdot X_\tau) + e_\tau, \qquad H_{\text{BS}}:\; \beta^{\text{int}} = \mathbf{0}. \tag{14}$$

Because Section 6.1 shows the projection is driven by a handful of crisis observations, the conventional Wald statistic is not to be trusted here, and we report a wild-bootstrap p-value alongside the heteroskedasticity-robust one.

Table 4 reports both specifications, and they part company. The interaction block of (14) is jointly significant: $W = 17.15$ on six restrictions with a wild-bootstrap p of 0.030, and the fit rises from $R^2 = 0.148$ to 0.203. The constant-gap null is rejected: how the surprise relates to pre-announcement information does depend on the ex-ante financial state.

The piecewise test of (13) does not deliver the same verdict. $\hat{\gamma}_- = 0.133$ ($t = 1.35$) against $\hat{\gamma}_+ = 0.026$ ($t = 0.30$): the point estimates are in the predicted direction and their ratio, about five to one, is close to Cieslak and Vissing-Jorgensen's (2021) 5.35 against -1.80 in the funds rate. But neither loading is individually significant and equality cannot be rejected, $p = 0.507$. The equity channel alone is too imprecisely estimated in the surprise to carry the hypothesis.

The honest reading distinguishes two claims that the Fed-put literature usually bundles. That the reaction gap is *state-dependent* (that a single β is the wrong object) is established at conventional levels. That the state dependence takes the specific *Fed-put shape*, an asymmetry identified off the equity loading, is directionally supported and statistically inconclusive on this sample. We therefore carry the first claim forward as a finding and the second as an interpretation. Everything in Section 5.5 that concerns whether a pooled projection is misspecified rests on the first. The economic story (that the systematic component is a one-sided easing response to financial deterioration) rests

Table 4: Testing the constant-gap null on an ex-ante state. Panel A: the piecewise specification (13). Panel B: the interaction test (14), with the interaction block tested jointly. Standard errors are HC1; the bootstrap column is a wild bootstrap with Rademacher weights.

	coefficient	HC1 t	p	wild-boot. p
Panel A: piecewise in the pre-meeting equity return				
γ_- (negative returns)	+0.1327	+1.35	0.176	—
γ_+ (positive returns)	+0.0264	+0.30	0.767	—
$H_0: \ \gamma_- = \gamma_+$	—	—	0.507	—
Panel B: full interaction with the ex-ante state				
$H_0: \ \beta^{\text{int}} = \mathbf{0}$ (6 restrictions)	—	$W = 17.15$	—	0.030
R^2, no interaction / with	0.148 / 0.203			

on the second, and would be settled by a sharper measurement of the state: our rx_τ is a sixty-six-day pre-meeting window, while Cieslak and Vissing-Jorgensen's (2021) object is the intermeeting return, and the mismatch can only attenuate $\hat{\gamma}_-$. Three refinements would sharpen the test, and we flag them as the natural next step rather than claiming them here. The first is to measure rx_τ over the true intermeeting window, from the previous announcement to the current one, which is Cieslak and Vissing-Jorgensen's (2021) own object and removes the attenuation. The second is to replicate their piecewise regression on the *realised* funds-rate change on our sample before running it on the surprise, so that a failure to reject on the surprise can be attributed to the surprise rather than to the sample or the window. The third is to test the kink where their evidence is strongest, on the easing side and outside the lower-bound period, rather than pooling states in which the funds rate could not move. The interaction test of (14) already rejects a common slope on the full control vector, so the piecewise specification is underpowered rather than uninformative, and these three changes all act on power.

Splitting instead on the sign of the realised surprise, easing surprises are far more predictable from the BS controls than tightenings: the in-sample R^2 of $MPS \sim$`BS6` is 0.35 for easings against 0.09 for tightenings, and leave-one-out leaves easings robustly predictable ($R^2_{\text{LOO}} = 0.20$) and tightenings not (-0.02). Two objections attach. The split conditions on the sign of the *dependent* variable, which truncates ε_τ and so does not deliver the population ϕ_- and ϕ_+; and, by (6), a common ϕ combined with a larger news variance in stress states generates the same R^2 ordering, so the comparison is biased *towards* our own hypothesis. Appendix D.2 works through both. Section 5.5 shows these are not hypothetical concerns: the headline consequence the sign split implies does not survive being re-derived on the ex-ante state.

What the by-sign evidence does establish, without either problem, is where the pooled coefficient comes from. The pooled fitted values correlate 0.78 with an easing-only fit and only 0.27 with a tightening-only fit, and the purge removes *fourteen times* more variance from easing surprises than from tightenings ($R^2 \times \text{Var}$ of 0.9×10^{-3} against 0.06×10^{-3}). Those are statements about the estimated projection rather than about the conditional distribution of the surprise, so the truncation objection does not bite on them.

5.2 The raw bias is one-sided (T2)

H_{BS} predicts, through (7), an upward bias in the estimated output response on *both* halves of the sample. It is not there. Splitting the identifying variation by sign makes this visible (Figure 5, right). Identified from *tightening* surprises alone, the *raw* instrument already gives a textbook contraction (industrial production falls to about -1.6%) with no puzzle to fix and hence no detectable upward bias. Identified from *easing* surprises alone, the raw instrument gives the entire puzzle: output *rises*, to about $+0.8\%$ at a year, and orthogonalisation fixes exactly this, collapsing the purged easing-only response to $+0.1\%$. The upward bias Bauer and Swanson (2023a) identify is real, and their remedy earns its keep, but it lives in one half of the sample, which is what H_{FP} predicts and H_{BS} does not.

Three checks discipline this comparison, and we report them rather than assert robustness. The two half-sample instruments differ in strength and in normalisation, so the sub-sample first stages and 25 bp slopes are reported alongside the responses. The easing-only identification is disproportionately weighted by 2008–09 and 2020, both of which carry exogenous dummies in the VAR, so the exercise is repeated excluding those windows. And because easings, crises and the zero lower bound are collinear, Table 6 cross-tabulates the sign cut against the regime cut, and the two are entered jointly, so that the reader can judge whether the two cuts are separately identified at all. These are Tables 5 and 6 of Section 5.3.

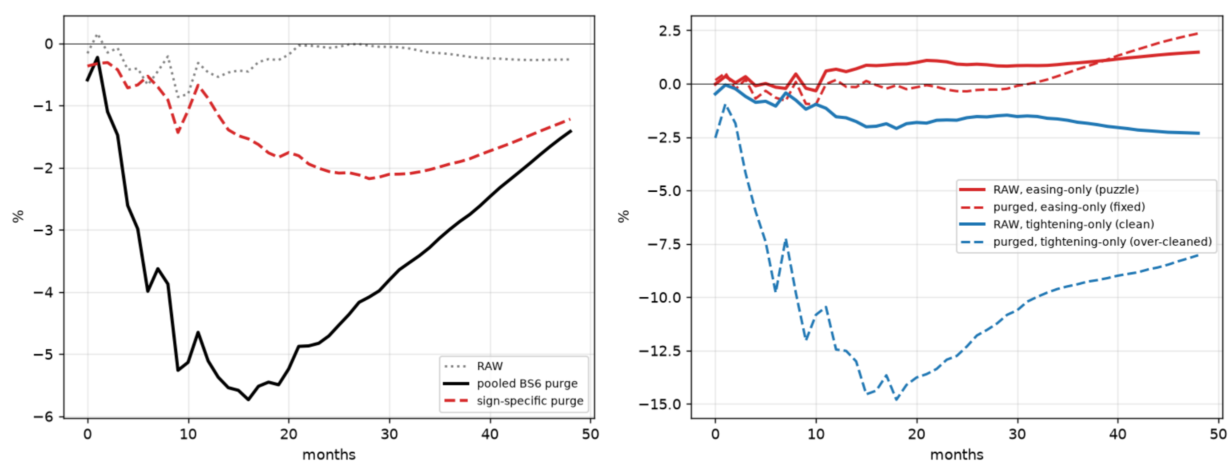

Figure 5: **The purge cleans a one-directional contamination.** *Left:* industrial production response to the raw, pooled-BS6-purged, and state-specific-purged surprise. *Right:* the identifying variation split by sign: the raw output puzzle lives entirely in the easing surprises (red solid) and is removed by the purge (red dashed), while the tightening surprises are already correctly signed (blue solid) yet are *over*-cleaned by the symmetric purge (blue dashed).

5.3 Three checks on the half-sample comparison

The comparison behind (T2) identifies the shock from one half of the surprises at a time, and three features of that exercise could generate the asymmetry without any state dependence in ϕ. We report each rather than assert robustness.

The two half-sample instruments are not equally strong, which matters for reading the comparison above. Zeroing out one sign leaves an instrument with a different first stage and a different normalisation slope, so the two responses are not automatically on a common scale. Table 5 reports, for each half, the number of events, the impact ι of (9), the announcement-window slope, and the resulting peak and twelve-month output responses. The relevant question is whether the tightening half is correctly signed *and* adequately strong; a correctly-signed response identified off a degenerate instrument would not support the argument.

Table 5: The half-sample identification behind (T2).

Identified from	n	ι (bp)	peak IP	IP h=12	ρ_6
Easing, raw	133	3.49	-0.38	0.64	0.68
Easing, purged	133	2.47	-0.60	0.33	0.11
Tightening, raw	227	2.29	-2.84	-1.95	4.17
Tightening, purged	227	0.53	-4.06	-3.41	4.79
Easing, raw, excl. 2008–09/2020	116	2.67	-1.50	-1.30	1.52

Easing surprises cluster in 2008–09 and in 2020, both of which carry exogenous dummies in the VAR of Section 3. If the raw easing-only output puzzle is generated entirely by those windows, the finding is about two crises rather than about a systematic response, and the last row of Table 5 reports the easing-only response with them excluded. We regard this as the single most important robustness check on (T2), because the alternative reading (that the Fed put is a crisis phenomenon rather than a standing feature of the reaction function) would materially weaken the case for a *state-dependent* rather than an *episode-specific* correction.

Easings cluster in crises, which cluster at the zero lower bound, so the contrast between the sign cut and the regime cut developed in Appendix D.6 is only identified if the two cuts have independent variation. Table 6 cross-tabulates them and reports the state-specific projection slope in each cell. If the off-diagonal cells (tightenings at the ZLB, easings outside it) are thin, the horse race between the two cuts is not identified and the claim that "misspecification lives in the sign, not in the calendar regime" cannot be sustained on this sample.

Table 6: Collinearity of the state and regime cuts: announcement counts and the within-cell projection slope on `BS6`. The four-cell specification is estimated jointly and the sign and regime interactions tested separately.

	pre-ZLB	ZLB (2008–2015)	post-ZLB
Easing ($MPS < 0$)	86 (+0.152)	26 (+0.099)	21 (-0.051)
Tightening ($MPS > 0$)	137 (+0.109)	41 (-0.140)	49 (-0.072)

A fourth check follows from Section 4.3 and applies to the half-sample responses as it does to the full-sample ones: the ρ_6 column of Table 5 records whether each half-sample shock traces a coherent policy path. A comparison of output magnitudes between two shocks that imply different rate paths inherits the problem Table 2 documents, so the (T2) contrast should be read as a statement about the *sign* of the output response (which is what the argument requires) rather than about the -1.6% against $+0.8\%$ magnitudes.

5.4 The predictability is not real-time (T3)

Persistent underestimation is a standing forecast error, so it should be detectable in real time. It is not: an expanding-window projection on real-time vintages has essentially no out-of-sample power, and the in-sample R^2 of 0.147 for the BS controls falls to -0.062 under five-fold cross-validation, averaged over twenty random partitions (Table 8). As Section 7 sets out, this is not evidence against contamination (the cross-validated R^2 is measured far too imprecisely to rule out a small population R^2), but it is decisive against a *detectable* one. What it does say is that agents were not leaving a persistent, symmetric, detectable bet on the table. The predictability that survives resampling is the state-contingent, easing-side component. As flagged in Section 2.3, this margin is suggestive rather than decisive on its own; (T1) and (T2) carry the argument.

5.5 What follows from state dependence, and what does not

Rejecting $\beta^{\text{int}} = \mathbf{0}$ activates Propositions 1 and 2: a pooled projection fitted across two states under-cleans one and over-cleans the other, and the resulting impulse response inherits a bias that vanishes only if the reaction gap is state-invariant. This subsection separates the consequences the data support from one that does not survive scrutiny.

The cleanest evidence is normalisation-free and does not require comparing impulse-response magnitudes at all. Purging removes systematic variation from both halves of the surprise, but not in the same proportion. The impact of the instrument on the two-year rate, ι of (9), falls from 3.49 to 2.47 basis points on the easing half, a reduction of 29%, and from 2.29 to 0.53 basis points on the tightening half, a reduction of 77% (Table 5). A single projection applied to both states destroys three-quarters of the tightening half's relevance while leaving most of the easing half's intact. That is what over-cleaning *is*, measured on the object the purge is supposed to preserve, and it is invariant to every scaling choice in the paper.

In the units an applied reader screens on, the same fact reads as follows. Regressing the announcement-window two-year change on the surprise within each half, the first-stage F falls from 372 to 131 on the tightening half and from 231 to 179 on the easing half. The pooled purge costs the tightening half two-thirds of its first stage and the easing half a fifth. Neither number crosses a conventional weak-instrument threshold at the announcement window (that problem arrives only after monthly aggregation, as (9) shows), but the *asymmetry* is the point, and it is not something a practitioner would discover from the pooled regression they actually run.

It also supplies the sharpest illustration of the normalisation argument of Section 3.1. The purged tightening-only instrument has the weakest first stage anywhere in this paper, $\iota = 0.53\,\text{bp}$, so the impact convention multiplies its response by roughly 45 against about 14 under the common slope, turning a twelve-month output response of -3.9% into -12.5%. The series this section identifies as *most* over-cleaned is mechanically the one that convention inflates *most*, which is why the paper does not use it.

Comparing like with like under the common-slope convention, identifying from tightening surprises alone gives a peak output response of -2.84% raw and -4.06% purged, so purging the near-clean half pushes its response further from zero rather than leaving it alone. The easing half behaves in the opposite direction relative to its own baseline, and Table 5 reports both at the twelve-month horizon alongside the crisis exclusion of Section 5.3. The direction predicted by Proposition 2 is in the data.

A natural next step is to ask whether correcting the misspecification moves the estimated magnitude, and a state-specific purge appears at first to moderate it substantially. That appearance depends entirely on *which state* the projection conditions on, and it does not survive the substitution that Section 5.1 argues for on independent grounds.

Conditioning on the *realised sign* of the surprise, the state-specific purge is far more moderate than the pooled one: a peak of -0.89% against -5.11%, a ratio of about one to six. Conditioning on the *ex-ante financial state*, and disciplining the split through the penalised regression (15), the state-dependent purge gives -2.58% against a pooled -2.27%: no moderation, a marginal deepening. Same estimator, same sample, same data: the only difference is whether the state is the sign of the left-hand variable or a variable dated before the announcement.

The gap between those two answers is exactly the selection problem of Section 5.1 and Appendix D.2, and it is larger than the moderation it appears to produce. Splitting on $\text{sign}(MPS_\tau)$ truncates ε_τ within each half; the resulting residual is smaller by construction, and the shock built from it is correspondingly attenuated. The moderation is a property of the conditioning variable, not of the Fed put. The same objection disqualifies any within-VAR decomposition of the pooled response into shock and leftover-contamination components computed from that split: it inherits the same defect, and we do not report one.

Two statements survive, and they are the ones we make.

The disciplined form of the state-dependent purge writes it as a single penalised regression,

$$\min_{\beta,\,\beta^{\text{int}}} \sum_{\tau} \Big(MPS_\tau - \beta' X_\tau - \big(\beta^{\text{int}}\big)' (s_\tau \cdot X_\tau) \Big)^2 \;+\; \kappa \, \big\|\beta^{\text{int}}\big\|_2^2, \tag{15}$$

which nests the pooled purge at $\kappa \to \infty$ and the unrestricted split at $\kappa = 0$, and penalises only the interaction block so that the common slope is estimated freely. Choosing κ by cross-validation on the surprise removes the mechanical in-sample advantage that fitting two projections would otherwise confer.

Table 7: The penalised state-dependent purge (15). κ is chosen by five-fold cross-validation on the surprise; the shrinkage ratio compares the fitted interaction block with its unrestricted ($\kappa = 0$) counterpart, so a value near zero would mean the data do not support a state-dependent slope and a value near one that the penalty does not bind.

Quantity	Value
CV-selected penalty κ^*	0.1425
Cross-validated R^2, penalised state-dependent purge	0.0681
Cross-validated R^2, pooled purge	-0.0108
Shrinkage $\lVert\hat{\beta}^{\text{int}}\rVert / \lVert\hat{\beta}^{\text{int}}_{\kappa=0}\rVert$	0.800
Peak industrial-production response	-2.58
Industrial production at $h = 12$	-2.41

First, the state-dependent projection is better *out of sample*, which no artefact of conditioning on the dependent variable can deliver. Penalising the interaction block and choosing κ by cross-validation on the surprise (Table 7), the selected $\kappa^* = 0.14$ leaves 80% of the unrestricted interaction intact and raises the cross-validated R^2 from -0.011 for the pooled purge to 0.068. A pooled projection has no out-of-sample predictive power for the surprise on this sample; a state-dependent one has some.

It is worth converting that into the quantity a central banker would ask about. Monetary policy surprises on FOMC dates would be about 11% less volatile if markets had priced the Fed's state-dependent response to pre-announcement conditions, against 8% for a response that does not vary with the state. Both numbers are small, and we draw the conservative conclusion rather than the flattering one: most of what moves at an FOMC announcement is not the systematic component this literature is trying to strip out, whether or not the stripping allows for the state. The three percentage points between them are what the Fed put is worth, and they are the whole subject of this section. That is a genuine improvement in the object the purge is built from, and it is immune to the criticism just made of the sign split.

Second, the consequence for applied work is a statement about specification rather than about magnitude. A practitioner should not fit one slope across states, because the data reject that restriction and because the pooled coefficient demonstrably strips three-quarters of the relevance from one half of the sample. What the practitioner should *not* expect is that fixing this delivers a materially different, and in particular a more moderate, estimate of the output response. Section 4.2 explains why: the sampling error around any of these responses is larger than the differences among them. The Fed put changes what the right estimator is; on this design it does not change the answer, because this design does not deliver a precise answer to change.

This also sharpens Bauer and Swanson (2023a) in a way their own framework anticipates. The upward bias they remove is not a diffuse, real-time under-appreciation of the rule (which the ex-post evidence of Section 5.4 shows markets could not have detected in real time), but a state-contingent response, and their constant-β purge is the $\phi_- = \phi_+$ corner of (32) rather than a rival account. Appendix D shows a single state-dependent coefficient generates the easing-side predictability, the asymmetric over- and under-cleaning, and the failure of a linear projection to represent a kinked conditional mean, and makes explicit the sense in which the Hack et al. (2024) critique applies. What we cannot claim, and no longer do, is that correcting the misspecification moves the estimated magnitude.

6 Where else the contamination lives

The reaction gap varies along three further dimensions that are observable and therefore, by Proposition 1, correctable: which *observations* carry the projection, which *component* of the surprise carries the contamination, and which *regime* it sits in. None of them substitutes for the sign, but each disciplines the purge.

6.1 The projection is outlier-driven

Two properties shape how the contamination should be removed. First, the nonlinearity is a *threshold* rather than a smooth curvature: a RESET test (adding squared and cubed fitted values) is insignificant ($p = 0.28$), so the contamination switches on for large, easing and crisis events rather than bending continuously with the controls, which is what the kinked response of Appendix D implies and a smooth misspecification would not. Second, the projection is *outlier-driven*: a robust Huber projection shrinks every BS coefficient towards zero, and by uneven amounts (Figure 21 in Appendix E). The standardised coefficients on the two payroll controls fall to between two-fifths and one-half of their OLS values, those on the slope and commodity controls to about seven-tenths, and the coefficient on the three-month equity return (the control through which the Fed put would operate) to about a quarter, from $+0.0077$ to $+0.0019$. The estimated "response to news" is therefore carried disproportionately by a handful of large, crisis-era observations, and the control most exposed to them is precisely the one Section 5.1 is about: the same observations that, by (8), dominate the variance weights w_-.

"A handful" is meant literally, but the count has to be done carefully, because one observation dominates every influence statistic in this sample. Ranked by Cook's distance, the single most influential meeting is **10 June 2020**, with a leverage of 0.937 against a sample mean of 0.019, about fifty times the average, and a Cook's distance of 7.40 when the next largest is 0.12. That one meeting is 87% of the total influence on a projection fitted to 360 observations.

Its provenance is not mysterious. Among the six standardised controls at that meeting, the nonfarm-payrolls surprise stands at $+18.3$ standard deviations and trailing payroll growth at -6.3; the other four are inside one and a half. The May 2020 employment report, released five days earlier, had come in roughly ten million jobs above consensus, and no other observation in thirty-five years is remotely comparable. Two features make this the textbook case rather than merely a large number. It is an outlier in the *controls*, not in the surprise (the surprise that day was 0.005, essentially zero), so a leverage of 0.94 means the fitted surface passes almost exactly through the point regardless of what the other 359 meetings say. And it is not holding the fit up but weighing it down: dropping it *raises* the in-sample R^2, from 0.147 to 0.156. It is a point the pooled slope is dragged toward, not one it explains.

Excluding it changes what the leverage diagnostic says, and the corrected version is the one the paper uses. The top ten then account for 49% of influence rather than 93%, and they are the meetings one would have named in advance: July 1995, December 1991, October 1998, April 2001, March 2008, September 2008, October 2008, December 2008, March 2020 and July 2020. Half were unscheduled or inter-meeting actions, and all but the first two sit inside a financial-stress episode. Dropping them takes R^2 from 0.156 to 0.125 and cuts the coefficient on the three-month equity return by 38%.

The composition of that remaining influence is the result that matters for Section 5.1. Meetings in the easing state, those preceded by a negative excess equity return, are 32% of the sample and carry 68% of the influence on the fitted coefficients, an over-representation of more than two to one. The Fed put is not merely present in these data; it is estimated almost entirely off the state in which it operates. Reassuringly for Section 5.1, the June 2020 observation is not what delivers this: removing it moves the equity coefficient by less than a tenth, from $+0.0067$ to $+0.0061$. The results of the previous section do not rest on the single most influential meeting in the sample, and we checked.

That the influence lands where it does is not a coincidence, and the reason ties the two properties of this subsection together. A Fed put is a response to financial stress; stress is by construction rare and large; so the meetings at which the committee eases hardest into a falling market are the meetings at which the equity control takes its most extreme values. Least squares weights an observation by the square of its distance from the mean of the regressors. The episodes that make the Fed put visible are therefore the same episodes that determine the coefficient meant to remove it. A threshold nonlinearity and concentrated leverage are not two findings but one seen from two sides, which is why a robust estimator is not a refinement in this setting but the first-order fix: of the two corrections it is the one that addresses the observations actually doing the work.

There is also a lesson here that generalises past this paper. The uncorrected statistic, "the ten most influential meetings carry 93% of the influence", is arithmetically true and substantively misleading, because a single high-leverage observation makes every aggregate share computed alongside it uninterpretable. Any orthogonalisation exercise should report the maximum leverage in its projection, which costs one line of code, before reporting anything that aggregates across observations.

The two corrections combine naturally: orthogonalise each surprise with a Huber regression estimated *within its own regime*. Figure 6 traces the resulting shock against the full-sample OLS and Huber purges. The regime split first localises the contamination: the standardised S&P response-to-news coefficient is about thirteen times larger during the ZLB ($+0.013$) than before it ($+0.001$) or after (-0.002), so the response to news the purge removes is overwhelmingly a ZLB phenomenon, and the regime-specific robust purge accordingly removes more variance than the full-sample Huber (0.14 against 0.11), concentrated there. Its impulse response, however, nearly coincides with the full-sample Huber and is far more moderate than OLS: the peak output response is about -0.5% rather than -1.9%. Robustness is thus the first-order correction (it is what moves the response from the OLS magnitude to a moderate one), while the regime split refines it and, more usefully, pinpoints *when* the contamination the purge targets actually occurs.

Two prescriptions follow. Downweight influential events: a robust projection is not a refinement but the first-order fix, since it is what moves the response from the OLS magnitude to a moderate one and roughly halves the identified set. And disclose leverage: report how much of $\hat{\beta}$, and of the resulting impulse response, rests on the handful of large, crisis-era meetings, because that is where the purge does almost all of its work.

6.2 Target versus path

A monetary announcement moves the whole expected path of rates, not just the current one. Following Gürkaynak et al. (2005) we decompose the statement-window surprise into a *target* factor, the surprise to the current policy rate, and an orthogonal *path* (forward-guidance) factor, extracting the first two principal components of the fed-funds and Eurodollar futures changes (USMPD; Swanson and Federal Reserve Bank of San Francisco Center for Monetary Research, 2024) and rotating so the path factor has zero loading on the current-month future; the loadings are textbook. Lewis (2021) and Jarociński (2024) pursue related multi-dimensional decompositions.

Predictability rises with the forecast horizon and the two components are not equally contaminated (Figure 7). The predictable share climbs from 7% at the current-month contract to about 15% at the six-month-to-one-year horizon, the contracts on which forward guidance operates, before easing at the long end, and the standardised path factor is roughly *twice* as predictable as the target: $R^2 = 0.109$ against 0.055 under the BS controls ($p = 6.5 \times 10^{-5}$ against $p = 0.027$), and 0.101 against 0.054 under a cross-validated selection of six principal components of the extended panel. The path factor is the forward-looking one: forward guidance revises the expected trajectory of policy in response to, and reveals information about, the economic outlook, and the BS controls are themselves largely forward-looking asset prices and macro trends that embed the same outlook. The target factor, the surprise in the immediate decision, is more idiosyncratic to the meeting and far less forecastable. The predictable, non-policy contamination therefore lives disproportionately in the path: the very component through which the Fed information effect operates.

This has a direct consequence for the impulse responses, since a shock identified from the

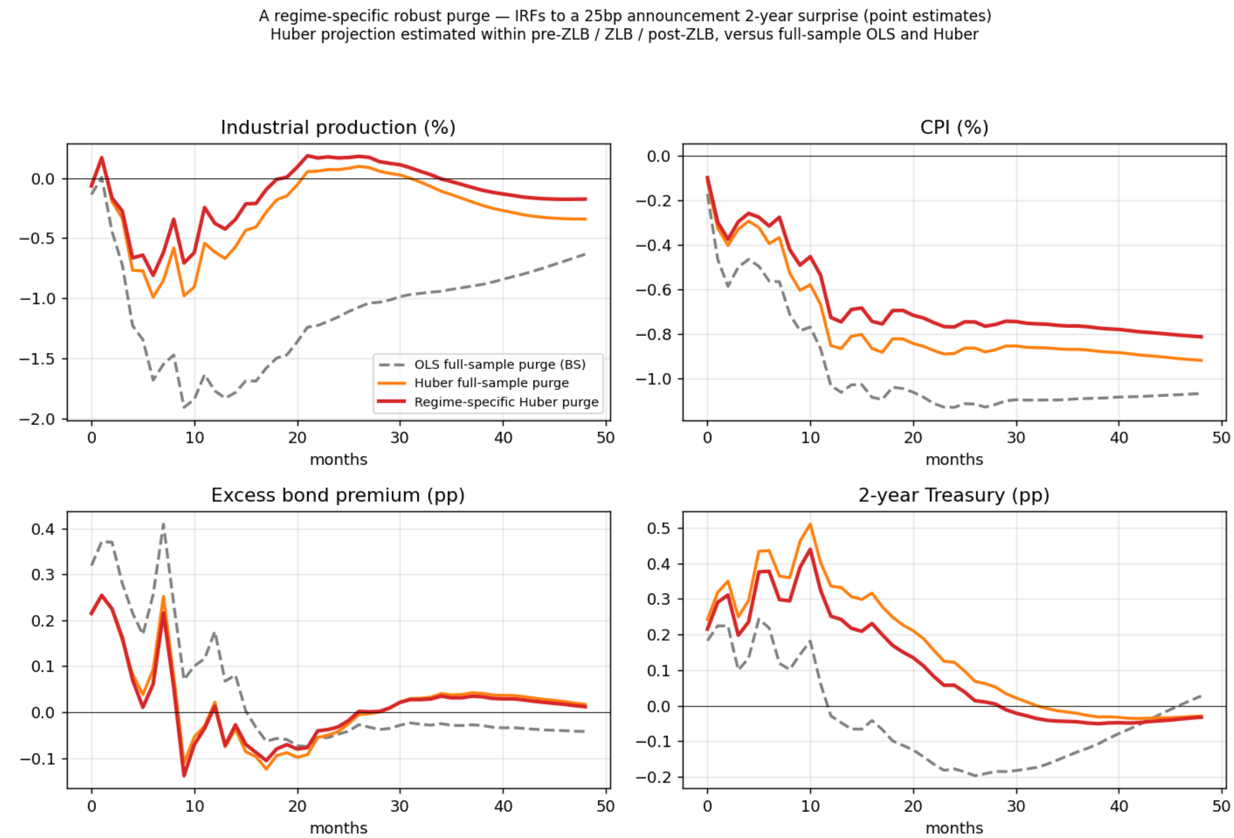

Figure 6: **Regime-specific robust purge.** A Huber projection estimated within pre-ZLB / ZLB / post-ZLB, versus the full-sample OLS (BS6) and Huber purges. Both robust purges give a far more moderate output response than OLS and nearly coincide; the regime split removes more where the S&P response-to-news coefficient is concentrated.

more-predictable path factor inherits the contamination (Figure 24 in Appendix E). The target factor delivers a clean contraction; the path factor does not (output *rises* after the tightening (+0.24% at twelve months), the information-effect signature, even as prices fall), and orthogonalising the path on the BS controls removes the wrong-signed output hump, taking the twelve-month response from +0.24% to −0.22%. This is the lesson of Section 8 seen through the factor decomposition: the predictable component is where the sign problem lives. The implication is to decompose *before* purging: report target and path separately and treat the path with more suspicion, since a purge applied to the blended surprise inherits a contamination it could have isolated.

The same exercise applied to the target factor illustrates why the impact diagnostic of Section 3.1 has to be reported alongside any such comparison. Purging the target factor appears to deepen its output response from −1.17% to −2.26%, which would invite the reading that the target too was contaminated toward zero. It was not: the purged target factor has an impact on the two-year rate of −0.0013, so it does not raise the policy rate on impact and its impulse response is a sign-flipped artefact rather than a monetary experiment. The purge here has removed enough of a component that was already the less predictable of the two to leave nothing usable behind. Purging

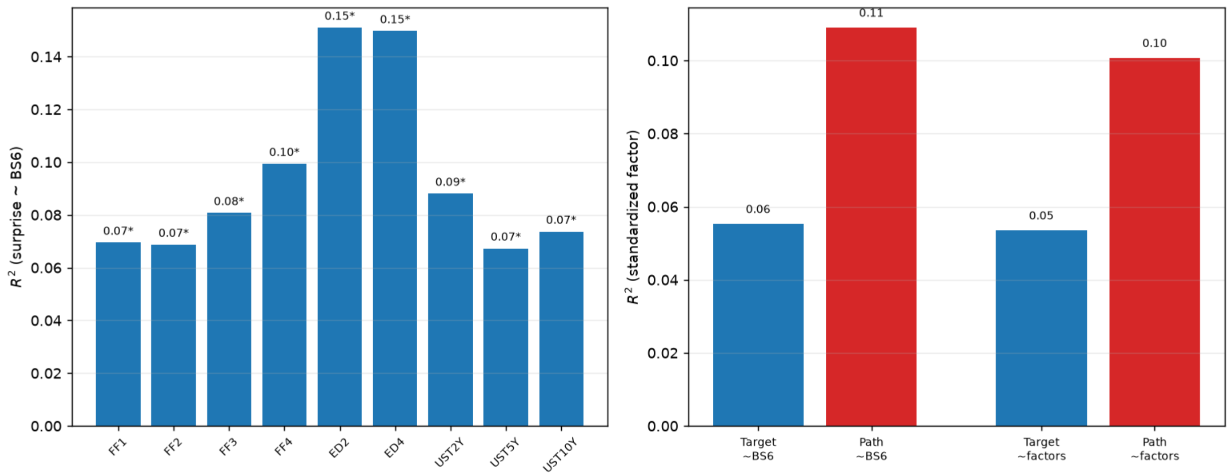

Figure 7: **Predictability by component.** Left: R^2 of each futures/yield surprise on the BS controls, by horizon ($*$ = significant at 5%); predictability peaks at the six-month-to-one-year (forward-guidance) horizon. Right: the standardised path factor is about twice as predictable as the target, under both the BS and the factor conditioning sets.

a component that is close to clean is not free: this is the over-cleaning of Proposition 2 appearing in the first stage rather than in the response, exactly as it does on the tightening half in Section 5.5.

6.3 Regime-dependence of the projection

The purge estimates a single β over the whole sample and applies it to every meeting. The coefficient vector is statistically stable, but the predictable *share* is not. A sup-Wald (QLR) test (Andrews, 1993) for an unknown break in the full coefficient vector of *MPS* $\sim$`BS6` does not reject stability (sup-$F = 2.8$, wild-bootstrap $p = 0.38$): there is no sharp break in the Fed's response *rule*. Yet the rolling 70-meeting R^2 ranges from about 0.07 to 0.6, and the regime-specific R^2 climbs from 0.12 before the ZLB to 0.47 during it, eases to 0.26 in 2016–2021, and returns to 0.42 after 2021 (Figure 25). The split by component is sharper still: the *target* factor's predictability spikes to 0.65 at the ZLB (when the current-rate surprise is tiny and dominated by the systematic response), then collapses to 0.045 in 2016–2021, while the *path* factor's rises steadily to 0.51 after 2021. The reconciliation is that the response rule is roughly constant but the *signal-to-noise* of the surprise shifts with the regime, because the predictable component is concentrated in specific episodes.

This has a direct identification consequence: re-identifying the impact vector from each regime's residual covariance, the target shock is clean and contractionary before the ZLB (-1.28% output at twelve months, impact on the two-year rate $+0.085$) but fails in both directions outside it (Figure 26). At the ZLB, where the funds rate is pinned and the target factor is nearly degenerate, the output response is wrong-signed and large ($+2.40$%, with prices also rising $+1.59$%). After the ZLB the output response returns to a plausible sign, but for the wrong reason: the impact of the target factor on the two-year rate is itself *negative* (-0.028; the raw first-stage coefficient carries the same sign), so the "contraction" is a sign-flipped artefact of a shock that does not raise the policy rate on impact at all. The apparently reassuring post-2015 number is the least trustworthy of the three. The lesson is not that the target factor breaks down *at* the lower bound but that its identification is secure only in the pre-ZLB sample, where the current-rate surprise still carries independent variation; and it is a warning against reading a plausible-looking impulse response without first checking the sign of the impact on the policy variable, which is exactly the diagnostic Section 3.1 recommends reporting.

The calendar splits impose the regime dates by hand. A stronger test lets the data choose them: for each series $y_t \in \{MPS, \text{target}, \text{path}\}$ we form the full-sample predictable index $c_t = \hat{\beta}' X_t$ and fit a two-state Markov-switching regression $y_t = a_{S_t} + b_{S_t} c_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2_{S_t})$, with S_t a hidden Markov chain; the regime with the larger implied $R^2 = b_S^2 \operatorname{Var}(c) / [b_S^2 \operatorname{Var}(c) + \sigma_S^2]$ is the contaminated state, and nothing about the ZLB or crisis dates is imposed. Because the *loading* switches, not just the variance, the model separates episodes in which the surprise tracks predictable news from those in which it does not. The estimated regimes recover the episodic structure of the hand-dated splits without being told the dates (Figure 8). For the headline surprise the high-predictability state carries $R^2 = 0.17$ against 0.03, is persistent (expected duration ≈ 8 meetings) and is active about half the sample. The component decomposition is the sharper result: the *path* factor is in its high-predictability regime three-quarters of the time, with the longest duration of the three (≈ 27 meetings, $R^2 = 0.13$ against 0.01), whereas the *target* factor's contaminated regime is shorter-lived (≈ 4 meetings) and occupies about a third of the sample.

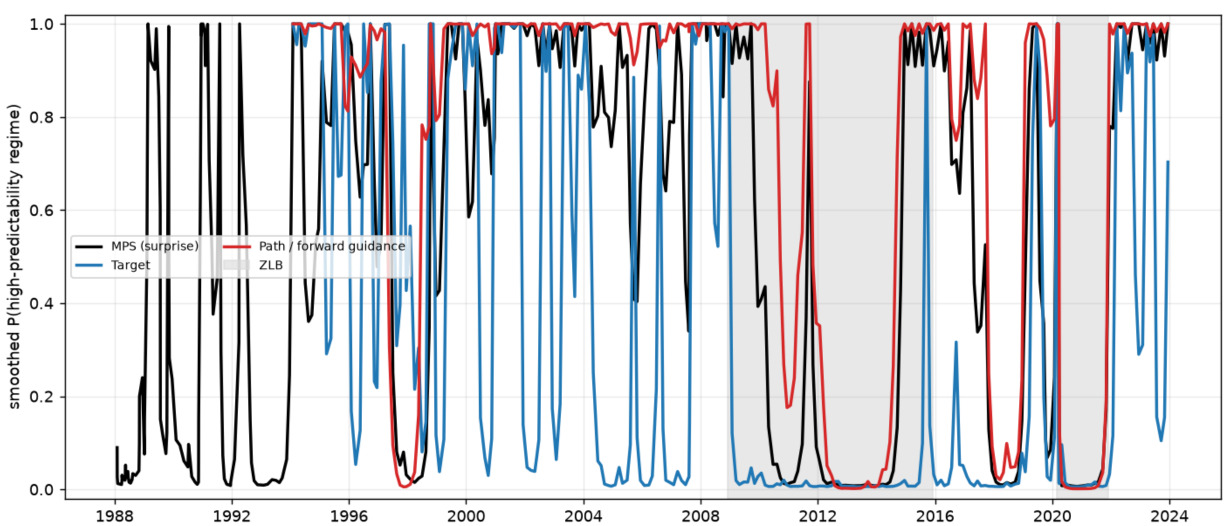

Figure 8: **Endogenous (Markov-switching) time-varying predictability.** Smoothed probability of the high-predictability (contaminated) regime for the surprise and its target/path components; the regime is estimated, not imposed. Shaded: ZLB and COVID. The path component sits in its contaminated regime far more persistently (≈75% of the sample, ≈27-meeting duration) than the target (≈36%, ≈4-meeting duration).

A purge is therefore not a set-and-forget transformation: the same constant β is too aggressive in calm years and too timid in crises. The projection should be estimated regime by regime or, at a minimum, reported alongside its rolling first-stage strength, so a user can see *when* the instrument is weak rather than assuming a constant signal-to-noise. Because the contaminated regimes are estimated rather than imposed, the same diagnostic can be run on any central bank's surprises without hand-picking crisis windows. One caution carries over from Section 5, and Appendix D develops it: a regime split changes *how much* systematic response there is, but not the *direction* of the projection (the pooled fit correlates 0.93–0.97 with the regime-own fits against only 0.27 with the tightening-own fit), so correcting regime by regime barely moves the aggregate response. Correcting *state* by state changes the projection's direction rather than its scale, which is why Section 5.1 finds the state interaction significant where a regime dummy is not, though, as Section 5.5 shows, changing the direction of the projection does not translate into a materially different impulse response on this design. The regime tells you where the contamination is concentrated; the state tells you that a single symmetric slope is the wrong object.

7 How to purge: what the diagnostic can and cannot license

The magnitude spread of Section 4 traces to two features of the purge: it leaves a weak instrument, and the projection is non-unique. This section addresses both: first by establishing what the predictability diagnostic actually measures, then by building projections that remove the degrees of freedom the hand-picked six controls leave open.

7.1 The predictability is small in the population, not illusory

The purge in (1) is justified whenever the surprise is correlated with pre-announcement information *in the population*, i.e. $\beta \neq 0$. Table 8 shows that the raw in-sample fit overstates that population quantity: for the six BS controls the in-sample R^2 of 0.15 falls to -0.06 under random five-fold cross-validation, and the broad 29-control panel over-fits outright (in-sample 0.18, five-fold -0.09); only ridge and lasso, by shrinking the overfit, recover a small positive cross-validated R^2 (0.03–0.05), and lasso keeps just six of 29 controls.

Those cross-validated figures must themselves be read with the sampling error attached, and this is where the diagnostic turns out to have the same defect as the impulse responses of Section 4. Re-running the `BS6` cross-validation over twenty independent random partitions gives a mean of -0.062 with a standard deviation *across partitions* of 0.171. The entire column of Table 8, from the panel's -0.085 to `A1`'s 0.120, spans about 1.2 partition-to-partition standard deviations of a single one of its entries. A single-partition cross-validated R^2 is not a statistic one should rank control sets on, and the differences among the rows below should not be read as differences in population predictability. We report the column because it is what the literature reports, and the twenty-partition figure because it is what the column is worth.

The right reading of this table is more careful than "the predictability is fragile." If $\beta \neq 0$ in the population, the cross-validated R^2 converges to the *population* R^2, not to zero. A cross-validated R^2 of -0.06 with a cross-partition standard deviation of 0.17 is therefore not evidence that $\beta = 0$; it is consistent with any population R^2 from zero up to roughly a tenth, and the diagnostic cannot separate those. What it does rule out is a population R^2 anywhere near the in-sample 0.15. In-sample predictability of the surprise on pre-announcement controls is real: the interaction test of Section 5.1 rejects at $p = 0.030$ on the same sample, but essentially none of it survives to a held-out fold. What is informative is that the gap between the in-sample 0.147 and the cross-validated -0.062 exceeds the degrees-of-freedom correction $k/n \approx 0.02$ by an order of magnitude; that excess is the leverage documented in Section 6.1: a handful of crisis meetings that different folds fit differently. Three conclusions follow, and they are the ones that matter for practice.

First, a small orthogonalisation R^2 is not a clean-shock certificate but the *weak-instrument* regime. Contamination is a population fact: even a small β biases the impulse responses if the predictable

Table 8: In-sample versus cross-validated predictability of the surprise. Random five-fold CV measures how much of the in-sample fit is population predictability rather than over-fitting. Event level, each set on its complete-case sample. The CV column is a *single* random partition, which is the convention in this literature; for `BS6` the mean over twenty partitions is -0.062 with a cross-partition standard deviation of 0.171, so differences between rows of this column are not informative.

Control set	in-sample R^2	5-fold CV R^2
`BS6` (six BS controls)	0.182	0.058
`A1`	0.153	0.120
`A2`	0.146	0.029
`A4`	0.107	0.069
`B1`	0.127	0.077
`B3`	0.113	0.061
Panel (29 controls), OLS	0.179	-0.085
Panel, ridge (CV penalty)	0.133	0.049
Panel, lasso (CV, 6/29 kept)	0.128	0.028

component is correlated with non-policy shocks, and indeed that bias is what produces the sign flips of Section 4. "Little predictability" and "little contamination" are different statements, and the diagnostic measures the first. Second, the orthogonalisation is an *ex-post* identification device, justified by the population projection, and one should not ask it to double as a real-time forecast, which is what (T3) of Section 5.4 establishes. Third, and consequently, a debiased (double-ML) cross-fit purge is inappropriate here: it forms the residual from an out-of-fold prediction that is essentially the sample mean, so the cross-fitted residual is 0.89-correlated with the raw surprise and discards the sign correction altogether. The purge should be estimated in sample, on the full data.

The gap between the in-sample and cross-validated R^2 has a decomposition that makes the reading above precise. For a projection on k controls estimated on n events, the expected in-sample fit exceeds the population fit by approximately the degrees-of-freedom correction,

$$\mathbb{E}\left[R^2_{\text{in}}\right] \;\approx\; R^2_{\text{pop}} + \frac{k}{n}\left(1 - R^2_{\text{pop}}\right), \tag{16}$$

so with $k = 6$ and $n = 360$ the mechanical inflation is about 0.018. The observed gap for `BS6` is $0.147 - (-0.062) = 0.209$, giving $\Delta = 0.192$: roughly eleven times what (16) accounts for. That excess is not degrees of freedom; it is instability of $\hat{\beta}$ across resamples, which is the leverage documented in Section 6.1. The two facts are therefore one fact seen twice: a projection carried by a handful of crisis observations fits its own sample well and a held-out fold poorly, and the size of the shortfall measures how concentrated that leverage is.

This yields a diagnostic worth reporting alongside any orthogonalisation:

$$\Delta \;\equiv\; \left(R^2_{\text{in}} - R^2_{\text{CV}}\right) - \frac{k}{n}\left(1 - R^2_{\text{CV}}\right), \tag{17}$$

the *excess* of the in-sample advantage over what the parameter count explains. $\Delta \approx 0$ says the projection is stable and the in-sample fit can be taken at face value; Δ large says a few events are doing the work, and the purge those coefficients imply will be correspondingly fragile. Table 8 reports R^2_{in} and R^2_{CV} for every control set; Table 9 reports Δ alongside the two fits and the degrees-of-freedom correction it nets out.

To read $\Delta = 0.192$ against a mechanical inflation of 0.018 concretely: a projection with this much instability behaves as though it had been fitted with about seventy free parameters rather than six, on three hundred and sixty observations. That is not what a practitioner believes they are doing when they regress a surprise on six controls, and it is the reason the purged series built from those coefficients differs so much more across specifications than the coefficients themselves suggest it should.

Table 9: The in-sample advantage, and how much of it the parameter count explains. Δ is defined in (17): it is the part of the in-sample-versus-cross-validated gap that the degrees-of-freedom correction does *not* account for, and therefore measures instability of $\hat{\beta}$ across resamples rather than overfitting in the ordinary sense.

Control set	n	R^2_{in}	R^2_{CV}	Δ
`BS6`	360	0.147	-0.062	0.192

7.2 Minimal, regularised and factor projections

If the goal is to remove the genuine ex-post contamination with the least attenuation, the natural estimator is a *minimal* projection: greedily add pre-FOMC controls from the broad full-coverage pool and stop as soon as the next control's HC1-robust t-statistic falls below 2. On our sample this selects just **two** controls: the three-month Treasury change and the commodity-index return ($R^2_{\text{in}} = 0.116$). A ridge projection on the whole 29-control pool (CV-chosen penalty, $R^2_{\text{in}} = 0.129$) is a regularised alternative that shrinks the overfit component instead of hard-selecting. Both reproduce the qualitative pattern `RAW` lacks, a persistent output decline and a sustained rise in the excess bond premium, but at a *tighter and more moderate* magnitude than the fixed six-control set: peak industrial production of about -1.3% against -2.0% for `BS6` (Figure 23 in Appendix E). Crucially the minimal and ridge purges, built from different pools and rules, agree with each other. Tying the projection to the contamination diagnostic, rather than to an arbitrary six controls, keeps the sign correction while collapsing much of the control-set magnitude spread. The minimal set is, however, sensitive to the candidate pool and the t-threshold: a stricter or looser cutoff yields one to four controls, so the *stability* of the selection, for instance its bootstrap frequency, is itself worth reporting.

A way to remove the "which controls?" freedom entirely is to summarise the whole pre-FOMC information set by principal components and purge on the factors, letting cross-validation choose only *how many*. We take every control with at least 200 meetings of coverage (58 series spanning the macro surprises, the Bloomberg financial block and the FRED additions), then standardise them, extract principal components, and forward-select the components by five-fold CV predictive power; the purge then uses full-sample coefficients on the selected factors. On the common post-1998 sample this keeps **six** components with a stable cross-validated R^2 of about 0.12, against roughly 0.04 for the six BS controls on the *same* sample: the sharpest available evidence that a real if small predictable component exists once the macro surprises are included. The intuition is coverage: pre-announcement contamination is the footprint of *many* underlying non-policy shocks, and each principal component is a linear combination of all 58 series, so a handful of factors span more of them than an equal number of raw controls.

That extra coverage is better *spanning* of the same contamination, however, not a source the BS controls miss. A Frisch–Waugh–Lovell partial test finds no reliable predictability *beyond* the six BS controls, stacking the factors on `BS6` overfits rather than improves the cross-validated fit, so the broad panel and the hand-picked six load on the same predictable factor. This is exactly what the impulse responses show (Figure 9): the factor purge lands essentially on top of `BS6` and the minimal purge for output and the excess bond premium, while delivering a somewhat larger and more persistent price decline. The exercise also makes a sample point visible: on this common window `BS6` itself is far less persistent than on the full sample, so the deep, permanent output decline associated with the fixed BS purge is substantially a pre-1998 sample feature rather than a property of the purge.

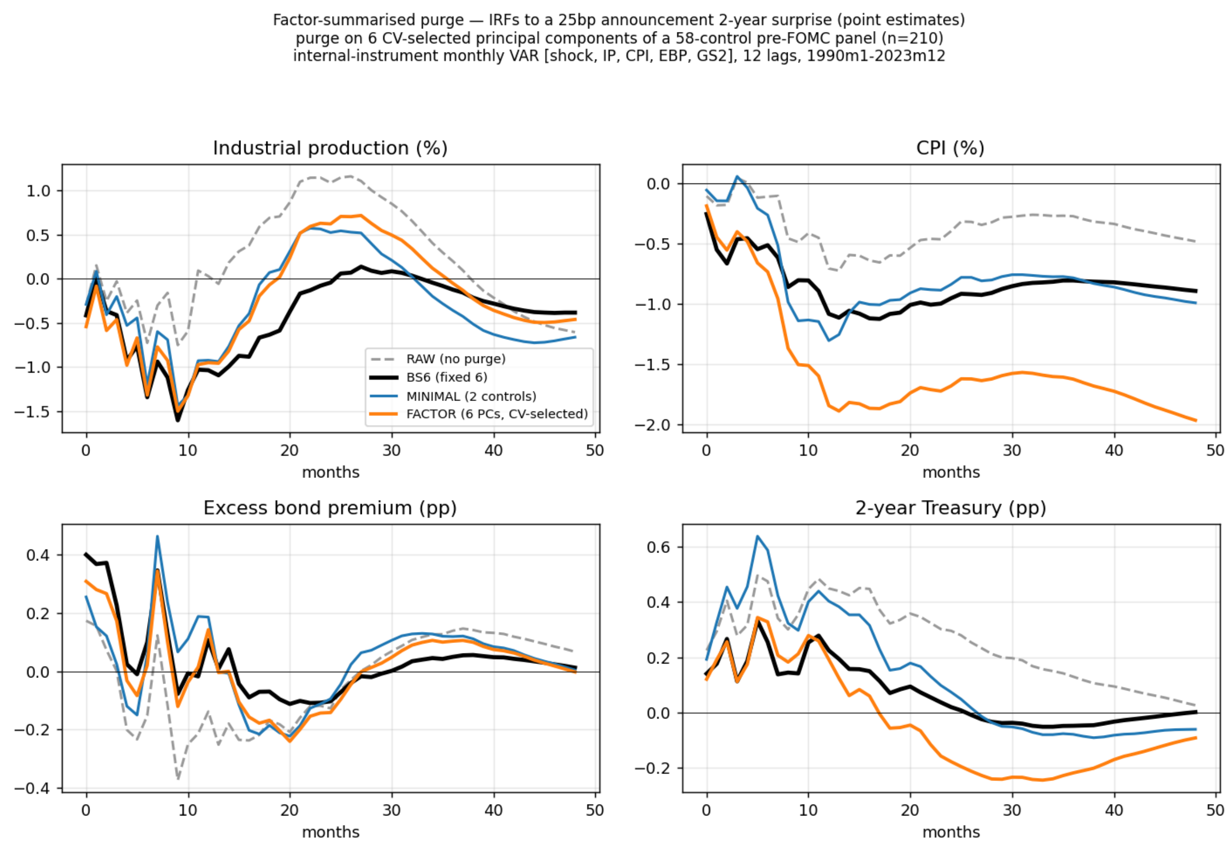

Figure 9: **Factor-summarised purge.** Purge on six CV-selected principal components of a 58-control pre-FOMC panel (CV $R^2 \approx 0.12$), versus `RAW`, `BS6` and the minimal purge, all on the common 1998–2023 sample. The factor purge removes the control-subset degree of freedom and reproduces `BS6` for output and credit, with a cleaner price decline.

7.3 What the factors are, and the one control always to include

Labelling the factors requires no sign restriction. A structural restriction is needed only to give a factor a *causal* name; to *describe* a statistical factor by the observable variables it is built from, one reads its loadings. Grouping the 58 controls into economic blocks and measuring each factor's variance share by block (Figure 10) names them at once. The factor that carries the predictability, the leading component `PC0` (single-factor $R^2 = 0.10$), loads on high-yield and corporate credit spreads, the five-year Treasury change and the Bloomberg financial-conditions index: a *financial-conditions factor*. The next predictive component mixes industrial-production surprises with commodities, and the component that loads on payrolls and the ADS activity index, a *real-activity* factor, barely predicts ($R^2 = 0.02$). This is naming by composition, exactly as the target and path factors of Gürkaynak et al. (2005) are named by their loadings, and it carries no causal claim.

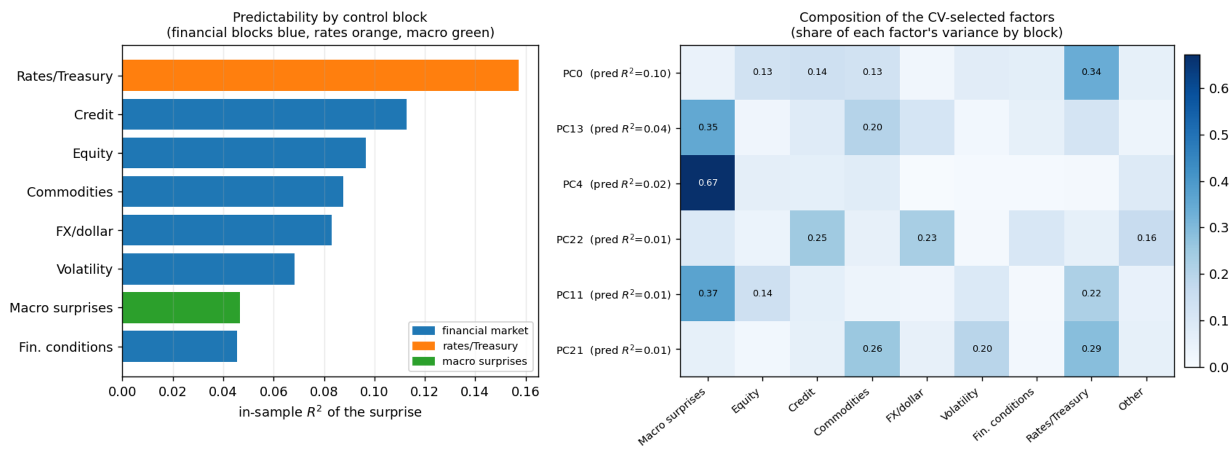

Figure 10: **Labelling the predictive factors.** *Left:* in-sample R^2 of the surprise by control block: the financial-market blocks and Treasuries carry the predictability, the macro-surprise block the least. Equity is also the most robust *cross-validated* predictor; the large in-sample R^2 of the twelve-variable rate block is mostly overfit. *Right:* each CV-selected factor's share of variance by block, with its single-factor predictive R^2.

The labels deliver a concrete prescription: *always include a financial-market control.* Block by block, the predictability lives in the financial and rate blocks (equity, credit, commodities and Treasuries), while the macro-surprise block is weakest; equity alone is the single most robust predictor, and a macro-surprise-only set carries almost none of the cross-validated contamination. The BS controls' predictable component correlates 0.73 with `PC0` and only 0.06 with the macro factor: the six BS controls are, in effect, a hand-built proxy for the financial-conditions factor, which is why the financially constructed Group A sets recover them and the macro-surprise Group B sets do not. This is the Chen (2025) finding that financial conditions carry most of the predictability, and it echoes the "proxy zoo" of Huang and Neri (2026), who document that monetary-policy proxies are significantly correlated with identified *financial* shocks. It is also what the Fed put of Section 5 implies: if the systematic component is an easing response to financial deterioration, the conditioning set must contain the financial state.

A caveat sharpens the rule against the closest alternative. Chen (2025) makes the case for financial conditions with a *single* index, the level of the OFR Financial Stress Index, which he reports predicts the surprise about as well as all other predictors combined. Placing a single index alongside the other financial proxies in the common 66-day-change form, it is informative but the *weakest* of them: on the common post-1998 sample the Bloomberg financial-conditions index predicts with a cross-validated R^2 of 0.04 and the Chicago Fed NFCI 0.03, against 0.09 for the S&P 500 three-month return alone and 0.07 for the financial-conditions factor.[5] A single index is not a *different* signal, the Bloomberg index change correlates 0.81 with `PC0`, but a smoother, and here weaker, proxy for it, so the sharpest *single* control is the equity return. For a practitioner the lesson is that *how* one summarises the information set matters more than *which* handful of controls one picks, provided the number of components is chosen by cross-validation rather than by in-sample fit, since the in-sample-versus-CV gap is exactly the overfit one wants to leave out.

8 The information effect: a second, complementary restriction

The purge conditions on *pre*-announcement information. A complementary restriction uses the *simultaneous* high-frequency co-movement of the surprise with the stock market (Jarociński and Karadi, 2020). A pure monetary tightening raises rates and lowers stock prices; a central-bank *information* shock raises both. The "poor-man's" sign restriction therefore keeps a surprise as pure monetary policy only when the rate and the statement-window S&P 500 move in *opposite* directions. We take the surprise (`pc1`, the first principal component of the interest-rate futures) and the statement-window asset moves from the updated Jarociński–Karadi shocks and the SF Fed US Monetary Policy Event-Study Database, and compare four shocks in the same VAR: `RAW`; `ORTH` (`pc1` purged of BS predictability); `INFO` (the JK sign-restricted `MP\_pm`); and `ORTH+INFO`.

The two corrections address distinct contamination (Figure 11). The raw surprise gives a *wrong-signed* real response: industrial production *rises* after a tightening (+0.7% at two years). The BS purge alone fixes the sign (IP $\approx -1.2\%$, excess bond premium up): the predictable component was the binding contamination for the sign. The information-effect correction *alone* does *not* fix it (output still rises), because dropping the same-sign events leaves the predictable component intact. Only when both are applied is the sign robustly negative at a disciplined magnitude. The two orderings differ by at most 0.2 across all horizons, so the order is immaterial, as separability in Section 2.1 implies; it is nonetheless more coherent to purge the predictable contamination first and check the residual's sign second. This does not hinge on the poor-man's hard classification: repeating the exercise with JK's preferred *median-rotation* shock, which splits every event into monetary and information components using the full-sample covariance sign restriction, the `INFO` shock is still non-negative on its own, and the median rotation combined with the purge yields the cleanest contraction of all the variants (Figure 27 in Appendix E), which is the recommended estimator.

8.1 No pre-announcement purge removes the information effect

That the purge does not itself remove the information effect is visible directly in the surprises. Orthogonalisation does not move the distribution of events across quadrants of the surprise/statement-window-equity plane at all. On the Jarociński and Karadi (2020) event sample the share in the

[5] These cross-validated R^2s are computed on the common post-1998 complete-case sample and are not comparable with the full-sample figures of Table 8; the ranking *among* financial proxies is the object. The `BS6` cross-validated R^2 on this window is reported in Table 9, computed as the mean over twenty random five-fold partitions. The OFR index is not in our panel and Chen conditions on its *level*, which need not have the same strength as a change; Chen (2025) notes the Bloomberg conditions variables are a subset of the OFR index.

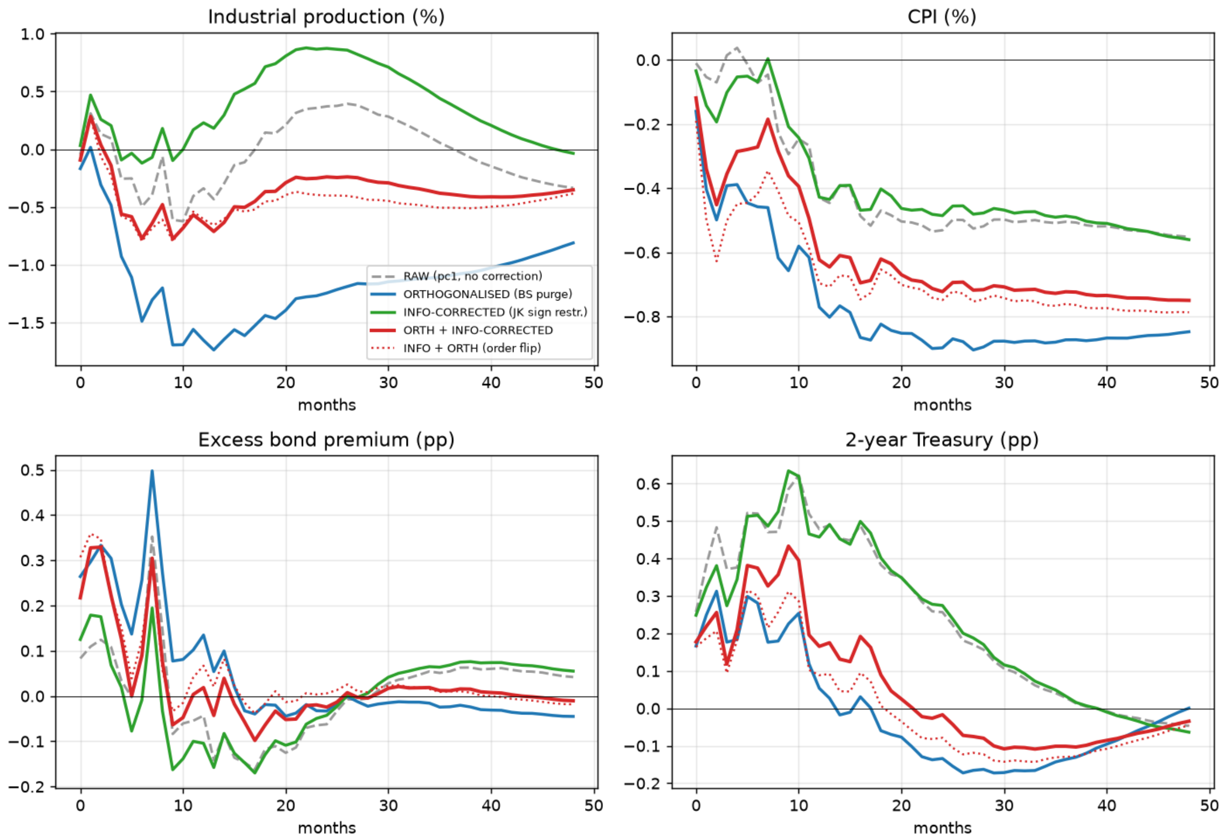

Figure 11: **Purge versus information-effect correction.** IRFs to a 25 bp announcement two-year surprise. The BS purge (`ORTH`) recovers the contractionary sign; the JK sign restriction alone (`INFO`) does not; both together (`ORTH+INFO`) do, and the ordering is immaterial (dotted).

"wrong" same-sign (information) quadrant is 34.7% before the purge and 35.0% after it (303 events with a nonzero equity move), even though projecting the first principal component of the intraday surprises on the Bauer and Swanson (2023a) controls removes a statistically comfortable $R^2 = 0.134$ of its variance. The purge is not weak on this sample; it is simply orthogonal to the dimension the quadrant lives in. Conditioning on the Fed's *own* forecasts does no better, if anything slightly worse. On the subsample with a contemporaneous Greenbook (Figure 12) the raw surprise has 36% of events in the wrong quadrant, the BS purge 32%, and a Miranda-Agrippino and Ricco (2021)-style purge on the Greenbook forecasts 37%.

Two benchmarks are needed to read these shares, and we supply both rather than let the reader supply the wrong one. Under independence of the two signs the expected share is 50%, so roughly a third is already well below chance before any purge is applied; the claim is that the purge leaves the residual information content where it found it, not that a third of events are information shocks. And quadrant counts are dominated by events near the origin, where the sign of either variable is noise. We therefore also report the share of $|\Delta i| \times |r|$ *weight* in the wrong quadrant and the count excluding the inner decile. Table 10 reports the count share, the magnitude-weighted share, and the count excluding the inner decile of $|\Delta i| \cdot |r|$, against the independence benchmark, and the three columns do not tell the same story. The table is computed on the paper's own monthly surprise rather than the Jarociński and Karadi (2020) event panel, so the levels differ slightly from those above; the pattern across columns is what matters. By *count* the wrong-quadrant share falls from 36% to 33%, which is the number usually quoted. Weighted by $|\Delta i| \cdot |r|$ it falls from 11% to 10%: the events in the information quadrant are systematically the *small* ones, and account for about a tenth of the economic weight rather than a third of it. Dropping the inner decile barely moves the count (34% to 32%), so this is a size gradient across the whole distribution rather than a cluster of near-zero events.

That qualification cuts in a specific direction and we state it plainly. It does not rescue the purge (the share the purge removes is small on every measure, which is the point of this section), but it does temper how much of the surprise's variance the information channel can plausibly be carrying. A reader who takes "a third of orthogonalised surprises lie in the information quadrant" as a statement about economic magnitude will overstate the channel by roughly a factor of three. The count is the right statistic for asking whether the purge *reaches* the information channel, which it does not; the weighted share is the right one for asking how much is at stake, and it is smaller than the count suggests.

Table 10: Wrong-quadrant shares, three ways. The count share is the raw fraction of events in which the rate and the statement-window equity return move together; the magnitude-weighted share weights each event by $|\Delta i| \cdot |r|$; the last column drops the inner decile of that product, where the sign of either variable is noise. Under independence of the two signs every column would be 0.5, so the object of interest is the movement from raw to purged, not the level.

Series	count share	magnitude-weighted	excl. inner decile
RAW	0.364	0.107	0.336
BS-orthogonalised	0.331	0.102	0.321
Independence benchmark	0.500	0.500	0.500

The intuition is that being "informationally robust" in the sense of Miranda-Agrippino and Ricco (2021) is not the same as being free of the information effect in the sense of Jarociński and Karadi (2020): the two invoke different notions of information. The information effect is *contemporaneous*, at the announcement the market infers the Fed's assessment from the decision. Both purges condition only on *pre-announcement* information, but the co-movement that defines the wrong quadrant is driven by what the announcement reveals *relative to expectations*, which is orthogonal to any pre-announcement set by construction. Conditioning on the Greenbook cannot help even in principle: it is *unobserved* by markets (released with a five-year lag), so the information effect exists *because* the market must infer the Fed's view at the announcement, and the Greenbook is a set of *macro* forecasts, orthogonal to the financial dimension the quadrant lives in. The BS controls, which include pre-announcement financial variables, overlap marginally more with that dimension, which is why the BS purge nudges the share down a touch while the Greenbook purge does not. A forecast-based "informationally-robust" instrument still needs the high-frequency sign restriction to shed the information effect. The stock market remains the identifying asset: a tightening and a positive information shock both appreciate the dollar, so an FX-based classification agrees with the stock-based one only 56–64% of the time, barely above chance (Figure 28 in Appendix E).

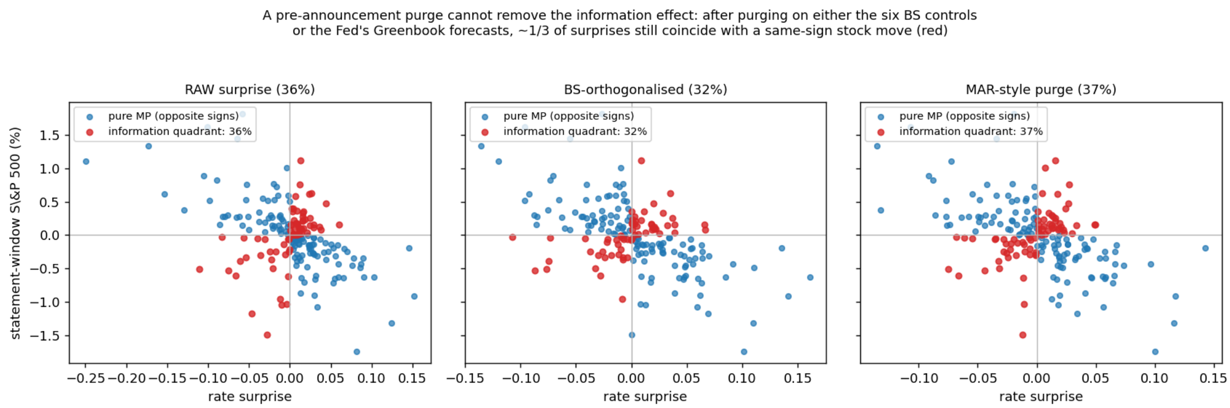

Figure 12: **No pre-announcement purge removes the information effect.** The surprise versus the statement-window S&P 500 on the contemporaneous-Greenbook subsample: raw (left), BS-orthogonalised (middle), and a MAR-style purge on the Greenbook forecasts (right). Red points (rate and stocks moving *together*) are the information quadrant; their share is essentially unchanged: 36%, 32%, 37%.

8.2 The Greenbook and the BS controls are complementary

A different conditioning set points the same way. Miranda-Agrippino and Ricco (2021, "MAR") construct an informationally-robust instrument by projecting the surprise on the Greenbook forecasts and *revisions* of output growth, inflation and unemployment, and on twelve of its own lags. To place the two purges on an equal footing we apply this same conditioning set to the *same* surprise the BS purge cleans, at meeting frequency.

Does the MAR-purged surprise still carry the predictability the BS controls target? Because the Greenbook forecasts and the BS controls are themselves correlated, the right question is whether the BS controls add predictive power *beyond* the Greenbook set: a partial test, not a regression of the MAR residual on the *raw* BS controls, which by Frisch–Waugh–Lovell ignores their shared component and understates the overlap. Adding the six BS controls to the full MAR conditioning set raises the R^2 of the surprise from 0.26 to 0.30, and the increment is significant: $F(6, 245) = 2.3$ ($p = 0.035$; heteroskedasticity-robust Wald $p = 0.035$), a partial R^2 of 0.05 (Figure 13, left). The naive sequential test reports only $R^2 = 0.02$, understating the overlap exactly as Frisch–Waugh–Lovell warns. The result is robust to dropping the longest Greenbook horizons ($p = 0.01$) and to restricting to single-meeting months ($p = 0.04$).

What explains this residual predictability? Not a *timing mismatch*. The Greenbook is finalised roughly a week before the decision, so one might expect the BS controls, measured up to the day before the announcement, to absorb the intervening week's news. Splitting the three-month S&P 500 window at the Greenbook date and residualising *both* the surprise and the two segments on the full Greenbook set before comparing them, the post-Greenbook return adds *nothing* ($t \approx 0$, $p = 0.98$), and neither does the pre-close segment ($t = 1.2$).[6] The operative mechanism is a *different-spaces* gap: the Greenbook set is the Fed's macro *point forecasts*, while the BS controls include slow-moving business-cycle and financial state variables those forecasts do not span. Consistent

[6] A naive sequential regression of the published MAR instrument on the *raw* pre- and post-close returns does make the post-Greenbook return look marginally significant ($t = -2.1$); this is the same Frisch–Waugh–Lovell artefact, and it vanishes under the partial test.

with this, the partial predictability is carried almost entirely by trailing year-over-year payroll growth (`NFP\_12M`, robust $t = 2.6$, $p = 0.01$), a business-cycle indicator known well before the Greenbook rather than a late-arriving surprise, with the remaining controls contributing jointly but not individually. This is the business-cycle predictability that Cieslak (2018) documents and Chen (2025) ties to financial conditions.

It is also the check the Fed put predicts. Cieslak and Vissing-Jorgensen (2021) find that Greenbook growth revisions absorb most of the intermeeting equity return's predictive power for policy, so if the contamination is the Fed put, the predictability surviving a Greenbook projection should *not* be carried by the equity control. It is not: the S&P segments are insignificant and payroll growth carries the increment. The two literatures agree, and the agreement is not mechanical.

Serial predictability, by contrast, is not the binding problem here. MAR motivate the twelve lags by autocorrelation in the surprise; in this sample that motivation does not bind (Figure 13, right). The raw surprise, its BS-orthogonalised version and the MAR-style residual all show own-lag R^2 at the overfitting floor ($12/n$), and the cross-sectional purge does not reduce it: orthogonalising on contemporaneous controls and on past surprises are different operations. Only the published MAR instrument, which removes the lags explicitly, sits at zero.

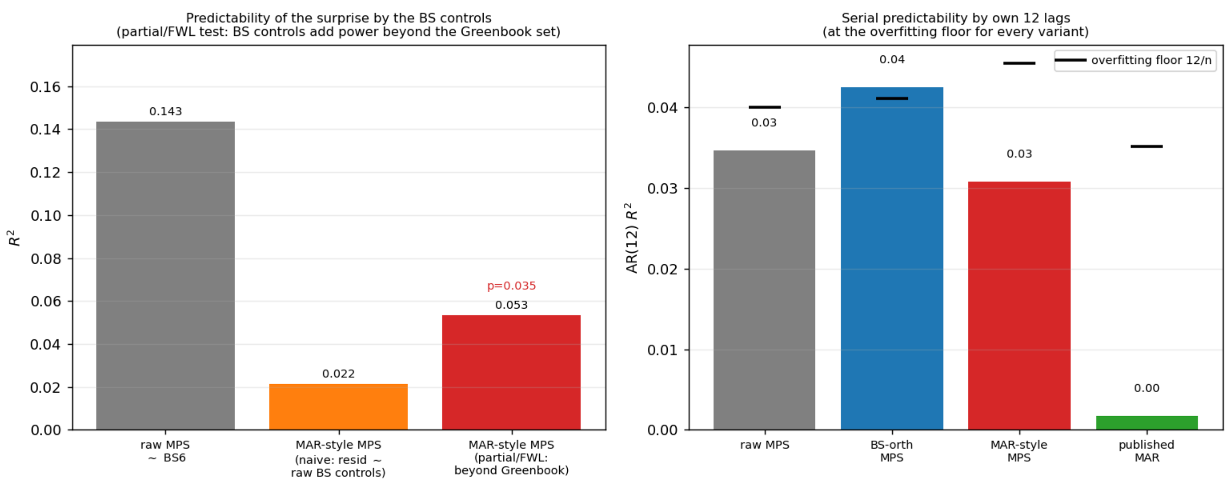

Figure 13: **Adapting the MAR conditioning set.** Left: predictability of the surprise by the six BS controls. The raw surprise is strongly predictable; the naive sequential test, the MAR residual on the *raw* BS controls, understates the overlap (0.02); the partial/FWL test, with the BS controls residualised on the Greenbook set, shows they add significant power *beyond* the Greenbook ($R^2 = 0.05$, $p = 0.035$). Right: serial predictability by own twelve lags is at the overfitting floor $12/n$ for every variant.

For applied work the recipe is two-step and complementary: orthogonalise on pre-announcement controls to fix the sign, then apply the high-frequency sign restriction to strip the information effect the purge cannot reach, conditioning on the Greenbook forecasts as well where they are available, since the financial and forecast-based controls clean different contamination. Neither correction substitutes for the other, and the order does not matter.

9 Aggregation and the timing of the control window

Every VAR above used a *purge-then-aggregate* (PTA) monthly instrument: each meeting's surprise was orthogonalised on its own pre-announcement controls and summed to the month. This is the last and subtlest cost of purging. At the monthly frequency one must both purge and aggregate, and when a month contains more than one surprise the order matters. *Aggregate-then-purge* (ATP, the ordering used by Bauer and Swanson, 2023b) sums the month's surprises and orthogonalises the total on one control vector dated before the *first* surprise; it keeps all the shock variation but conditions on stale information that misses news arriving between announcements: biased, which is Amodeo (2026)'s point. PTA, which he recommends, orthogonalises *each* surprise on its own controls and sums. The catch is that the BS controls are *asset prices* measured over multi-week windows: in a two-surprise month the window ending just before the second announcement *contains* the first, so those controls have already moved in response to the first monetary shock, and purging the second surprise on them can remove genuine monetary variation.

Making that precise requires only the timing. Index the two announcements in month m by $t = 1, 2$, write each surprise as a predictable response to news plus a structural shock, $MPS_{tm} = \beta' X^-_{tm} + s_{tm}$, and let the monthly object of interest be the sum $S_m = s_{1m} + s_{2m}$. The first control is predetermined, $X^-_{1m} = W_{1m}$. The second is measured over a window that closes just before Announcement 2 and therefore *opens before Announcement 1*, so its asset-price entries have already responded to the first shock,

$$X^-_{2m} = W_{2m} + \Lambda\, s_{1m}, \tag{18}$$

where W_{2m} is genuine inter-announcement news and Λ collects the high-frequency responses of the controls to a unit monetary shock: nonzero for an equity index or a yield curve, zero for a payroll surprise dated before the month began. The second announcement may itself re-price the first shock, so we allow

$$MPS_{2m} = \beta' W_{2m} + a\, s_{1m} + s_{2m}, \tag{19}$$

with $a \in [0, \beta'\Lambda]$ the *re-pricing coefficient*. Take a single asset-price control ($k = 1$, $\lambda \equiv \Lambda$), genuine-news variance σ^2_w, normalise $\sigma^2_s = 1$, write ρ for the within-month shock correlation, and let q be the share of months carrying a contaminated later surprise.

The one feature that a naive treatment gets wrong is that β is estimated *once*, by pooling every meeting, so the contaminated controls, which appear only for later within-month surprises, have their effect on $\hat{\beta}$ diluted by the many clean single-surprise months. Carrying that through gives the result.

Proposition 3 (Control-window leakage under purge-then-aggregate)**.** *Under* (18) *and* (19), *the pooled least-squares purge coefficient satisfies*

$$\hat{\beta} \xrightarrow{p} \frac{\beta\, \sigma^2_w (1+q) + q\, \lambda\, (a+\rho)}{\sigma^2_w (1+q) + q\, \lambda^2}, \tag{20}$$

and the purge-then-aggregate instrument in a two-surprise month is

$$\tilde{z}_m = \underbrace{\left(1 + a - \hat{\beta}\lambda\right)}_{\textit{loading on the earlier shock}} s_{1m} + s_{2m} + (\beta - \hat{\beta})\,(W_{1m} + W_{2m}). \tag{21}$$

The net leakage *is therefore*

$$L \equiv \hat{\beta}\lambda - a, \tag{22}$$

the amount of the first shock the purge removes in excess of what the second announcement had already re-priced. Two cases bracket it. Under cancellation, *$a = \beta\lambda$, we have $\hat{\beta} = \beta$, $L = 0$, and $\tilde{z}_m = S_m$ exactly. Under* innovation, *$a = 0$, the purge subtracts a component that was never in MPS_{2m}, $\hat{\beta}$ is attenuated by classical errors in variables, and $L = \hat{\beta}\lambda > 0$.*

Proof. The event-2 residual is $\hat{u}_{2m} = MPS_{2m} - \hat{\beta}\, X^{-}_{2m} = (\beta - \hat{\beta})W_{2m} + (a - \hat{\beta}\lambda)s_{1m} + s_{2m}$, and the event-1 residual is $\hat{u}_{1m} = (\beta - \hat{\beta})W_{1m} + s_{1m}$. Summing gives (21). Equation (20) is the pooled normal equation with a share $q/(1+q)$ of observations carrying $\mathrm{Var}(X^{-}_{2m}) = \sigma^2_w + \lambda^2$ and $\mathrm{Cov}(X^{-}_{2m}, MPS_{2m}) = \beta\sigma^2_w + \lambda(a + \rho)$. Appendix F gives the full environment, the aggregate-then-purge comparison, and a simulation that reproduces every expression to three decimals. □

The two poles have clean economic readings. Under cancellation the second surprise re-prices the first shock exactly as the Fed's systematic rule would, so purging on a window that contains it removes nothing genuine: this is implicitly the world of Amodeo (2026), in which PTA is clean and its weak first stage reflects only the removal of predictable news. Under innovation markets priced the implications of s_{1m} at Announcement 1, so Announcement 2's surprise is orthogonal to it, yet the econometrician still purges on a control that carries it. The magnitudes are not small. With $a = \rho = 0$, $\sigma^2_w = \lambda = \sigma^2_s = 1$ and $q = 0.12$, (20) gives $\hat{\beta} \approx 0.63$, so the loading on the earlier shock in (21) falls to about 0.37: nearly two-thirds of the first shock is purged away.

Two properties of the pooled treatment temper this, and both cut in the conservative direction. Because β is estimated by pooling, the aggregate first stage falls only about 8% in that calibration, more with larger λ or q, since s_{2m} is intact and single-surprise months are clean. And by (21) the estimator stays *consistent* for θ so long as the injected news term $(\beta - \hat{\beta})(W_{1m} + W_{2m})$ is uncorrelated with the macro error; it becomes biased only through that term, which is the aggregate-then-purge channel. Leakage is therefore a *relevance* phenomenon, not an asymptotic bias, and the per-event calculation that sets $\hat{\beta}$ from the contaminated moment alone would overstate L. Pooling is what both Bauer and Swanson (2023b) and Amodeo (2026) actually do.

Table 11: ATP versus PTA: two poles of a single bias–variance trade-off.

	ATP (aggregate-then-purge, BS)	**PTA** (purge-then-aggregate)
Controls for later surprise	pre-first-surprise W_{1m} (stale)	event-specific $X^{-}_{2m} = W_{2m} + \Lambda s_{1m}$
Omitted intra-month news W_{2m}	retained $\Rightarrow$ bias $+\beta' \,\mathrm{Cov}(W_{2m}, e)$	removed
Shock variation S_m	fully kept $\Rightarrow$ strong	leaked iff $a < \hat{\beta}\lambda$: loading $1 + a - \hat{\beta}\lambda$
First stage / weak-IV bias	strong; no weak-IV bias	weak by $L = \hat{\beta}\lambda - a$
Consistency of $\hat{\theta}$	inconsistent (news bias)	consistent (bias only via retained news)

Everything therefore turns on one estimable quantity, and suggests a direct test: regress the later within-month surprise on the earlier same-month shock, using the first surprise's purged residual

$\hat{u}_{1m}$ as a proxy,

$$MPS_{2m} = c + a\,\hat{u}_{1m} + \text{error}, \tag{23}$$

on the subsample of multi-surprise months. Across the 56 within-month "later" announcements in the 1988–2023 sample the re-pricing coefficient is small and insignificant, $\hat{a} = -0.15$ ($t = -1.1$, $p = 0.28$), and of the six BS controls only the three-month S&P 500 change significantly loads on the earlier shock ($\hat{\lambda} = -0.39$, $t = -2.4$, $p = 0.02$) while the slope, commodity, payroll and skewness controls do not (Figure 14). The second finding is exactly the theoretical prediction: only the shock-sensitive asset price embeds the earlier announcement, so the S&P window control is the leakage vector.

The first finding must be reported for what it is. Failing to reject $a = 0$ is not evidence *for* the innovation pole; what matters is whether the confidence interval for $\hat{a}$ excludes the cancellation value $\hat{\beta}\hat{\lambda}$. With 56 later events, only 11 of them after 1998, the test is underpowered, and we report the interval and the comparison rather than a verdict. Table 12 reports the interval and the cancellation value $\hat{\beta}\hat{\lambda}$ side by side, so the reader can see directly whether the sample discriminates between the two poles or merely fails to reject either.

Table 12: The leakage hinge, with its sampling uncertainty. The question is not whether $\hat{a}$ differs from zero but whether its confidence interval excludes the cancellation value $\hat{\beta}\hat{\lambda}$; an interval containing both poles means the sample is uninformative about the hinge rather than supportive of the innovation pole.

Quantity	Value
Within-month later announcements	56
Re-pricing coefficient $\hat{a}$	-0.1542 (0.1314)
$\text{CI}_{95}(\hat{a})$	$[-0.4118, +0.1034]$
S&P loading on the earlier shock, $\hat{\lambda}$	-0.3848 (0.2315)
Purge weight on the S&P control, $\hat{\beta}_{\text{SP}}$	+0.0871
Cancellation pole $\hat{\beta}\hat{\lambda}$	-0.0335

The implied net leakage is in any case modest here, an earlier-shock loading around 0.87, because the purge places only a small weight on the S&P control ($\hat{\beta}_{\text{SP}} = 0.09$) and multi-surprise months are rare after 1998. The mechanism is present and identifiable; its magnitude would sharpen in the multi-surprise-rich 1980s–1990s, which a longer replication of the controls would reach. It is a real *additional* channel alongside Amodeo's own explanation (the removal of the predictable, news-driven covariance that made the raw instrument strong), not a replacement for it.

The trade-off is an artefact of the conditioning set, not fundamental: ATP conditions on information that is clean but stale, PTA on information that is fresh but dirty. The remedy is to condition each within-month surprise on information updated with genuine news yet orthogonal to prior within-month shocks, $X_{2m}^{-\perp} = X_{2m}^{-} - \Lambda\hat{s}_{1m} = W_{2m}$. Operationally this is what a variable event window achieves: measure each control from the day after the previous announcement to the day before the next, so the window never spans a monetary announcement. Starting the window after the last announcement is the single operative invariant: it removes both contemporaneous leakage and the pervasive *lagged* contamination of BS's constant windows, in which every meeting's control reaches across the previous meeting (Appendix F). Because inter-announcement gaps differ, normalise each change to a per-unit-time rate or standardise by realised volatility, carry

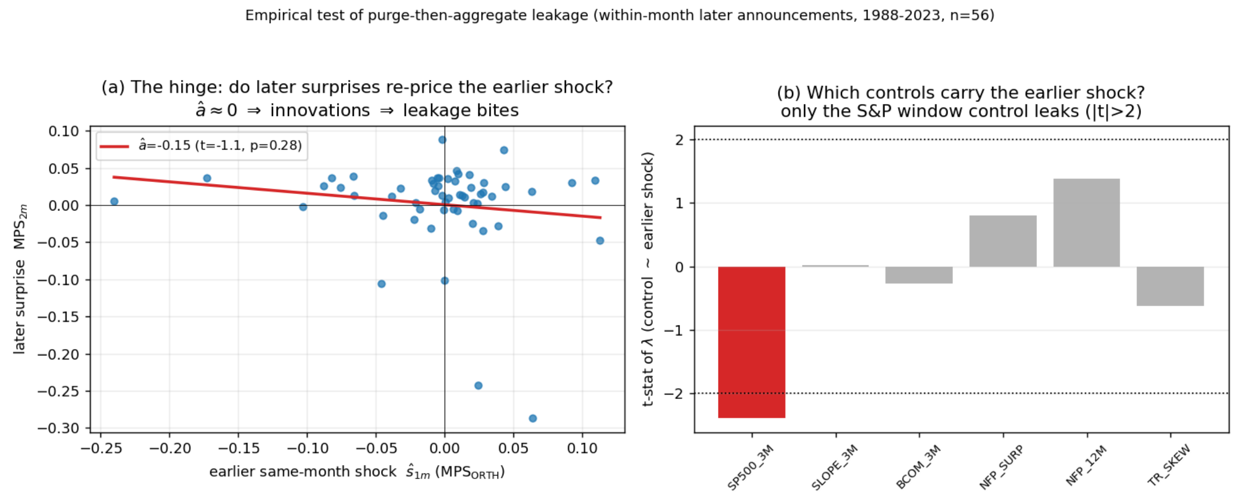

Figure 14: **Empirical leakage test** (56 within-month later announcements, 1988–2023). (a) The hinge: the re-pricing coefficient is small and imprecise, so the sample does not discriminate sharply between the two poles. (b) t-statistics of λ: only the S&P window control significantly carries the earlier same-month shock, so it is the control through which purging leaks.

the log window length as an auxiliary regressor, and use heteroskedasticity-robust inference: the identifying invariant is the window's *start*, not its length. Where windows cannot be reconstructed, identifying β on single-surprise months, where the controls are automatically predetermined, and applying it out of sample avoids the errors-in-variables attenuation entirely. The practitioner's takeaway is narrow but concrete: at the monthly or lower frequency, do not purge each surprise on a fixed multi-week window and then sum.

10 Conclusion

Orthogonalising monetary policy surprises on pre-announcement information is the right response to a real problem (the surprises are contaminated by a systematic response to news), but this paper shows it is a diagnostic, not an identification device. The purge is necessary: it is what recovers the correct signs of the impulse responses, which the raw surprise gets wrong. It is not sufficient, and in a stronger sense than the literature's set-identification language conveys. Across control sets equally defensible by the predictability diagnostic the peak output response varies by a factor of two, but that spread is smaller than the sampling error around any one of them: a between-set variance of 0.13 against a mean within-set variance of 0.90. The honest statement is not that the magnitude is set-identified but that this design does not estimate it informatively, and the reason is visible in two numbers any user of the method can report: an instrument that moves the policy rate by one to three basis points on impact, and candidate shocks whose implied policy paths differ by a factor of five hundred in persistence, six of ten wrong-signed at a year.

The maintained assumption behind the standard purge is nonetheless testable, and it fails. Bauer and Swanson read the contamination as a persistent, state-invariant underestimation of the Fed's reaction function. Conditioning on an *ex-ante* financial state rather than on the realised sign of the surprise, a common slope is rejected ($W = 17.15$ on six restrictions, wild-bootstrap $p = 0.030$), and a penalised state-dependent projection gives the surprise out-of-sample predictability that the pooled projection lacks entirely. The misspecification is visible where it matters most, in the first stage: one pooled slope strips 77% of the instrument's relevance from tightening surprises against 29% from easings. Their procedure is the corner of our result in which the reaction gap is state-invariant, and the contribution here is to show that we are not in it.

What does not follow is a different magnitude. A state-dependent projection appears to moderate the estimated response substantially, but only when it conditions on the realised sign of the surprise; substituting the ex-ante state gives -2.6% against a pooled -2.3%. The moderation is a property of conditioning on the dependent variable rather than of the Fed put, as is any within-VAR decomposition of the pooled response computed from the same split. The statement that remains is the sharper one: state dependence changes which estimator is correct without changing the answer, because on this design there is no precisely estimated answer to change.

Three further lessons discipline the procedure. The predictability the purge removes is small in the population and almost entirely ex-post, so it should be estimated as a full-sample projection, a debiased cross-fit un-purges the shock, and is best implemented as a minimal, regularised or factor projection that always includes a financial-market control. The purge does not address the information effect, which lives in the forward-guidance component and which only a high-frequency sign restriction removes; the two corrections are complementary, as is an informationally-robust conditioning set, since the financial and forecast-based controls clean different contamination. And the contamination is concentrated (in easing and crisis surprises, in the forward-guidance factor, and at the zero lower bound) with an outlier-driven coefficient, so robust and state-aware projections are not refinements but requirements.

Finally, the mechanics matter: purging each surprise before aggregating can leak genuine monetary variation whenever the control window spans an earlier announcement, and starting the window the day after the previous announcement removes both that leakage and the aggregation bias of the opposite ordering. In practice we recommend that applied work report the identified set under a robust, state-dependent purge, verify the residual is unpredictable by the union of financial, macro and forecast controls, pair the purge with a stock-price sign restriction, and use clean event windows.

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A The extended control panel

Table 13 documents the full set of pre-announcement controls used throughout the paper: the four blocks introduced in Section A, with each series' descriptive name, mnemonic, source, and coverage of the 361 FOMC meetings (1988–2023). The mnemonics match the **Controls** column of Table 14.

Two features of the panel bear on how it is used. Because the window grid is nested by construction (the one-month window sits inside the three-month window, which sits inside the six-month window), it is used with subset search or LASSO rather than a single joint F-test over the whole overlapping grid. And market prices such as yields, option-adjusted spreads, breakevens, term premia and the commercial-paper–bill spread are never revised, so their real-time and final vintages coincide; only the revisable indices and macro levels carry a distinct real-time column. Daily series are carried over non-trading days, so each anchor is the last available print on or before the anchor day, and each macro release enters at the latest observation whose reference date, plus a publication lag, precedes the meeting.

Table 13: Extended pre-announcement controls: the four blocks, with each series' descriptive name, mnemonic, source, and coverage of the 361 FOMC meetings (1988–2023). Each control is measured over a pre-meeting window ending the last trading day before the announcement (Section A); macro releases enter as a surprise and a standardised surprise, financial and public series as window changes and pre-meeting levels.

Series	Mnemonic	Source	Coverage
Block 1: Macro-announcement surprises (16 columns)			
Nonfarm payrolls	`NFP_SURP`, `NFP_NEWS`	MMS / Bloomberg survey	1997-08–2023
Industrial production	`IP_SURP`, `IP_NEWS`	MMS / Bloomberg survey	1997-08–2023
ISM manufacturing	`ISM_SURP`, `ISM_NEWS`	MMS / Bloomberg survey	1997-08–2023
Leading economic index	`LEI_SURP`, `LEI_NEWS`	MMS / Bloomberg survey	1997-08–2023
Initial jobless claims	`Claims_SURP`, `Claims_NEWS`	MMS / Bloomberg survey	1997-08–2023
Real GDP	`GDP_SURP`, `GDP_NEWS`	MMS / Bloomberg survey	1997-08–2023
Consumer price index	`CPI_SURP`, `CPI_NEWS`	MMS / Bloomberg survey	1997-08–2023
New home sales	`NHS_SURP`, `NHS_NEWS`	MMS / Bloomberg survey	1997-08–2023
Block 2: Bloomberg high-frequency financial (53 columns), window change			
S&P 500	`SP500`, `SP500_long`	Bloomberg	1988–2023
NASDAQ Composite	`NASDAQ_composite`	Bloomberg	1988–2023
Wilshire 5000	`Wilshire5000`	Bloomberg	1988–2023
US dollar index	`USD_DXY`	Bloomberg	1988–2023
Broad real dollar	`USD_broad_real`	Bloomberg	1988–2023
Euro / US dollar	`EURUSD`	Bloomberg	1988–2023
US dollar / yen	`USDJPY`	Bloomberg	1988–2023
Asia dollar index	`Asia_Dollar`	Bloomberg	2006–2023
CBOE volatility index	`VIX`	CBOE	1990–2023

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Table 13, continued

Series	Mnemonic	Source	Coverage
S&P 100 volatility	`VXO`	CBOE	1988–2023
Merrill option volatility	`MOVE`	ICE / BofA	1988–2023
VIX of VIX	`VVIX`	CBOE	2006–2023
CBOE skew index	`SKEW`	CBOE	1990–2023
NASDAQ-100 volatility	`VXN`	CBOE	2001–2023
Treasury yields, 3M–30Y	`UST_3M`, `UST_2Y`, `UST_5Y`, `UST_10Y`, `UST_30Y`	Bloomberg	1988–2023
Effective fed funds rate	`FedFunds_eff`	Bloomberg	1988–2023
10-year real yield	`Real_yield_10Y`	Bloomberg	2003–2023
OIS rates, ON–5Y	`OIS_ON`, `OIS_1Y`, `OIS_2Y`, `OIS_5Y`	Bloomberg	2002–2023
Breakeven inflation, 2Y & 10Y	`Breakeven_2Y`, `Breakeven_10Y`	Bloomberg	2005 / 1998–2023
Treasury futures, 2Y–bond	`UST_fut_2Y`, `UST_fut_5Y`, `UST_fut_10Y`, `UST_fut_bond`	Bloomberg	1988–2023 (2Y 1996+)
Investment-grade OAS	`CORP_OAS`	ICE / BofA	1996–2023
High-yield OAS	`HY_OAS`	ICE / BofA	1996–2023
1–3y corporate yield	`CORP_yield_1_3Y`	Bloomberg	1996–2023
CDX investment grade 5y	`CDX_IG_5Y`	Markit	2011–2023
CDX high yield 5y	`CDX_HY_5Y`	Markit	2012–2023
S&P GSCI total return	`GSCI_total_return`	Bloomberg	1988–2023
Gold spot	`Gold_spot`	Bloomberg	1988–2023
WTI crude oil	`WTI_crude`	Bloomberg	1988–2023
CRB commodity index	`Commodity_CRB`	Bloomberg	1988–2023
Bloomberg commodity, total return	`BCOMTR_Index`	Bloomberg	1988–2023
Bloomberg financial conditions	`BBG_FCI`	Bloomberg	1988–2023
Chicago Fed NFCI (Bloomberg)	`Chicago_Fed_NFCI`	Bloomberg	1988–2023
MBS master OAS	`MBS_master_OAS`	Bloomberg	1988–2023
MBS master yield	`MBS_master_yield`	Bloomberg	1988–2023
FNCL current coupon	`MBS_FNCL_curr_coupon`	Bloomberg	1988–2023
30-year primary mortgage	`Mortgage_30Y_primary`	Bloomberg	1988–2023
EUR cross-currency basis, 3M	`XCCY_basis_EUR_3M`	Bloomberg	2014–2023
Block 3: Public FRED series (window change and pre-meeting level)			
National financial conditions	`NFCI`	FRED (Chicago Fed)	1971–2023
Adjusted NFCI	`ANFCI`	FRED (Chicago Fed)	1971–2023
St. Louis financial stress	`STLFSI`	FRED (St. Louis Fed)	1993–2023
Weekly economic index	`WEI`	FRED (Lewis–Mertens–Stock)	2008–2023
5y5y forward inflation	`5y5y`	FRED	1990s–2023
5y / 10y breakeven inflation	`5y`, `10y`	FRED	1990s–2023
Treasury yields, 1y/7y/20y	`UST_1y`, `UST_7y`, `UST_20y`	FRED	1990s–2023
Kim–Wright term premium	`TP10`	FRED	1990–2023
Moody's Aaa / Baa yields	`Aaa`, `Baa`	FRED	full history
Commercial paper–bill spread	`CP_Tbill`	FRED	1997–2023
ICE BBB/CCC/EM OAS series are available only from 2023 under FRED's rolling licence and are dropped in favour of Moody's and the Bloomberg credit block.			
Block 3b: Other public sources			
ADS business conditions	`ADS`	Philadelphia Fed	1960–2023
OFR financial stress index	`OFR_FSI`	OFR	2000–2023

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Table 13, continued

Series	Mnemonic	Source	Coverage
ACM 10y term premium	`ACM_TP10`	NY Fed	1961–2023
SF Fed news sentiment	`NewsSent`	SF Fed	1980–2023
Block 4: Real-time (as-of-meeting) vintages of revisable series			
NFCI (real-time)	`NFCI_rt`	FRED / ALFRED	2011–2023
Adjusted NFCI (real-time)	`ANFCI_rt`	FRED / ALFRED	2011–2023
Weekly economic index (real-time)	`WEI_rt`	FRED / ALFRED	2020–2023
St. Louis stress (real-time)	`STLFSI_rt`	FRED / ALFRED	2011–2023
Claims & macro levels (real-time)	`Claims_rt`, ...	FRED / ALFRED	vintage-dependent

B The candidate shocks and control sets

Table 14 lists the shocks compared in the VAR of Section 3: the raw and BS-orthogonalised surprises, the five equally-predictive Group A control sets, and the three deliberately BS-distinct Group B sets, each with the R^2 of the surprise on the set, the canonical correlation with the six BS controls, the first month of coverage, and the event-level two-year normalisation slope.

Table 14: Shocks used in the VAR: R^2 of the surprise on the control set, canonical correlation with the six BS controls, first month of coverage, and the event-level two-year normalisation slope.

	Group	R^2	cancorr	cov. start	2y slope	Controls
`RAW`	ref.	,	,	1988-02	0.73	none (raw `MPS`)
`BS6`	ref.	0.18	1.00	1988-02	0.73	BS six predictors
`A1`	A	0.15	0.94	1988-07	0.74	`FedFunds_eff`, `NASDAQ`, `GSCI`, `MOVE`, `ADS`, `EURUSD`
`A2`	A	0.15	0.99	1994-05	0.78	`BCOMTR`, `FedFunds_eff`, `CRB`, `HY_OAS`, `NASDAQ`, `UST_fut_2Y`
`A3`	A	0.15	0.95	1989-10	0.73	`MOVE`, `CORP_OAS`, `UST_10Y`, `FedFunds_eff`, `NASDAQ`, `GSCI`
`A4`	A	0.11	0.99	1988-02	0.73	`BCOMTR_2`, `UST_5Y`, `UST_fut_10Y`, `UST_2Y`, `Gold`, `USD_DXY`
`A5`	A	0.13	0.94	1997-05	0.76	`GSCI`, `Real_yield_10Y`, `BBG_FCI`, `UST_2Y`, `MBS_yield`, `USDJPY`
`B1`	B	0.13	0.65	1998-07	0.77	`IP_SURP`, `LEI_NEWS`, `GDP_NEWS`, `GDP_SURP`, `ISM_SURP`, `UST_2Y`
`B2`	B	0.13	0.64	1998-07	0.76	`IP_SURP`, `ISM_NEWS`, `LEI_NEWS`, `LEI_SURP`, `NHS_SURP`, `UST_2Y`
`B3`	B	0.11	0.71	1998-07	0.77	`CPI_SURP`, `Claims_NEWS`, `Claims_SURP`, `NHS_SURP`, `UST_2Y`, `UST_3M`

Table 15: First-stage and stability diagnostics for every shock used in the note. ι is the impact response of the two-year rate to the ordered-first shock, per standard deviation of the instrument (9); ς is the share of the instrument's variance surviving as a reduced-form innovation (24); the last column is the largest modulus of the companion eigenvalues. T is the number of monthly observations after lags.

Shock	T	ι (bp)	ς	max root
`RAW`	396	3.03	0.761	1.0037
`BS6`	396	0.92	0.970	1.0035
`A1`	396	1.97	0.926	1.0033
`A2`	344	1.19	0.891	1.0032
`A3`	396	1.51	0.937	1.0034
`A4`	396	1.47	0.909	1.0034
`A5`	308	1.45	0.938	1.0046
`B1`	294	0.94	0.882	1.0059
`B2`	294	0.86	0.896	1.0058
`B3`	294	1.68	0.880	1.0051

Table 16: Design robustness. The `RAW` and `BS6` responses across lag length, sample and the treatment of the trending variables. With Δ log IP and Δ log CPI the output response is reported cumulated, so the columns are comparable with the levels rows. The final column reports the largest companion eigenvalue: the claim that the purge is *necessary* should not rest on rows where it exceeds one.

Specification	Shock	peak IP	IP h=12	max root
levels, 12 lags (baseline)	`RAW`	-0.75	-0.41	1.0037
	`BS6`	-2.27	-1.97	1.0035
levels, 6 lags	`RAW`	-0.62	0.08	1.0051
	`BS6`	-1.93	-1.90	1.0051
differences, 12 lags	`RAW`	-0.25	0.12	0.9583
	`BS6`	-1.88	-1.19	0.9580
differences, 6 lags	`RAW`	-0.16	0.41	0.9740
	`BS6`	-1.41	-1.39	0.9764
levels, 12 lags, 1990–2007	`RAW`	-1.13	-1.05	1.0031
	`BS6`	-1.83	-1.54	1.0032

B.1 Stationarity and the innovation share

With log industrial production and log CPI entering in levels, twelve lags and five variables imply sixty autoregressive coefficients per equation on roughly 420 monthly observations, and the estimated companion matrix has a largest eigenvalue modulus of 1.003–1.006 across the ten systems. The estimated dynamics are therefore at, or just past, a unit root: the fitted system contains no mechanism returning output to trend, so whatever a shock is doing at two years it is still doing at four. Two things follow. Level responses do not decay: a horizon-48 response carries a factor of $1.005^{48} \approx 1.27$ from the dominant root alone, so *permanent* output effects are a property of the specification and not evidence about any instrument. Sign reversals within a year, by contrast, require complex roots with a period of roughly two years and cannot be attributed to the dominant root; they are a symptom of the lag structure rather than of persistence. We report responses to 48 months because that is the convention in this literature, but read magnitudes beyond two years as uninformative.

A structural shock should be unpredictable from the system's own lags, so the share of the instrument's variance that survives as a reduced-form innovation,

$$\varsigma \;\equiv\; \frac{[\Sigma]_{zz}}{\mathrm{Var}(z_t)}, \tag{24}$$

should sit near one. What ς asks is whether the series entering the VAR is still a surprise *to the VAR*. Across the ten shocks it ranges from 0.76 to 0.97: for the weakest, roughly a quarter of the monthly instrument is explained by twelve lags of the macro-financial system. That is event-frequency-purged variation becoming predictable again after monthly aggregation, and it is the same aggregation channel Section 9 studies from the control-window side. It is also a reminder that a purge validated at meeting frequency is not automatically a purge at the frequency of the VAR.

Table 17: The choice of policy variable does not rescue the purged instrument. Each row re-estimates the internal-instrument VAR of Section 3 with a different variable in the policy slot, on the common 1990–2023 sample ($n = 408$). ι is the monthly first stage of (9) in basis points and F its heteroskedasticity-robust first-stage F; ρ_6 and ρ_{12} are the scale-invariant policy path of (11). The shadow rate is **?**, spliced with the effective funds rate after February 2022. The ten-year yield is included as a placebo: a monetary surprise should not move it, and the purged instrument does not, with an impact of -0.08 bp and $F = 0.00$.

	RAW				BS6			
Policy variable	ι	F	ρ_6	ρ_{12}	ι	F	ρ_6	ρ_{12}
Effective funds rate	1.95	7.03	4.56	4.47	0.44	0.26	4.34	-1.89
3-month bill	2.03	4.27	2.71	2.62	0.60	0.28	1.90	-0.89
1-year Treasury	3.09	10.16	1.88	1.33	1.09	1.05	0.41	-3.32
2-year Treasury (baseline)	3.03	5.79	1.92	1.39	0.92	0.54	0.01	-4.67
5-year Treasury	2.42	3.39	2.21	2.00	0.51	0.17	-1.53	-6.74
10-year Treasury *(placebo)*	1.23	1.03	3.42	3.16	-0.08	0.00	2.11	27.37
Wu–Xia shadow rate	2.89	5.15	1.80	1.67	1.11	0.78	-0.57	-3.14

C The identified set: construction and supporting figures

The set of Section 4.1 is built by repeatedly drawing six-control subsets from a pre-announcement pool, keeping those that satisfy a stated predictability criterion, purging the surprise on each and running it through the monthly VAR of Section 3, and shading the pointwise minimum–maximum and inter-quartile envelope of the resulting impulse responses with `BS6` and `RAW` overlaid. Two criteria are used. The *equally-predictive* set (Figure 2) draws from the full-coverage 1988+ pool and keeps subsets whose in-sample R^2 is at least the `BS6` level; 150 subsets qualify, and the twelve-month industrial-production response spans $[-0.7\%, -2.5\%]$ with median -1.6%. The *BS-distinct* set draws from the broader 1998+ panel containing the macro surprises and keeps subsets with canonical correlation below 0.75 against `BS6` and $R^2 \geq 0.08$; only 44 qualify, and the resulting band is both shallower ($[-0.2\%, -1.5\%]$) and thinner (Figure 17). That genuinely BS-distinct yet predictive control sets are scarce is itself consistent with the finding that, on the full sample, nothing beats BS6.

Selecting on *in-sample* R^2 selects the overfit subsets, which is the fragility documented in Section 7.1, so part of the band's width may measure overfitting rather than identification ambiguity. Table 18 rebuilds the set on a *cross-validated* criterion and reports the sensitivity to the subset size.

Table 18: The identified set under a cross-validated selection criterion. Draws are kept when the five-fold cross-validated R^2 of the surprise on the subset exceeds the threshold, rather than the in-sample R^2, so the set is not populated by the overfit subsets of Section 7.1. The three rows vary the number of controls per draw.

Subset size	N	band	median	IQR
$\lvert X\rvert = 3$	150	$[-1.94, -0.65]$	-1.18	0.54
$\lvert X\rvert = 6$	150	$[-2.15, -0.56]$	-1.32	0.48
$\lvert X\rvert = 10$	150	$[-2.34, -0.44]$	-1.39	0.46

The observation that all 150 qualifying sets contain a financial-market control is a property of the draw pool as much as of the data, and is reported as such.

D The state-dependent decomposition: derivations

This appendix derives the asymmetric reaction gap of Section 2.2 from a kinked policy rule, proves the two propositions of Section 2.4, and shows that the same algebra organises three empirical regularities of the body as three faces of one object: a systematic response coefficient that is not constant. It is deliberately in the reduced-form spirit of Chen (2025), and it makes precise the sense in which we agree with Hack et al. (2024) that projecting the surprise on expected fundamentals cannot, by itself, deliver a clean shock.

D.1 A rule that generates the Fed put

The asymmetry in ϕ can be derived rather than assumed. Let the Fed set the expected path in response to its growth outlook, $i_\tau = a_g\, g_\tau + \varepsilon_\tau$, and let the outlook respond to the intermeeting equity return with a kink,

$$\frac{\partial g_\tau}{\partial rx_\tau} = \kappa_-\, \mathbf{1}\{rx_\tau < 0\} + \kappa_+\, \mathbf{1}\{rx_\tau \geq 0\}, \qquad \kappa_- > \kappa_+ \geq 0. \tag{25}$$

The kink is Cieslak and Vissing-Jorgensen's (2021) *driver* view: equity declines transmit to consumption through the wealth effect and to investment through balance sheets, and both channels are convex in the decline, so a 10% drawdown downgrades the outlook by more than a 10% rally upgrades it. It is also what their FOMC-minutes evidence shows directly, with negative mentions of the stock market predicting cuts and positive mentions predicting nothing. Composing, $a(s) = a_g \kappa_s$, so $a_- > a_+$: the rule itself is asymmetric.

Now let agents price a *linear* rule. Futures prices and survey point forecasts are linear in observables, so the priced slope is the projection of $a(s)$ on a constant, $\bar{a} = \sum_s a_s \omega_s$ with ω_s the

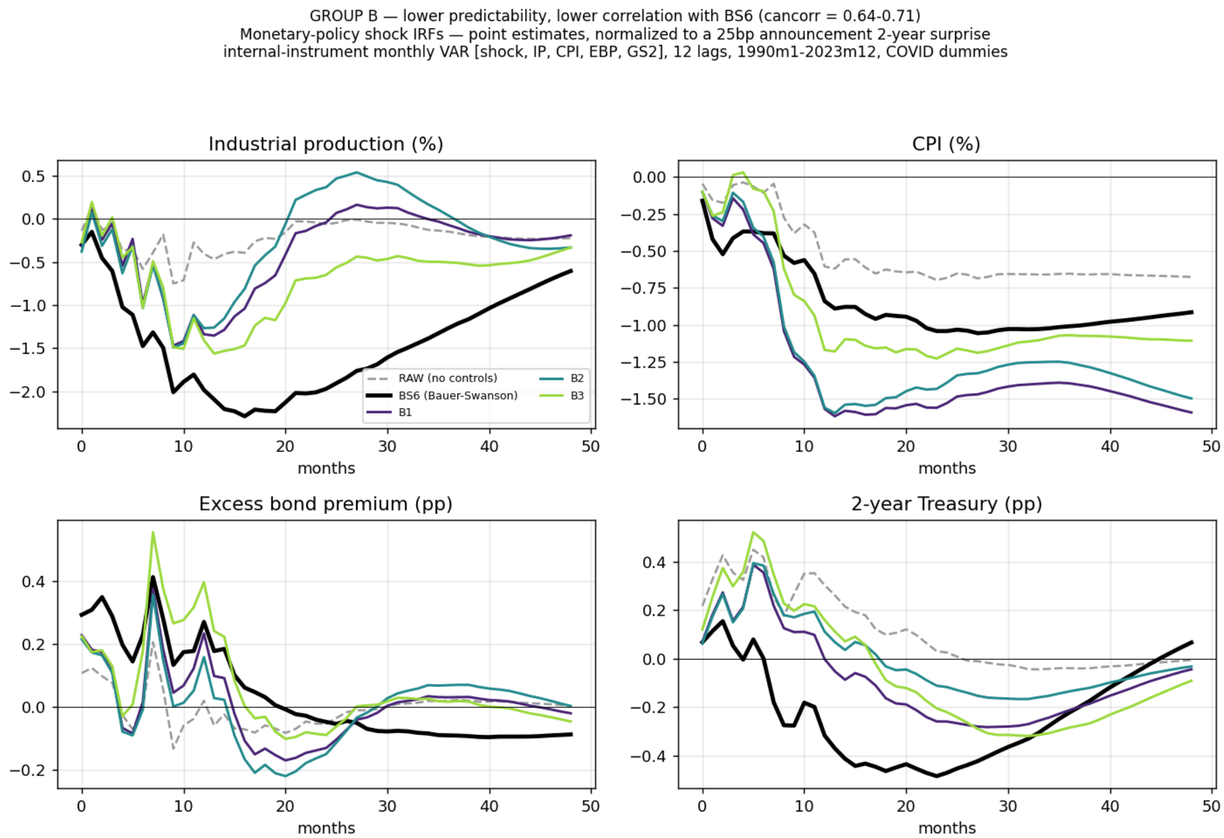

Figure 15: **Group B: lower predictability, lower correlation with BS** (canonical correlation 0.64–0.71). Removing a *different* predictable component yields a shallower, far less persistent output decline and a larger, more persistent price decline than the BS shock.

variance weights of (8). Then

$$\phi_- = a_- - \bar{a} = (a_- - a_+)\,\omega_+ \;>\; 0 \;>\; -(a_- - a_+)\,\omega_- = \phi_+, \tag{26}$$

so the reaction gap inherits the kink exactly, with ϕ_- and ϕ_+ of opposite sign and magnitudes in inverse proportion to the states' news variances. Two features are worth noting. First, no irrationality is required: agents who know a_- and a_+ but must quote a single slope produce (26) mechanically, so "underestimation of the rule" need not mean agents are mistaken about the rule, only that the instrument through which they express it is linear. Second, because $\omega_- \gg \omega_+$ (stress states carry most of the news variance), we have $|\phi_+| \ll \phi_-$: the tightening-side gap is small, which is why $\phi_+ \approx 0$ is a good approximation and why the tightenings look clean. The Bauer and Swanson (2023a) case is $\kappa_- = \kappa_+$ with an additive pricing wedge, which returns $\phi_- = \phi_+ = \delta$.

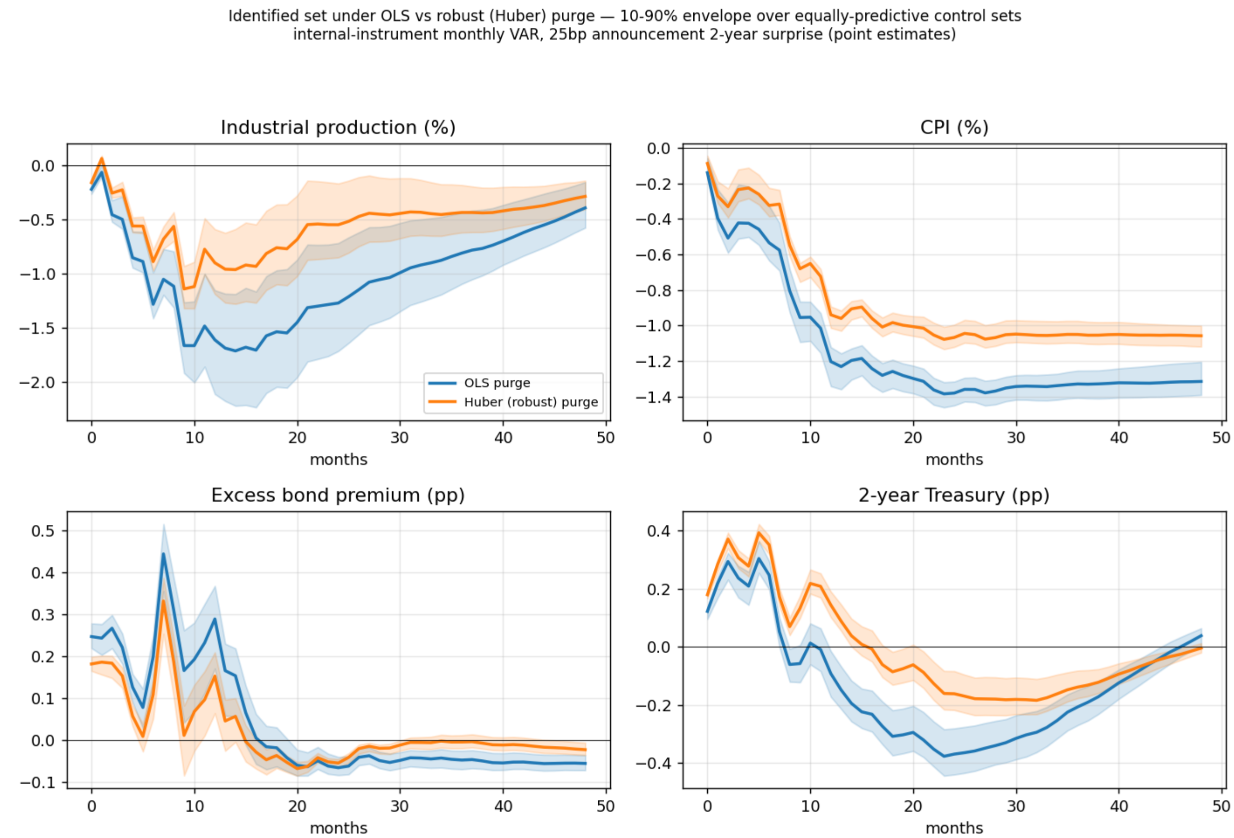

Figure 16: **Identified set under OLS versus a robust (Huber) purge.** 10–90% envelope (shaded) and median (line) of the IRFs across equally-predictive control sets. The robust purge shifts the band toward more moderate magnitudes, median twelve-month industrial production from -1.6% to -0.9%, and roughly halves its width, from an interquartile range of 0.50 to 0.24. Much of the set-identification width under OLS therefore came from a handful of influential events that different control subsets fit differently.

D.2 Why the state-conditional fit is not a test

Writing $MPS_\tau = \phi(s_\tau)x_\tau + \varepsilon_\tau$ and conditioning on the state gives (6), and hence

$$\left.\frac{R^2(-)}{R^2(+)}\right|_{\phi_-=\phi_+} = \frac{\mathrm{Var}(x \mid -)}{\mathrm{Var}(x \mid +)} \cdot \frac{\sigma_\varepsilon^2(+) + \phi^2\,\mathrm{Var}(x \mid +)}{\sigma_\varepsilon^2(-) + \phi^2\,\mathrm{Var}(x \mid -)},$$

which exceeds one whenever the stress state carries the larger news variance, *even under* H_{BS}. The state-conditional R^2 is therefore not a test of state dependence in ϕ; the interaction slope of (13) is. A second and independent problem attaches to conditioning on $\mathrm{sign}(MPS_\tau)$ rather than on the ex-ante state: since $\mathrm{sign}(MPS_\tau) = \mathrm{sign}(\varepsilon_\tau + \phi(s_\tau)x_\tau)$, the split truncates ε_τ, so $\mathbb{E}[\varepsilon_\tau \mid MPS_\tau < 0] < 0$ and the subsample slopes are not the population ϕ_- and ϕ_+. Both problems are avoided by (13), which is why the body treats the sign split as description and the ex-ante interaction test as evidence.

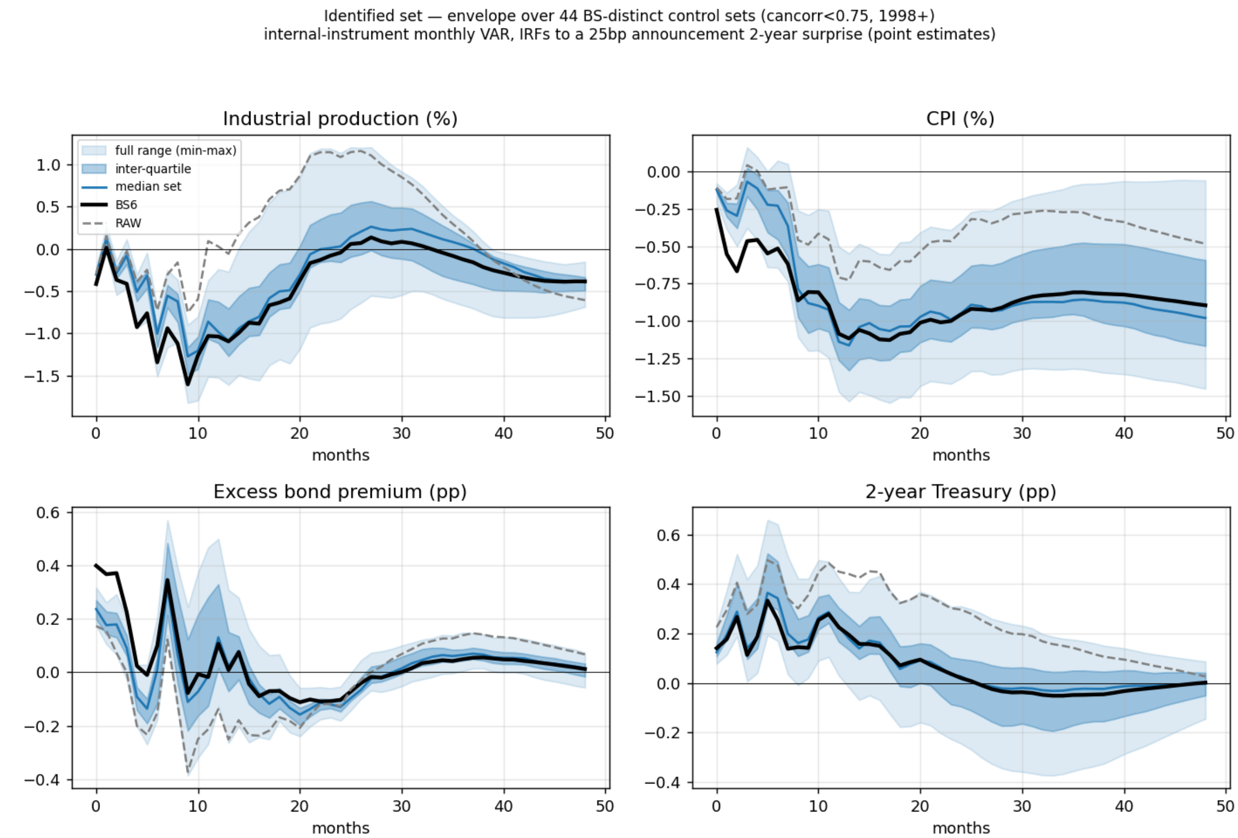

Figure 17: **Identified set, BS-distinct purges** (44 six-control sets, canonical correlation with `BS6` < 0.75, 1998+). `BS6` and `RAW` are shown on the same post-1998 window. The band is shallower and thinner than the equally-predictive set.

D.3 The constant-coefficient purge, and Proposition 1

Pooled OLS returns the single slope b of (8), a variance-weighted average with $\phi_- > b > \phi_+$, so the purged surprise is

$$\tilde{e}_t \;\equiv\; MPS_t - b\,x_t \;=\; \varepsilon_t \;+\; \big[\phi(s_t) - b\big]\,x_t. \tag{27}$$

The residual is *not* ε_t: it carries the deviation of the state's true response from the pooled average. This is precisely the $\tilde{\phi}_t' x_t$ that Hack et al. (2024) show survives a regression on expected fundamentals: removing the *average* systematic response leaves the *state-specific* part. Replacing the single slope with a state-specific one, $b(s_t) = \phi(s_t)$, removes the systematic term exactly,

$$MPS_t - b(s_t)\,x_t \;=\; \varepsilon_t, \tag{28}$$

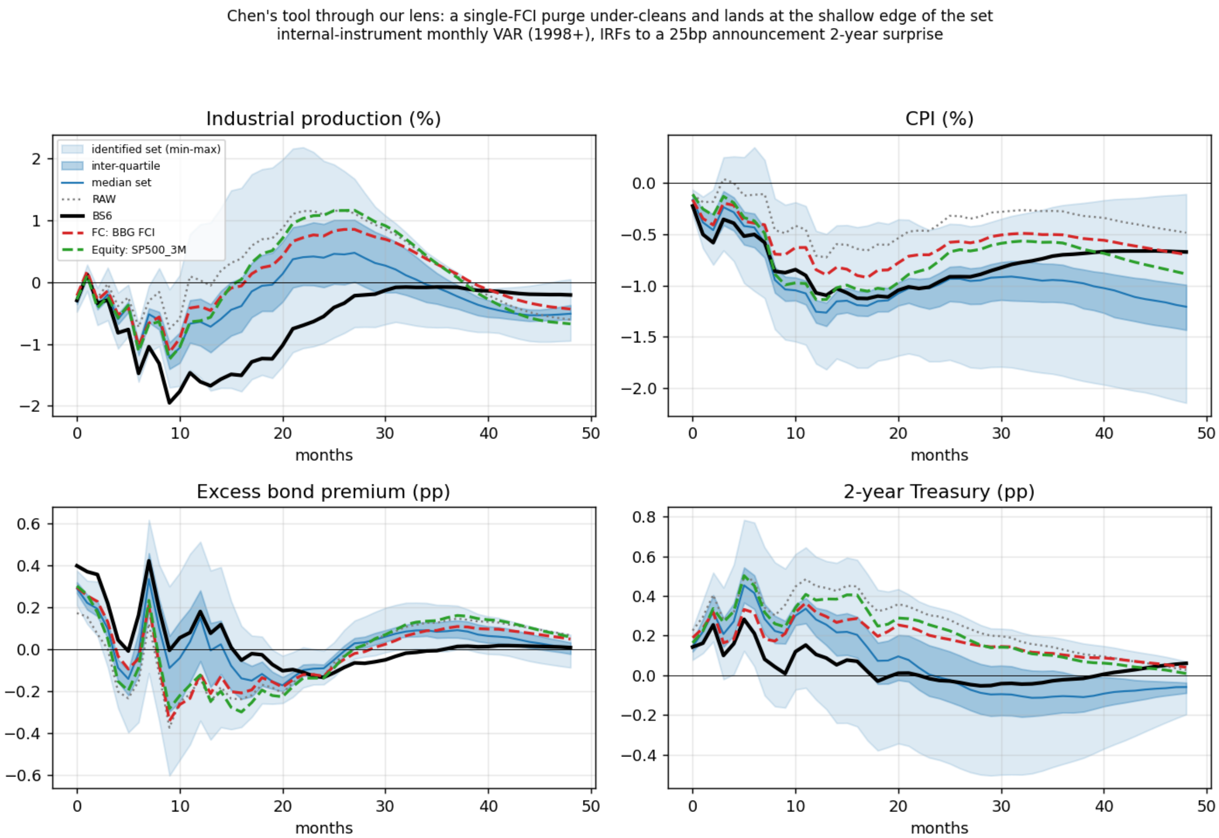

Figure 18: **Chen (2025)'s tool through the identified set.** Purging the surprise on a single financial-conditions index (FC), on the equity return, and on the six BS controls, all through the same internal-instrument VAR (1998+). All three flip the output sign and land in the identified set; depth is monotone in how much each proxy removes ($R^2 = 0.05, 0.10, 0.19$), so a lightly-cleaned single-index purge sits near the median and BS6 at the deep edge. Both purges chase the financial-conditions factor of Section 7.3; how deep a contraction one reports is a function of how completely the chosen proxy spans it.

which is Proposition 1. Residual (27) collapses to ε_t if and only if $[\phi(s_t) - b]x_t = 0$ almost surely, which requires either

$$\text{(i)}\ \ \phi(s_t) \equiv \phi\ \ \text{(no state dependence, so } b = \phi), \qquad \text{or} \qquad \text{(ii)}\ \ x_t \equiv 0\ \ \text{(no intermeeting revisions).} \tag{29}$$

These are exactly the two knife-edge cases of Hack et al. (2024). Neither holds in the data (the Fed put is state dependence in ϕ, and there are genuine intermeeting revisions), which is why the purge is necessary but not sufficient. Operationally (28) is the state-specific purge of Section 5.5: interact the controls with the ex-ante financial state and project within regime. It nests the pooled purge, and reduces to it precisely under (29)(i).

Because $\phi_- > b > \phi_+$, the sign of the leftover contamination in (27) flips with the state. In easing and stress states the bracket is positive, the purge *under-cleans*, and a predictable non-policy component remains, which is where ex-post predictability survives leave-one-out and where the raw output puzzle lives. In tightening and normal states the bracket is negative, the purge *over-cleans*: it subtracts more systematic response than is present and injects a component of the opposite sign, which is the over-cleaned tightening response of Section 5.5 (raw -1.6%, pooled-purged -3.9%). Both are quantitatively what the data show, because $\mathbb{E}[x^2 \mid \text{stress}]$ is large, so w_- dominates (8) and b is fit almost entirely to the easing observations. The same object delivers the functional-form failure: since $\phi(\cdot)$ is a step (or smooth threshold) function of the state, the conditional mean $\mathbb{E}[MPS_t \mid x_t] = \phi(s(x_t))\, x_t$ has a kink, and no linear projection (BS or MAR, both linear) can represent a kinked mean, so a RESET-type test on the linear purge rejects. Sign asymmetry, over-cleaning and nonlinearity are one non-constant coefficient viewed three ways. This misspecification is also itself a source of the set-identification: a projection that under-cleans one state and over-cleans the other traces out different magnitudes depending on how the sample weights the two, which is why an outlier-robust purge, by down-weighting the high-leverage easing observations that pin down $\hat{\beta}$, moves the estimated response so far.

D.4 Proposition 2: the impulse-response bias

Embed the surprise in the proxy VAR and let the outcome respond both to the structural shock and, because x_t is genuine news, directly to the state,

$$y_{t+h} \;=\; \theta_h\, \varepsilon_t \;+\; \psi_h(s_t)\, x_t \;+\; u_{t+h}, \qquad u_{t+h} \perp (\varepsilon_t, x_t), \tag{30}$$

where $\psi_h(s_t)$ is the transmission of the news itself, allowed to differ across states because the news that accompanies easing surprises (financial deterioration) need not move output like the news behind tightenings (strong activity). Reading $\phi(s_t) = \gamma(s_t) - \tilde{\gamma}(s_t)$ as the Bauer and Swanson (2023a) gap between the Fed's true reaction coefficient and the one agents price, their *persistent underestimation of the rule* is $\phi > 0$ on average, and the asymmetric predictability says that this underestimation is itself state-dependent.

Normalise the shock by its impact on the policy indicator $p_t = MPS_t$, so the proxy-IV estimand for an instrument z_t is $b_h(z) = \mathrm{Cov}(y_{t+h}, z_t) / \mathrm{Cov}(p_t, z_t)$. For the raw surprise, the pooled purge $\tilde{z}_t^{\text{pool}} = MPS_t - b\, x_t$, and the state-specific purge $\tilde{z}_t^{\text{state}} = MPS_t - b(s_t)x_t = \varepsilon_t$,

$$b_h^{\text{raw}} = \frac{\theta_h \sigma_\varepsilon^2 + \mathbb{E}[\psi_h(s)\phi(s)x^2]}{\sigma_\varepsilon^2 + \mathbb{E}[\phi(s)^2 x^2]}, \qquad b_h^{\text{pool}} = \frac{\theta_h \sigma_\varepsilon^2 + (\psi_h^- - \psi_h^+)\, W}{\sigma_\varepsilon^2 + V_\phi}, \qquad b_h^{\text{state}} = \theta_h, \tag{31}$$

where $W = (\phi_- - b)\, \mathbb{E}[x^2 \mid s{=}-]\, \Pr(-) > 0$ is the state-conditional covariance the pooled purge leaves uncleaned (equal and opposite across states, because the pooled normal equations set $\mathbb{E}[(\phi(s) - b)x^2] = 0$), and $V_\phi = \mathbb{E}[\phi(s)^2 x^2] - b^2 \sigma_x^2 \geq 0$ is the x^2-weighted variance of the reaction gap. The cross terms in εx drop because $\varepsilon \perp x$, and the state-specific residual is ε_t exactly because $b(s_t) = \phi(s_t)$ removes $\phi(s_t)x_t$ within each state. Hence

$$b_h^{\text{pool}} - \theta_h \;=\; \frac{(\psi_h^- - \psi_h^+)\, W \;-\; \theta_h\, V_\phi}{\sigma_\varepsilon^2 + V_\phi}, \tag{32}$$

which is Proposition 2. The bias has two pieces: a *transmission-asymmetry* term $(\psi_h^- - \psi_h^+)W$, nonzero only when *both* the reaction gap is state-dependent and the news transmits asymmetrically; and an *attenuation* $-\theta_h V_\phi$ from the extra variance the uncleaned residual carries. Both vanish when

ϕ is state-invariant ($W = V_\phi = 0$): that is Bauer and Swanson's (2023a) benchmark, in which a single β suffices and the pooled purge returns θ_h. Their result is the $\phi_- = \phi_+$ corner of (32).

Imposing $\phi \equiv \delta$ throughout (so that the instrument carries no state dependence and ψ_h enters only through its mean) the raw-surprise expression in (31) collapses to (7) of the body, whose sign is that of ψ_h for small δ. This is the formal content of implication (T2): under H_{BS} the raw output response is biased *upward*, and by the same amount in both states, so the output puzzle should not be confined to one of them.

D.5 From sign to pure time-variation

The two-state threshold is the sharpest special case; the same algebra covers a fully time-varying coefficient. Let the systematic response be ϕ_t, indexed by time, so that

$$MPS_t \;=\; \varepsilon_t \;+\; \phi_t\, x_t, \qquad \tilde{e}_t \;=\; MPS_t - b\, x_t \;=\; \varepsilon_t \;+\; (\phi_t - b)\, x_t, \tag{33}$$

with $b = \mathbb{E}[\phi_t x_t^2]/\mathbb{E}[x_t^2]$ the same variance-weighted average. Residual (33) is the Hack et al. (2024) object in full generality. What differs across special cases is not whether contamination remains but whether it is *correctable by conditioning*. When $\phi_t = \phi(s_t)$ for an *observable* state (the intermeeting equity return, a financial-conditions threshold, an indicator for the zero lower bound or the forward-guidance regime) one can replace b with $\phi(s_t)$ and clean within regime, as in (28). The zero-lower-bound and path-factor concentration documented in Section 6 is precisely this. The harder case is a ϕ_t that drifts with an *unobserved* regime; then no static interaction identifies it and one needs a state model, a rolling or Markov-switching projection that lets $b_t = \hat{\phi}_t$ track the coefficient, which is why the endogenously-timed high-predictability state of Section 6.3 is the natural estimator for the drift the state split only approximates.

D.6 A regime split confirms the mechanism but not the misspecification

Figure 19 runs the exercise of Section 5.5 through the zero-lower-bound regime instead, as the observable-state generalisation predicts it should, and the contrast is the point. First, the raw output puzzle is even more concentrated at the effective lower bound than on the easing side: identified from ZLB meetings alone the raw instrument sends industrial production sharply the wrong way, the normal-times surprises are already correctly signed, and the pooled purge removes three times as much variance inside the ZLB regime as outside it (0.34 against 0.11). Second, and this is where the regime split differs from the state split, correcting it regime by regime barely moves the aggregate response (-1.9% against the pooled -2.1%, versus -0.7% for the state-specific purge). The reason is visible in the projection coefficients: the pooled fit correlates 0.93–0.97 with the regime-own fits, whereas it correlated only 0.27 with the tightening-own fit. The ZLB regime changes *how much* systematic response there is, but not the *direction* of the projection; the state changes the direction itself. Misspecification lives in the state, not in the calendar regime.

This contrast is only as good as its identification, and the two cuts are collinear: easings cluster in crises, which cluster at the zero lower bound. This contrast is only as good as its identification, and the two cuts are collinear: easings cluster in crises, which cluster at the zero lower bound. Table 6 in Section 5.3 reports the cross-tabulation and the four-cell specification; if the off-diagonal cells are thin, the horse race between the two cuts is not identified on this sample and the claim that misspecification lives in the state rather than the calendar regime cannot be sustained.

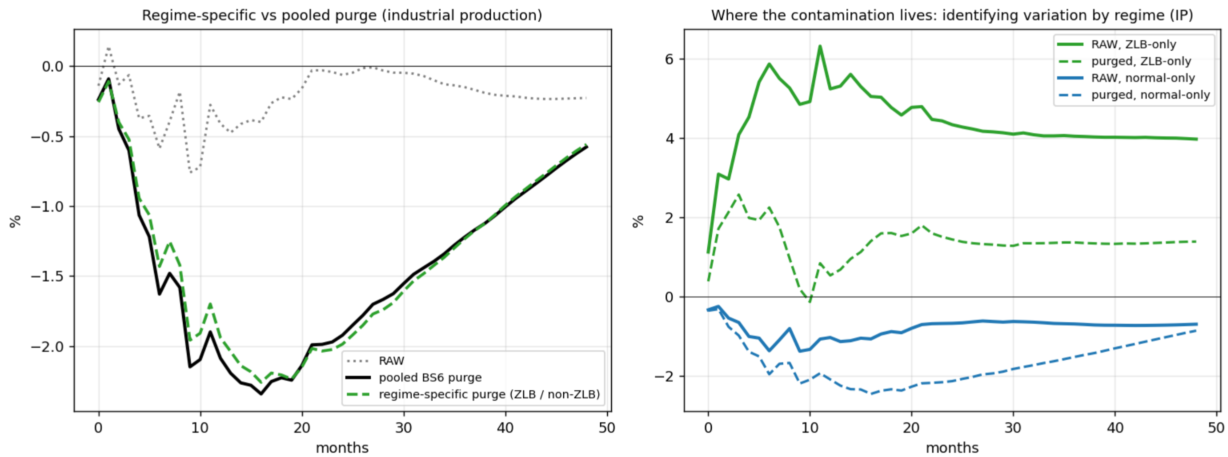

Figure 19: **The same exercise through an observable regime (the zero lower bound).** *Left:* industrial-production response to the raw, pooled-BS6-purged, and regime-specific-purged surprise; the regime-specific purge sits almost on top of the pooled one, unlike the state-specific purge of Figure 5. *Right:* the identifying variation split by regime: the raw puzzle is concentrated in the ZLB meetings and removed by the purge, while normal-times surprises are already clean and mildly over-cleaned.

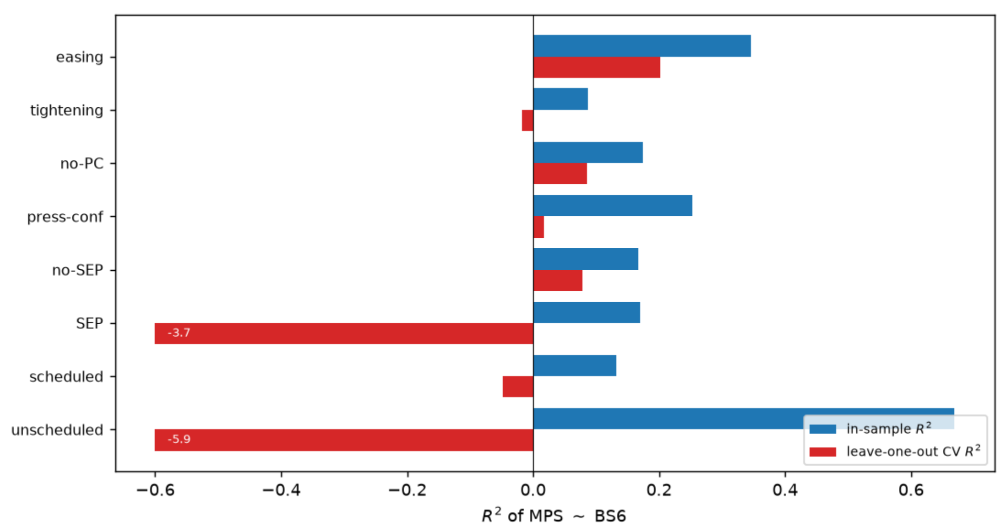

Figure 20: **Predictability by meeting type: only the Fed-put asymmetry survives.** In-sample R^2 (blue) versus leave-one-out cross-validated R^2 (red) of *MPS* ∼BS6 by subset. Easing surprises are robustly predictable out of sample; the high in-sample R^2s of unscheduled and SEP meetings collapse under cross-validation (values off-scale annotated). As §D.2 notes, these are descriptive.

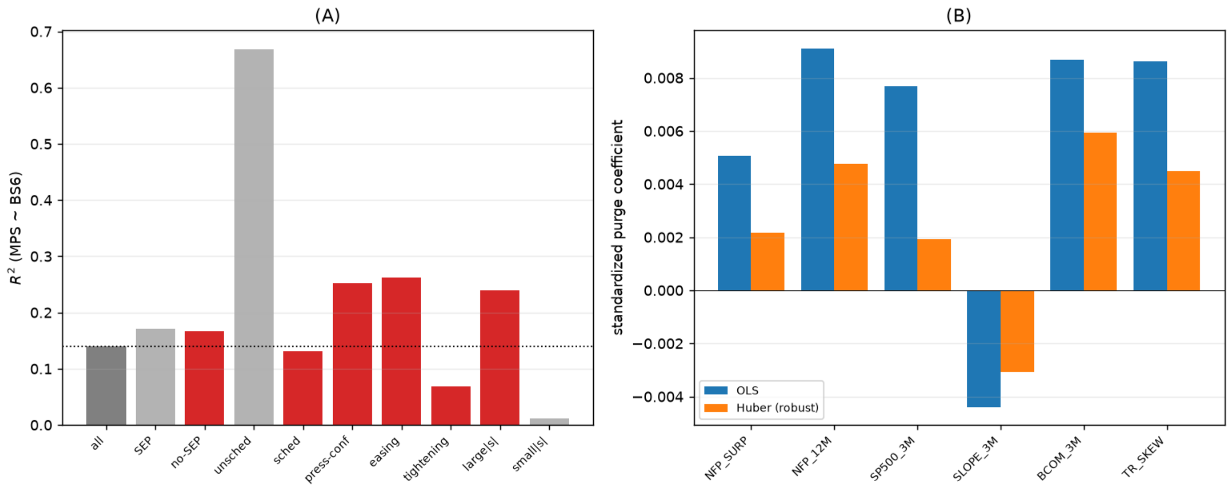

Figure 21: **Cross-section and robustness.** Left: *in-sample* R^2 of *MPS* $\sim$`BS6` by meeting type (red = significant at 5%; dotted line = full sample); these small-subsample R^2s are largely over-fit. Right: robust (Huber) purge coefficients run from about a quarter to seven-tenths of the OLS values, with the equity control shrunk most: the projection is driven by a few influential events; the smooth RESET nonlinearity test is insignificant ($p = 0.28$), consistent with the threshold form of (25).

Table 19: The meetings that fit the projection. Events are ranked by Cook's distance in *MPS* $\sim$`BS6` over the 360-meeting sample. ΔR^2 is the fall in the projection's R^2 when the event is dropped, so a *negative* entry marks an event the pooled slope is being dragged toward rather than one it is fitting. The June 2020 meeting is reported first because its leverage of 0.94, against a sample mean of 0.019, makes every aggregate influence share computed with it uninformative; the text reports the ranking both ways.

Meeting	Episode	*MPS*	Cook's *D*	ΔR^2
2020-06-10	–	0.005	7.400	-0.008
2008-12-16	move to the zero lower bound	-0.250	0.124	+0.028
1998-10-15	unscheduled 25bp cut eight days before the scheduled meeting	-0.276	0.061	+0.003
2008-10-08	coordinated unscheduled cut, post-Lehman	0.017	0.053	-0.010
2020-03-15	unscheduled cuts to zero as COVID hit	-0.213	0.052	+0.013
2001-04-18	unscheduled 50bp cut between meetings	-0.283	0.051	+0.008
2008-09-16	–	0.103	0.048	-0.014
1991-12-20	–	-0.287	0.040	+0.005
2008-03-11	Bear Stearns weekend	0.090	0.037	-0.004
2008-10-29	coordinated unscheduled cut, post-Lehman	-0.047	0.031	-0.003
These 10 of 360 events carry 93% of total influence.				

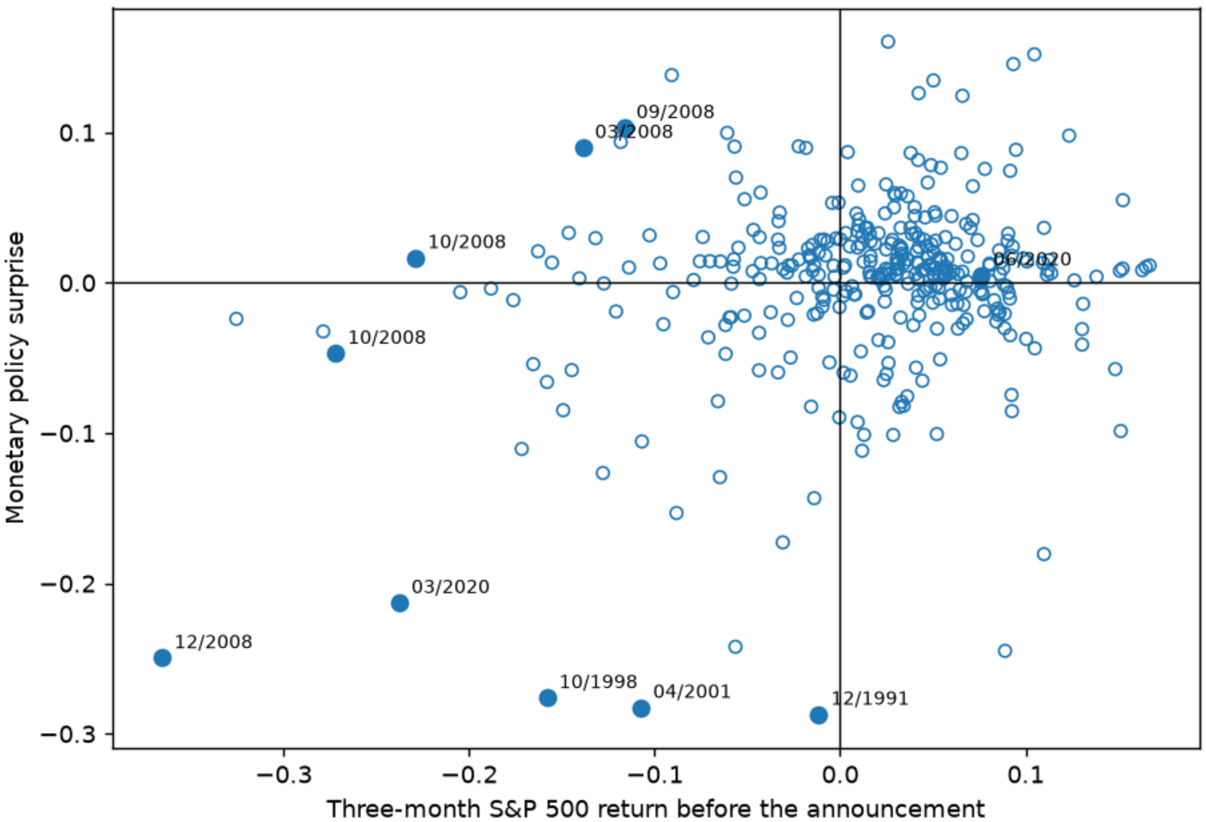

Figure 22: **Which meetings fit the projection** (Section 6.1). The monetary policy surprise against the three-month pre-announcement S&P 500 return, one circle per FOMC announcement. The ten observations with the largest Cook's distance in *MPS* ∼`BS6` are filled and labelled by month and year. June 2020 is among them and is not representative: its leverage of 0.94 is fifty times the sample mean and it carries 87% of total influence on its own, so the ranking should be read from the remaining nine.

E Supplementary results

F Leakage under purge-then-aggregate: derivations and simulation

This appendix formalises the mechanism of Section 9, states and proves the pooled leakage decomposition, reports the simulation that traces it, and gives the clean-window specification ready for a longer daily panel.

F.1 Setup

To make the mechanism precise, index months $m = 1, \dots, M$ and within-month announcements $t = 1, \dots, T_m$, and following Amodeo (2026) write each raw surprise as a predictable response to news plus a structural shock,

$$MPS_{tm} = \beta' X^{-}_{tm} + s_{tm}, \tag{34}$$

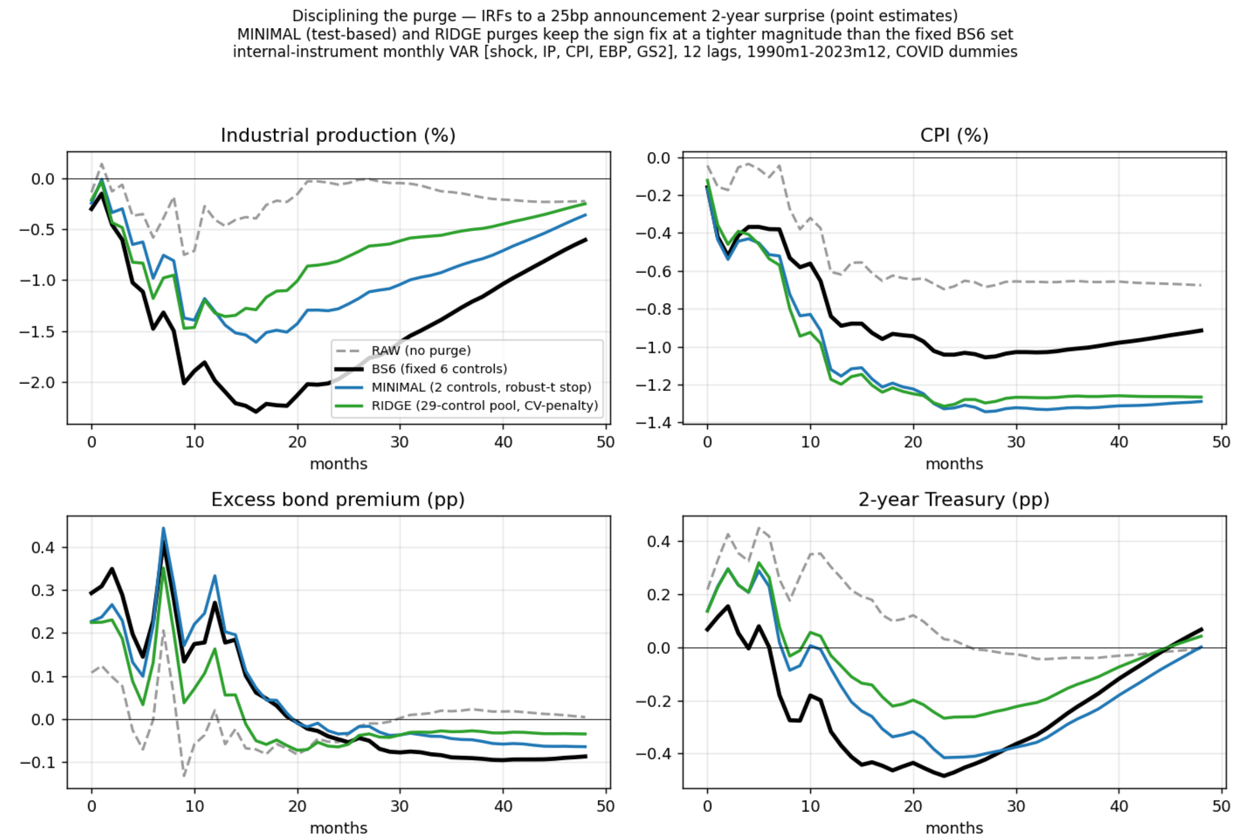

Figure 23: **Disciplining the purge** (Section 7.2). IRFs to a 25 bp announcement two-year surprise. The minimal (two-control, robust-t stop) and ridge (29-control pool, CV-penalty) purges keep the sign correction of `BS6` while delivering a tighter, mutually consistent magnitude (peak IP $\approx -1.3\%$ against -2.0%). The debiased / cross-fit residual is omitted: because the out-of-fold prediction is essentially the sample mean, that residual $\approx$ `RAW` and does not deliver a purge.

with $\mathrm{Var}(s_{tm}) = \sigma_s^2$, within-month dependence $\mathrm{Cov}(s_{1m}, s_{2m}) = \rho\sigma_s^2$, and monthly object of interest the sum of shocks $S_m = \sum_t s_{tm}$. The surprises instrument a monthly design in which a policy indicator has first stage $p_m = \pi S_m + \nu_m$, the structural equation is $y_m = \theta p_m + e_m$ with $\mathrm{Cov}(p_m, e_m) \neq 0$ and $\mathrm{Cov}(S_m, e_m) = 0$, and θ is the estimand. The first within-month control is predetermined with respect to every month-m shock, $X_{1m}^- = W_{1m}$ with $\mathrm{Cov}(W_{1m}, s_{jm}) = 0$; the second is measured after the first announcement, so its asset-price entries embed the first shock,

$$X_{2m}^- = W_{2m} + \Lambda\, s_{1m}, \qquad \mathrm{Cov}(W_{2m}, s_{jm}) = 0, \;\; \mathrm{Cov}(W_{1m}, W_{2m}) = 0, \tag{35}$$

where W_{2m} is genuine inter-announcement news and Λ collects the high-frequency responses of the controls to a unit monetary shock: nonzero for asset-price controls, zero for a pre-dated payroll surprise. Figure 29 places this timing on the calendar for October 2008.

With two within-month surprises the raw monthly total is $MPS_{1m} + MPS_{2m} = \beta' W_{1m} + \beta' W_{2m} + \beta' \Lambda s_{1m} + S_m$. Aggregate-then-purge forms $z_m = MPS_{1m} + MPS_{2m} - \gamma' X_{1m}^-$; since W_{2m}, Λs_{1m} and

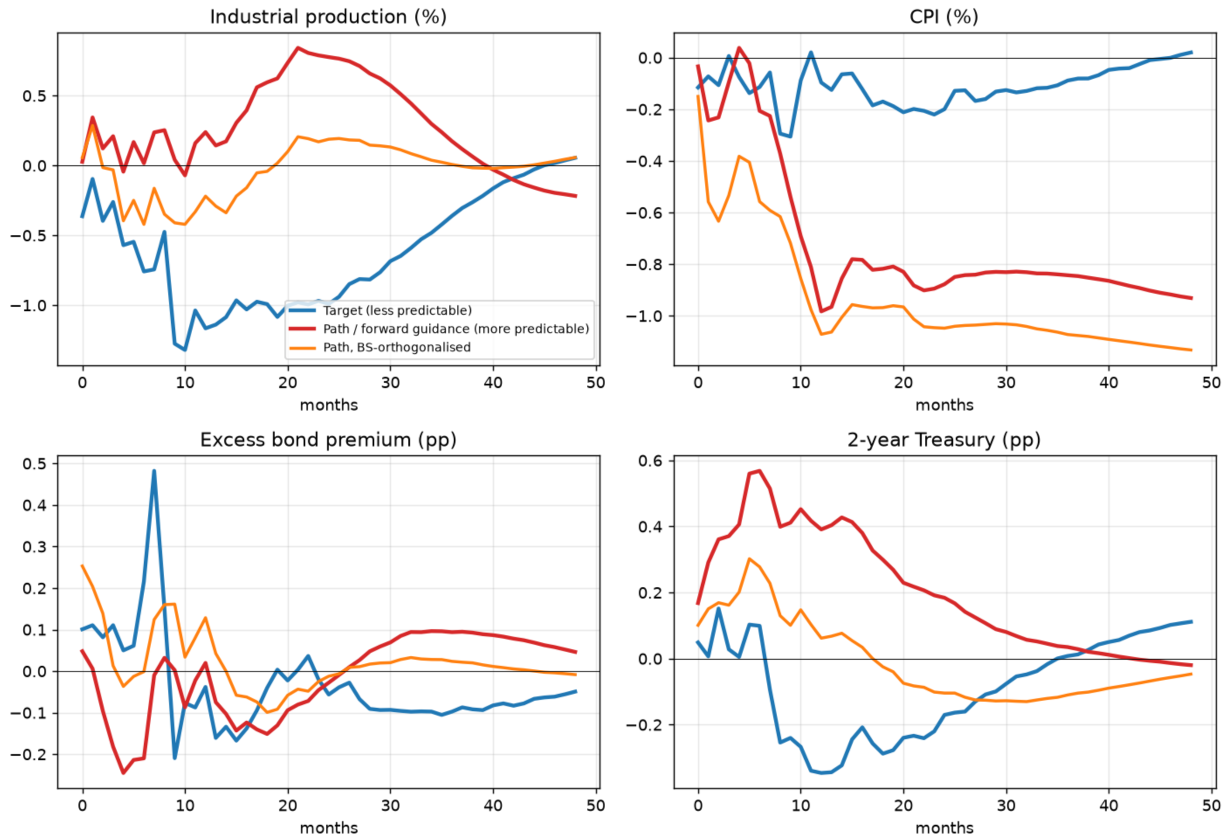

Figure 24: **Target versus path IRFs** (Section 6.2). The less-predictable target factor gives a clean contraction; the more-predictable path factor gives a wrong-signed output response; orthogonalising the path on the BS controls moves it back toward the target. The purged *target* factor is shown for completeness but should not be read as an experiment: its impact on the two-year rate is negative (-0.0013), so its apparent deepening to -2.26% is a sign flip. Magnitudes are less comparable across factors than signs, because each is normalised to a 25 bp two-year move it achieves through different loadings.

S_m are orthogonal to X_{1m}^-, the projection gives $\gamma = \beta$ and

$$z_m = S_m + \underbrace{\beta' W_{2m}}_{\text{predictable inter-event news}} + \beta' \Lambda s_{1m}, \qquad \mathrm{Cov}(z_m, e_m) = \beta' \,\mathrm{Cov}(W_{2m}, e_m) \neq 0. \tag{36}$$

ATP keeps *all* the shock variation (S_m enters with unit loading, so it is *strong*), but it is *biased* through the retained news term $\beta' W_{2m}$, exactly Amodeo's result. Purge-then-aggregate removes that term; the question is what else it removes.

The key correction to a naive treatment is that β is estimated *once*, by pooling all meetings, so the contaminated controls, which appear only for the later within-month surprises, have their effect on $\hat{\beta}$ *diluted* by the many clean single-surprise meetings. Allowing the later surprise to re-price the earlier shock, write for a two-surprise month

$$MPS_{2m} = \beta' W_{2m} + a\, s_{1m} + s_{2m}, \tag{37}$$

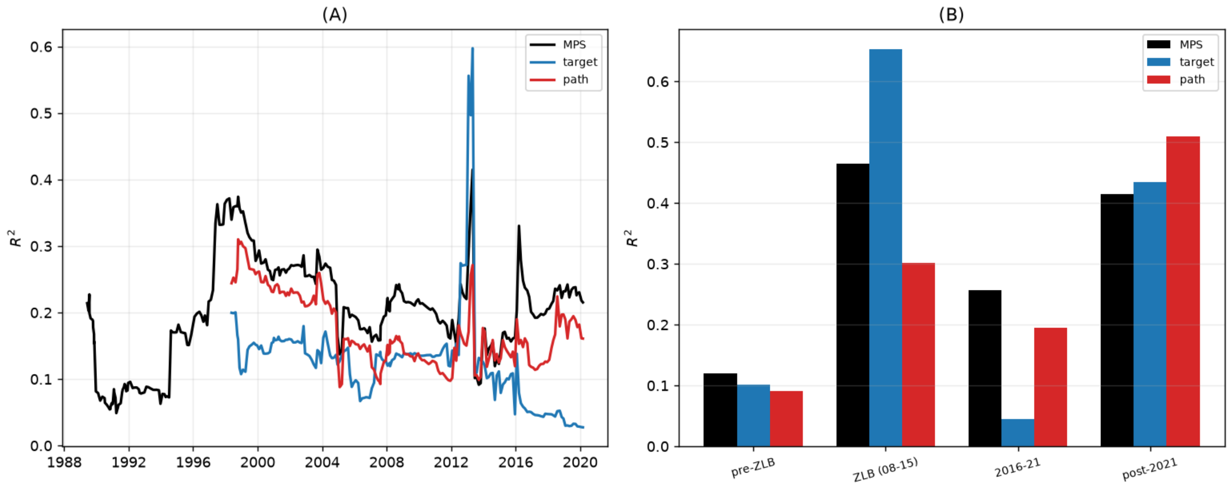

Figure 25: **Regime-dependence of the projection** (Section 6.3). Left: rolling 70-meeting R^2 of the surprise (and the target/path factors) on the BS controls: strongly time-varying. Right: regime-specific R^2; predictability peaks at the ZLB (target) and after 2021 (path), though a sup-Wald test finds no sharp break in the coefficients ($p = 0.38$).

where $a \in [0, \beta'\Lambda]$ is the *re-pricing coefficient*, the extent to which Announcement 2's surprise loads on the first shock. To keep the algebra transparent take a single asset-price control ($k = 1$, $\lambda \equiv \Lambda$), genuine-news variance σ_w^2, normalise $\sigma_s^2 = 1$, and let q be the share of months carrying a contaminated later surprise.

F.2 The pooled leakage decomposition

Proposition 3 in Section 9 states the result in the notation of (37); we restate the two displays here in the appendix's notation and record the consequences that the main text compresses. The pooled least-squares purge coefficient satisfies

$$\hat{\beta} \xrightarrow{p} \frac{\beta\,\sigma_w^2(1+q) + q\,\lambda\,(a+\rho)}{\sigma_w^2(1+q) + q\,\lambda^2}, \tag{38}$$

and the PTA instrument in a two-surprise month is

$$\tilde{z}_m = \left(1 + a - \hat{\beta}\lambda\right) s_{1m} + s_{2m} + (\beta - \hat{\beta})\,(W_{1m} + W_{2m}), \tag{39}$$

so the loading on the earlier shock is $1 + a - \hat{\beta}\lambda$ and the net leakage is $L \equiv \hat{\beta}\lambda - a$. The derivation is the one given with the proposition: the event-2 residual is $\hat{u}_{2m} = (\beta - \hat{\beta})W_{2m} + (a - \hat{\beta}\lambda)s_{1m} + s_{2m}$, the event-1 residual is $\hat{u}_{1m} = (\beta - \hat{\beta})W_{1m} + s_{1m}$, and summing gives (39); (38) is the pooled normal equation with a share $q/(1+q)$ of observations carrying $\mathrm{Var}(X_{2m}^-) = \sigma_w^2 + \lambda^2$ and $\mathrm{Cov}(X_{2m}^-, MPS_{2m}) = \beta\sigma_w^2 + \lambda(a+\rho)$. All expressions are verified by simulation to three decimals.

Two consequences follow, both *diluted* relative to a per-event calculation. First, the earlier shock is mismeasured: its loading is $1 + a - \hat{\beta}\lambda$, and with $a = 0$, $\rho = 0$, $\sigma_w^2 = \lambda = \sigma_s^2 = 1$ and $q = 0.12$, (38) gives $\hat{\beta} \approx 0.63$ and a loading ≈ 0.37: a large attenuation of s_{1m}. Second, relevance falls but

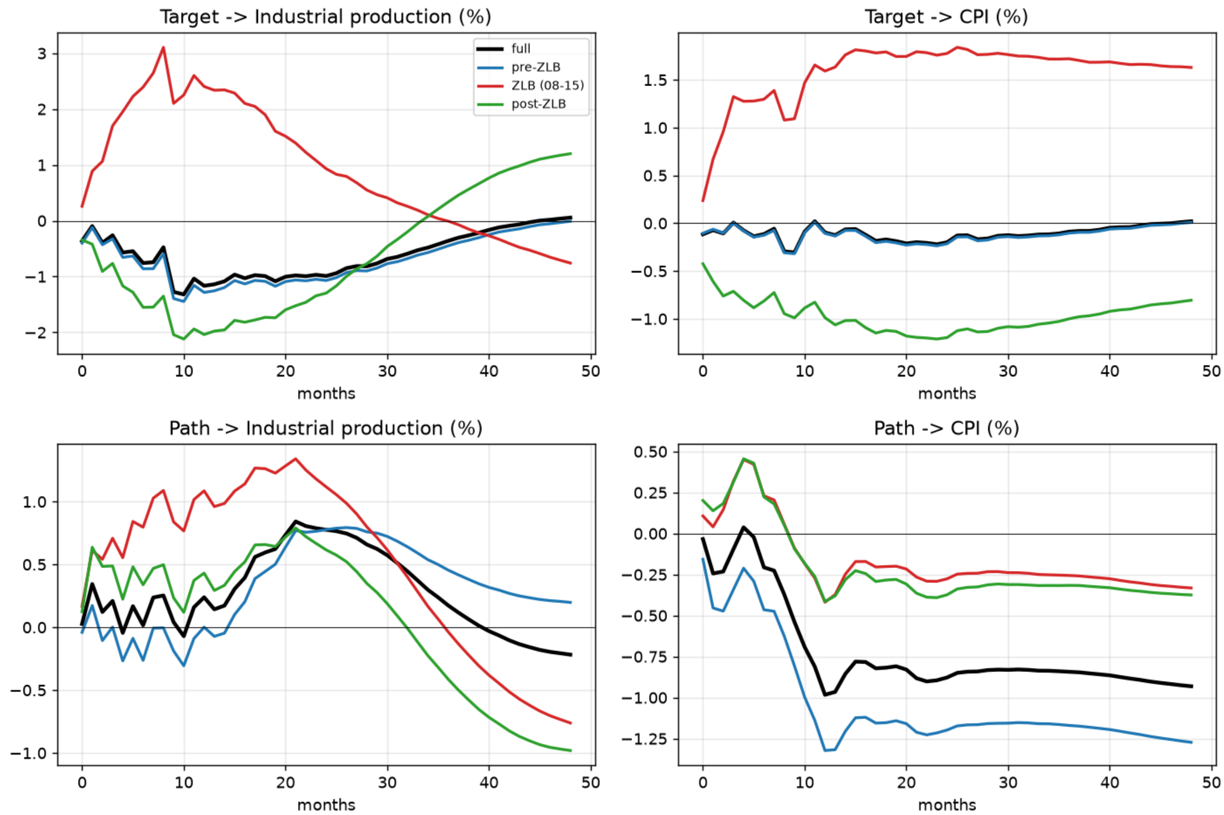

Figure 26: **Target and path IRFs by regime** (Section 6.3). Impact re-identified from each regime's residual covariance, dynamics full-sample, so this isolates changes in contemporaneous transmission rather than regime-specific propagation. The target shock is contractionary pre-ZLB but wrong-signed and explosive when identified from the ZLB (funds rate pinned ⇒ degenerate target factor), and post-ZLB it has a *negative* impact on the two-year rate, so its plausible-looking output response is sign-flipped rather than clean; the path is much more stable.

only modestly in the aggregate: because s_{2m} is intact and single-surprise months are clean, the aggregate first-stage F falls only a little from this channel alone (about 8% in that calibration, more with larger λ or q), and the estimator remains *consistent* for θ so long as the injected news term $(\beta - \hat{\beta})W$ is uncorrelated with e_m: it becomes biased only through that news term, the ATP channel of (36). Leakage is therefore primarily a *weak-instrument* phenomenon, not an asymptotic bias. This pooled treatment is more conservative than a per-event calculation, which would set $\hat{\beta}$ from the contaminated moment alone and overstate L; pooling is what BS and Amodeo actually do.

F.3 Simulation

Figure 30 confirms the mechanism by simulation: the earlier-shock loading $1 + a - \hat{\beta}\lambda$ rises from heavy attenuation under innovation ($a = 0$) to unity under cancellation ($a = \beta\lambda$), where $\mathrm{Corr}(\tilde{z}, S)$ in multi-surprise months tracks it (panel a); leakage grows with the controls' asset-price sensitivity λ in the innovation world (panel b); and conditioning on *clean* windows keeps the full first stage

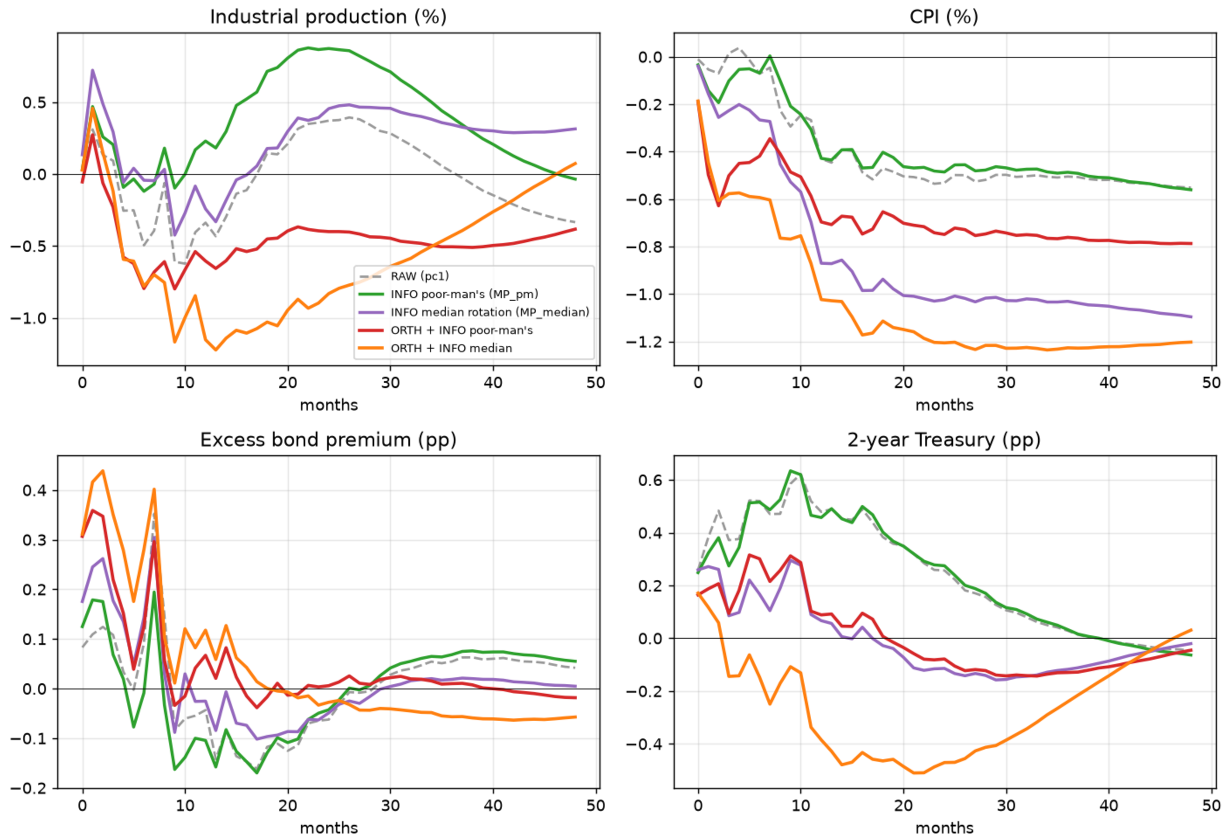

Figure 27: **Robustness to the JK identification** (Section 8). The information-effect correction under the poor-man's classification versus the median rotation. Neither `INFO` shock is contractionary on its own; both require the purge, and the purge combined with the median rotation gives the cleanest contraction.

while BS-style constant windows, which contaminate every meeting's control with the previous meeting's shock, cost the most (panel c). Throughout, the estimator stays consistent, so leakage is a *relevance*, not a bias, phenomenon in the canonical design.

F.4 Lagged contamination under constant windows

A natural conjecture is that only *contemporaneous* same-month contamination matters, while a control window reaching back into a *previous*-period announcement is harmless; the simulation refines it. On consistency the conjecture holds: when shocks are orthogonal to the macro error, both contemporaneous and lagged control-contamination leave $\hat{\theta}$ *consistent*, because a lagged shock $s_{1,m-1}$ injected into $\tilde{z}_m$ is orthogonal to the current target S_m and, with lag controls, to e_m (it is classical measurement error in the instrument), whereas contemporaneous leakage is the piece that mismeasures a *component of the current target* S_m and maps onto Amodeo's multi-surprise-month problem. On relevance, lagged contamination is not free: with BS's *constant* windows (a fixed three-month change) *every* meeting's control reaches across the previous meeting, so contamination is pervasive rather than confined to about 8% of months, attenuating the pooled $\hat{\beta}$ sharply (from

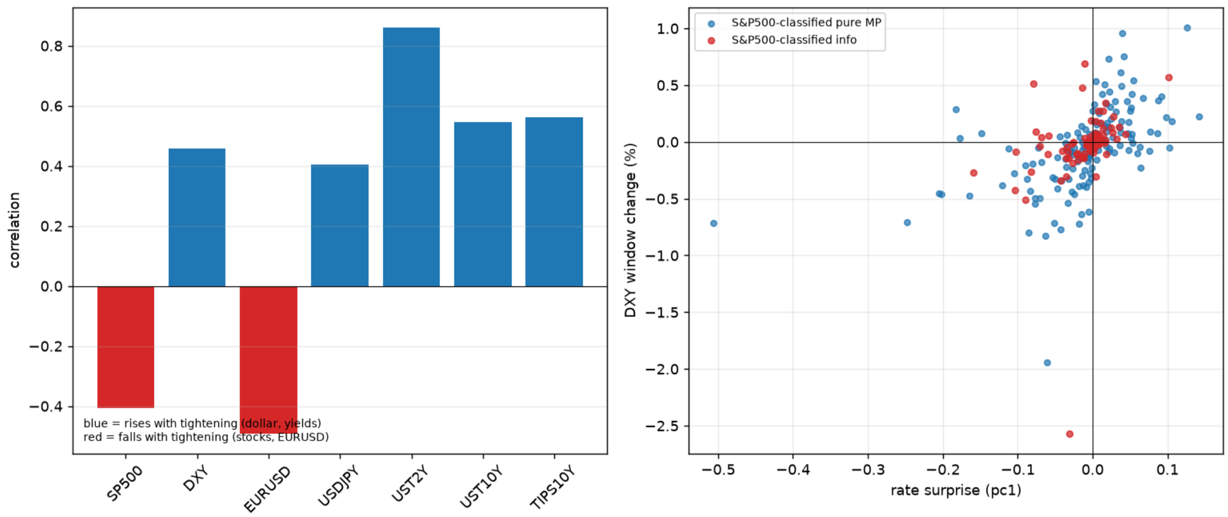

Figure 28: **Other assets** (Section 8.1). Left: co-movement of the surprise with each USMPD asset (blue rises with a tightening, red falls): negatively with the S&P 500 (-0.42) and the euro (-0.51), positively with the dollar index ($+0.49$), the yen ($+0.46$) and yields ($+0.87$ at two years), exactly as a dominant monetary component implies. Right: the dollar cannot *separate* the two shocks: the S&P-classified information events (red) do not occupy a distinct region in `pc1`–DXY space, because a tightening and a positive information shock both appreciate the dollar.

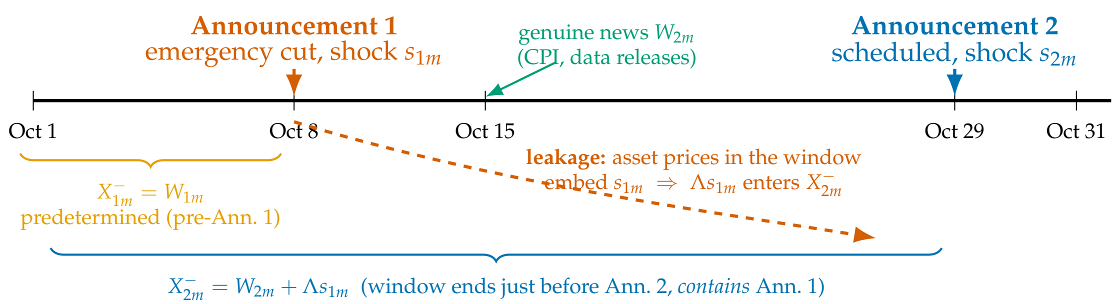

Figure 29: Timing within a two-surprise month (October 2008). The pre-first-surprise control X_{1m}^- predates every month-m shock; the second control X_{2m}^- ends just before Announcement 2 and therefore spans Announcement 1, so $X_{2m}^- = W_{2m} + \Lambda s_{1m}$. ATP purges the monthly total on X_{1m}^- only (missing W_{2m}); PTA purges Announcement 2 on X_{2m}^-, which contains s_{1m}.

0.70 to ≈ 0.29 in the calibration of Figure 30c) and lowering the aggregate first stage to about 0.64 of the clean-window value, a larger hit than contemporaneous leakage alone, while the attenuated $\hat{\beta}$ also *under*-purges and can reintroduce contamination bias if the retained news is correlated with the macro error. Lagged contamination is thus an efficiency-and-relevance cost and contemporaneous leakage a target-mismeasurement cost, and both are removed by the window fix below. One caveat on magnitudes: Amodeo's empirical first stage falls by roughly two orders of magnitude under PTA, and the leakage channel here, under realistic pooling and multi-surprise shares, accounts for only a modest part of that: the bulk is consistent with his own explanation, the removal of the predictable,

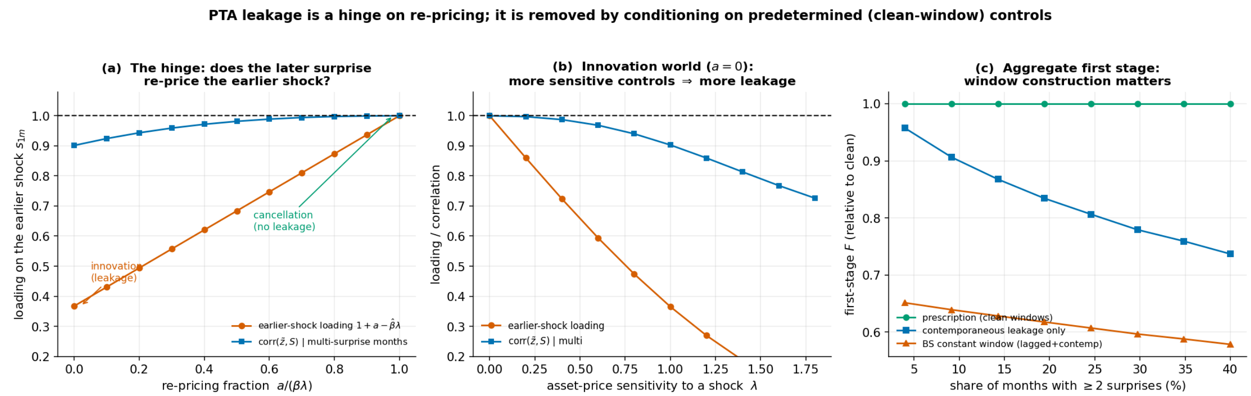

Figure 30: **Simulation of the leakage hinge.** (a) The earlier-shock loading $1 + a - \hat{\beta}\lambda$ rises from heavy attenuation under innovation ($a = 0$) to unity under cancellation ($a = \beta\lambda$). (b) In the innovation world, more asset-price-sensitive controls (larger λ) leak more. (c) Aggregate first-stage F relative to the clean-window prescription: predetermined windows keep the full first stage; contemporaneous leakage costs a little; BS-style constant windows cost the most.

news-driven covariance that made the raw instrument strong. Leakage is a real and identifiable *additional* mechanism, sharpened where λ is large, re-pricing a is small, and multi-surprise months are frequent.

F.5 The clean-window specification

The clean-window specification makes this operational for the extended sample. The direct test compares the two windows head to head. For announcement i (previous announcement $i-1$) and each asset-price control c with daily level $P_{c,d}$ ($g = \log$ for prices/indices, the identity for yields), define

$$X^{\text{const}}_{c,i} = g\big(P_{c,\tau_i-1}\big) - g\big(P_{c,\tau_i-1-66}\big), \qquad \text{(BS: fixed 66 trading-day window)} \tag{40}$$

$$X^{\text{clean}}_{c,i} = g\big(P_{c,\tau_i-1}\big) - g\big(P_{c,\tau_{i-1}+1}\big), \quad \text{length } L_i, \qquad \text{(starts the day } \textit{after} \text{ announcement } i-1). \tag{41}$$

Because L_i varies, normalise the clean change to a per-trading-day rate $\widetilde{X}^{\text{clean}}_{c,i} = X^{\text{clean}}_{c,i}/L_i$ (or a random-walk $/\sqrt{L_i}$) and carry $\log L_i$ as a nuisance regressor. Then, for each window $W \in \{\text{const}, \text{clean}\}$, run

$$\text{(predictability)} \quad MPS_i = \alpha_W + \gamma_W' \widetilde{X}^W_i + \phi_W \log L_i + e^W_i, \qquad R^2_W, \;\; u^W_i \equiv \hat{e}^W_i, \tag{42}$$

$$\text{(first stage)} \quad \Delta i^{2y}_i = \pi_W u^W_i + \nu_i, \qquad F_W = (\hat{\pi}_W/\text{se})^2, \tag{43}$$

$$\text{(channel)} \quad \widetilde{X}^W_{c,i} = \kappa + \lambda^W_c \hat{s}_{i-1} + \text{err} \quad \text{on within-month later events.} \tag{44}$$

Leakage is present if, at matched predictability ($R^2_{\text{clean}} \approx R^2_{\text{const}}$), the clean-window instrument is more relevant, $F_{\text{clean}} > F_{\text{const}}$; the channel regression corroborates it if $\lambda^{\text{const}}_{\text{SP500}} \neq 0$ while $\lambda^{\text{clean}}_{\text{SP500}} \approx 0$. On simulated daily prices with the real FOMC calendar the pipeline behaves as designed: the clean-window residual retains more of the true shock ($\text{Corr}(u, s) = 0.86$ versus 0.77). The current daily coverage begins only in the late 1990s, so this comparison is not yet powered on our sample; we report it as a specification rather than a result, and note that the 1980s–1990s, where multi-surprise months are frequent, is where it would bite.